

A Hybrid Deep Learning Approach for Dengue Fever Prediction in real-world case study

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Declaration

The project submitted herewith is a result of my own efforts in totality and in every aspect of the project works. All information that has been obtained from other sources had been fully acknowledged. I understand that any plagiarism, cheating or collusion or any sorts constitutes a breach of TAR University rules and regulations and would be subjected to disciplinary actions

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Abstract

Dengue remains a serious and worsening problem in subtropical regions. Statistical models have clear limits, and while deep learning is more accurate, health officials hesitate to trust it: not because it does not work, but because it is unclear how it reaches its predictions. This project set out to fix that. It combines an LSTM encoder with an additive temporal-attention mechanism, the Interpretable LSTM-Transformer or ILT, so that the attention weights come out of the same forward pass that produces the forecast: every prediction records which input days received the strongest attention, with feature-level importance derived from those weights. The model was trained on Singapore dengue and weather data (2013–2022, 3,652 daily records) and compared against SARIMA, a Random Forest and a plain LSTM at five horizons over thirty fixed seeds. It reduced RMSE against SARIMA by 42.5% to 52.0% at every horizon, and improved on the plain LSTM by 1.6%, a margin significant only at one day ahead and pooled. All twenty top-ranked attention features matched documented dengue risk factors (exact binomial $p = 7.98 \times 10^{-4}$), at 1.227 times the parameters of a plain LSTM and with sub-quadratic scaling in input length. Applied unchanged to an unseen 2023–2025 period, the frozen model held positive skill at one and seven days only; a two-parameter affine recalibration restored it at all five. The trade conventionally assumed between accuracy and explanation was negligible here; the useful question is whether the explanation is true, which this study shows can be tested.

Keywords: Attention mechanism, dengue forecasting, hybrid model, interpretable AI, LSTM-Transformer

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Chapter 1

Introduction

1.0 Introduction

Dengue fever is a problem for public health in warm places. It is getting worse because of cities getting bigger the weather changing and more people traveling. Dengue fever is spread by Aedes mosquitoes. To deal with this health authorities need to be ready for outbreaks. They need to know when an outbreak is likely to happen so they can prepare. This means they need to stock up on medicine and supplies spray insecticides to kill the mosquitoes and warn people.

1.1 Overview

Dengue fever is a disease that will not go away. It is carried by Aedes mosquitoes. This disease puts a lot of pressure on health systems in parts of the world. Between 2000 and 2019 the number of Dengue fever cases increased a lot globally. In Singapore there was an outbreak in 2020. The reasons for this are clear. Cities are getting bigger the weather is. People are traveling more. All these things create conditions for Aedes mosquitoes to breed and spread Dengue fever. So controlling the virus is not about getting rid of it but also about reducing its impact.

This study looks at how to predict when an outbreak of Dengue fever will happen. It looks at five different time frames: one day, one week, two weeks, three weeks and four weeks ahead. Each of these time frames is important for public health actions. If health authorities know that an outbreak is likely to happen they can take action. They can stock up on medicine and supplies spray insecticides or warn people.

For a time people used statistical models to try to forecast when outbreaks would happen. These models looked at what happened in the past to make predictions. They are good at seeing patterns that happen at the time every year and trends that happen over a few years.. These models have a hard time making good predictions when the data does not follow a simple pattern.

1.2 Motivation

So why do we need another forecasting model? The simple answer is that getting predictions right is not always the same as getting useful predictions. Some models like the deep learning kind, especially the ones called LSTMs, are really good at making predictions. They are often better than the old ways of doing things with statistics. But people do not always trust these new models. And that is fair. These models are like black boxes. You put information in. They give you an answer. But they do not tell you how they got that answer (Loh et al., 2022; Bharati et al., 2024). In technical terms, a black-box model is one whose internal parameters — the learned weight matrices and activation patterns — cannot be directly mapped to human-interpretable variables. Unlike a regression coefficient, an LSTM hidden state carries no inherent meaning that a practitioner can read or challenge.

In a public health setting this is a real problem. You cannot ask a health officer to spend a budget on things like fumigation just because a model says so, without knowing why the model is saying that. Some recent studies have pointed this out again and again: being able to understand how a model works is not just a nice thing to have, it is something that is absolutely necessary (Loh et al., 2022; Bharati et al., 2024). So the reason for doing this work is not just to build something that gets the job done.

1.3 Research Gaps and Problem Statement

1.3.1 Research Gaps

A research gap is a statement about the literature. It identifies what is not yet known, not yet tested, or inconsistently reported, and to carry academic weight it must be parameterised — stated relative to a defined body of work, a defined time window, and a defined methodological domain. Claiming that dengue forecasting is “under-explored” would fail that test, because the field is in fact mature and heavily published. The three gaps below are therefore stated as re-examinations within a mature literature rather than as empty spaces in it, and each is substantiated in Chapter 2, where it is traced to specific studies (Table 2.1) and mapped to the architectural component that addresses it (Table 2.3).

Gap 1 — Methodological and Transparency Gap. Extensive literature published between 2019 and 2026 has established that recurrent and hybrid deep learning models outperform statistical baselines such as SARIMA on multivariate dengue incidence forecasting in tropical and subtropical settings (Zhao et al., 2023; Chen & Moraga, 2025a; Cheng et al., 2026; Castellanos Garzón et al., 2026). Within that same body of work, however, interpretability is almost always supplied after the fact by perturbation-based tools such as SHAP and LIME (Loh et al., 2022; Joly et al., 2024; Prome et al., 2024), and those tools produce approximated attributions whose stability on sequential, multi-step forecasts is not established. What remains unknown is therefore not whether deep learning can forecast

dengue — that is settled — but whether a dengue forecasting architecture can obtain its explanations from its own forward pass rather than from an external approximator, without surrendering the accuracy advantage that motivated the move to deep learning in the first place.

Gap 2 — Evidence Gap on the Cost of Interpretability. Where intrinsic attention mechanisms have been combined with recurrent encoders for epidemic forecasting, the reported evaluation is confined to predictive error (Wen & Li, 2023; Hu et al., 2024; Majeed et al., 2025). The computational price of the interpretability — training time, inference latency, parameter count, and how the mechanism scales with input sequence length — is not quantified anywhere in that literature. The consequence is that a health agency has no published basis on which to judge whether an interpretable model is affordable to operate, and this study treats that omission as a measurable quantity rather than an assumption.

Gap 3 — Contextual and Temporal Gap. The relationship between lagged meteorological drivers and dengue incidence has been established largely on retrospective datasets whose training and test windows fall within the same epidemiological era (Rahman et al., 2025a, 2025b; Sang et al., 2025). Whether those learned relationships remain stable when a model fitted on a pre-2023 period is applied to a subsequent period — one in which case levels, vector-control policy and post-pandemic travel patterns have all shifted — has not been re-examined for Singapore. This is a question about the stability and continued applicability of existing findings, not about their original validity.

These three gaps are gaps in the literature. The operational difficulties that follow from them, for an agency that has to act on a forecast, are stated as the research problem in Section 1.3.2, and the correspondence between the two is set out in Table 2.3.

1.3.2 Research Problem

Where Section 1.3.1 states what the literature has not yet settled, this section states the specific issue this study sets out to solve: current dengue forecasting tools cannot be both accurate enough to be worth running and legible enough to be worth acting on. Two concrete problems follow.

Even with the recent shift toward deep learning, most dengue forecasting models still have problems. One problem is that they are too rigid. These models are designed with fixed architectures. They treat every input variable the same way. This means that rainfall from two weeks ago is treated the same as temperature from last month. But the truth is that dengue transmission is not that stable. The things that drive dengue transmission change. They change depending on the season, the location, and even what happened the year before. When the model is not able to make adjustments, the predictions of the model start to drift. They become less accurate (Chen et al., 2025; Riti et al., 2025). Without dynamic feature weighting, a fixed-weight model cannot shift its predictive emphasis from rainfall-driven signals (dominant during monsoon peaks) to case-autocorrelation signals (dominant during inter-outbreak lulls). The result is that the model's residuals correlate with season — a

diagnostic indicator of structural misspecification rather than random noise. Statistical models such as SARIMA assume linearity and constant seasonal structure, making them structurally incapable of representing the multiplicative interactions between temperature, humidity, and rainfall that drive dengue transmission. Deep learning models address this non-linearity but introduce a different rigidity: without an attention mechanism, every input time step contributes equally to the final prediction, regardless of whether it was epidemiologically significant.

The second problem is the black-box issue I just talked about. These models work well, but they do not tell us how they work. You get a prediction from the model. But you do not get any information about which features are most important or what time frame caused the alert to go off (Chen & Moraga, 2025b; Loh et al., 2022). If you are a public health official, this is not just annoying, it is also risky. Without a clear reason for what the model is doing, how can you explain why you are spending a lot of money on something? This is a central reason why interpretability has come to be treated as a prerequisite for deployment in health settings rather than an optional extra (Bharati et al., 2024). So the problem is not just about the technology. It is also about how to use these tools in a real world setting.

1.4 Research Objectives, Research Questions and Hypotheses

1.4.1 General Objective

This study is trying to create a system to predict when dengue will happen. The system should be good at guessing and also be easy to understand. To do this the study will use a method that looks at what happens over time and also gives more importance to some features than others when making predictions. The goal is to make a system that is good at predicting dengue and also easy to understand, without having to choose between these two things. Predictive quality is operationalised as improvement in RMSE and MAE over established statistical and machine learning baselines; interpretability is operationalised through domain-expert validation of attention outputs against documented dengue epidemiology (Section 3.6.3, tested in Section 5.1.1.3) and through the computational cost at which those explanations are obtained (Section 3.6.4, tested in Section 5.1.1.4)

1.4.2 Specific Objectives

The main objective is to build a hybrid LSTM-Transformer, referred to throughout as the Interpretable LSTM-Transformer (ILT), whose attention weights show which time lags and which weather signals drive each forecast, and then to evaluate that model on accuracy, domain validity, computational efficiency and real-world robustness. The LSTM encoder

captures dependencies across the full 28-day input window—long enough to span the complete Aedes mosquito lifecycle from egg to infectious adult—while the temporal attention mechanism identifies which subset of those 28 days carries the strongest predictive signal for a given forecast. That main objective is decomposed into the five specific objectives below, each of which is expressed as a research question in Section 1.4.3 and tested as a hypothesis in Section 1.4.5.

RO1 — Benchmark the accuracy of the ILT against the SARIMA statistical baseline on RMSE, MAE and R^2 at five forecast horizons ($h = 1, 7, 14, 21$ and 28 days) over thirty fixed seeds. Tested by H1 and reported in Section 5.1.1.1.

RO2 — Isolate the value of the temporal-attention module by benchmarking the ILT against a Standalone LSTM under an identical protocol, so that the two architectures differ only in the attention layer and the dual-path fusion. Tested by H2 and reported in Section 5.1.1.2.

RO3 — Cross-check the attention-derived feature importances against documented dengue risk factors, using the consensus ranking averaged over thirty seeds. Tested by H3 and reported in Section 5.1.1.3.

RO4 — Confirm that the interpretability is obtained at an acceptable computational cost relative to a plain LSTM, by benchmarking training time, inference latency and model size over thirty seeds, and by measuring how temporal-attention pooling scales with sequence length T against full self-attention. Tested by H4 and reported in Section 5.1.1.4.

RO5 — Evaluate the robustness and generalisation of the trained model on an independent 2023–2025 Singapore holdout that was not used at any point in training, and compare six adaptation strategies to establish whether adaptation on recent data restores accuracy at the longer horizons. RO5 was added during the revision of this study; it is specified in Section 4.3, tested by H5 and reported in Section 5.2.

1.4.3 Research Questions

Each specific objective is expressed as a research question with an explicit, quantified criterion. The criteria were fixed before any model was trained, in keeping with the positivist, deductive paradigm of Section 3.1, and each is tested in the section of Chapter 5 named beside it.

- RQ1 (RO1): Does the proposed model achieve at least 20% lower forecast error (RMSE) than SARIMA across all five horizons, at $\alpha = 0.05$ over 30 seeds? → Section 5.1.1.1
- RQ2 (RO2): Does the temporal-attention module reduce forecast RMSE relative to a Standalone LSTM under a paired test at $\alpha = 0.05$? → Section 5.1.1.2
- RQ3 (RO3): Do the model's top-20 attention-ranked features reach a domain-validation hit rate of at least 70% against documented dengue epidemiology? → Section 5.1.1.3
- RQ4 (RO4): Is the interpretability obtained within twice the standalone LSTM's training time, inference latency, and model size, with sub-quadratic attention scaling in sequence length T ? → Section 5.1.1.4
- RQ5 (RO5): Does the model retain positive predictive skill ($R^2 > 0$) on an unseen 2023–2025 holdout, and does adaptation on recent data recover the long horizons? → Section 5.2

1.4.4 Research Hypotheses

Each research question is tested as a pair of statistical hypotheses: a null hypothesis (H_0) that is assumed until the data reject it, and a directional alternative (H_a) that states the claim the research question makes; the subscript notation H_0/H_a is used for the null and alternative pair so that H1 to H5 refer unambiguously to the five research hypotheses. Every alternative is directional, so one-sided tests are used throughout, and where a hypothesis is tested at five forecast horizons the p-values are Holm–Bonferroni corrected across those five tests. The decision rule for each is stated in Section 4.2.5 and the outcome in Chapter 5. The narrative statement of each hypothesis is retained below and is followed by its formal pair.

H1: The Interpretable LSTM-Transformer achieves materially higher forecast accuracy than the SARIMA statistical baseline. The claim is quantified rather than left as a comparative adjective: a reduction in RMSE of at least 20% at every one of the five forecast horizons.

H2: The ILT also achieves lower RMSE than a Standalone LSTM that is architecturally identical apart from the attention layer and its dual-path fusion. The comparison isolates the attention component, so a difference in its favour is evidence that the added mechanism contributes predictive value rather than only architectural complexity.

H3: The attention weights are not merely mathematically well defined but epidemiologically meaningful: the features they rank most highly correspond to documented dengue risk factors, such as rainfall lagged by around three weeks or sustained overnight warmth, at a rate exceeding what the ranking would produce by chance.

H4: The interpretability is obtained at an acceptable computational cost. Additive temporal-attention pooling adds a bounded overhead that grows linearly rather than quadratically in the input sequence length, unlike full multi-head self-attention. Training time, inference latency and model size therefore remain within the same order as a plain LSTM, which is the condition under which the architecture is deployable in an operational setting.

H5: Generalisation beyond the training period is treated as a question rather than an assumption. Applied without adjustment to an independent 2023–2025 dataset, the model is expected to retain positive out-of-sample skill at the shorter horizons and to lose it as the horizon lengthens; where skill is lost, adaptation on recent data is expected to restore it.

1.4.5 Formal statement of each hypothesis

The five hypotheses are restated below in null and alternative form with the test and the decision rule that separates them. Two of the nulls differ from the narrative statements above, and the difference is not cosmetic. H1's null was originally “no difference in RMSE between the two models”, but RQ1 claims a reduction of at least 20%; a model improving by three per cent would have rejected that null without meeting the stated criterion, so the null is now written against the 20% margin. H5's null was originally “out-of-sample error does not differ from within-sample error”, but the criterion RQ5 states is $R^2 > 0$, so the null is now written against zero skill. Both now test the criterion their own research question states.

- H1 (RQ1) — $H_0: \mu_{\text{RMSE(ILT)}} \geq 0.80 \times \text{RMSE(SARIMA)}$, the improvement does not exceed 20%. $H_a: \mu_{\text{RMSE(ILT)}} < 0.80 \times \text{RMSE(SARIMA)}$. Test: one-sample one-sided t against the $0.80 \times \text{SARIMA}$ margin over 30 seeds, Holm-corrected across five horizons, Wilcoxon alongside. One-sample because SARIMA is a single deterministic fit with no per-seed distribution.
- H2 (RQ2) — $H_0: \mu(\text{RMSE_ILT} - \text{RMSE_LSTM}) = 0$, the attention module changes nothing. $H_a: \mu(\text{RMSE_ILT} - \text{RMSE_LSTM}) < 0$. Test: paired one-sided t per horizon, paired by seed, Holm-corrected across five horizons, plus a pooled test over all $150 \text{ seed} \times \text{horizon}$ pairs.

- H3 (RQ3) — $H_0: \pi \leq 0.70$, at most 70% of the top-20 attention-ranked features are domain-validated dengue drivers. $H_a: \pi > 0.70$. Test: exact one-sided binomial test on the consensus top-20 ranking against $p_0 = 0.70$, with a Clopper–Pearson interval; exact rather than normal-approximate because $n = 20$.
- H4 (RQ4) — H_0 : the cost ratio ($\text{ILT} \div \text{Standalone LSTM}$) ≥ 2 at some horizon, and/or the attention scaling exponent $\beta \geq 2$. H_a : the ratio < 2 at every horizon and $\beta < 2$. Test: one-sided t of the per-seed cost ratio against 2 at each horizon, Holm-corrected; log-log OLS of latency on sequence length with the slope tested against 2. Parameters and model size are deterministic and compared exactly.
- H5 (RQ5) — $H_0: R^2 \leq 0$ at horizon h , no out-of-sample skill. $H_a: R^2 > 0$ at horizon h . Test: moving-block bootstrap of the out-of-sample R^2 with block length 28 days and $B = 5,000$, giving a 95% interval and a one-sided p-value; blocks rather than independent draws because the residuals are serially correlated.

All tests are conducted at $\alpha = 0.05$ and every outcome is recorded as an explicit decision, “reject H_0 ” or “fail to reject H_0 ”, together with an effect size and a 95% confidence interval. The label H_a is used here for the alternative hypothesis; elsewhere in this thesis the five strands are named H1 to H5, so H_a is used within those discussions where a collision would otherwise arise.

Note on the definition of H4. In an earlier version of this study H4 was framed as a practitioner-validation hypothesis: that data science peers shown the attention visualisations would identify the model's key time steps and features at a recognition accuracy of at least 70% and rate the visualisations at a mean Likert score of at least 3.5 out of 5. That practitioner study was not conducted. H4 was redefined as the computational-efficiency hypothesis stated above and formalised in Section 1.4.5, which is the version tested in Section 5.1.1.4 and reported in the submitted results workbook. The change is recorded in Section 3.6.4.

1.5 Knowledge Contribution

This work sits right between two areas of study that do not talk to each other enough: deep learning forecasting and interpretable AI. The model that is being proposed here tries to fix the black-box problem that deep learning is known for. It does this by augmenting an LSTM encoder with a temporal attention pooling mechanism—a lightweight learned weighting function that identifies which of the 28 historical time steps most influenced each forecast, without requiring the full parameter overhead of multi-head self-attention. This way the model can still perform well without being completely impossible to understand. This places

the design between a fully opaque deep-learning predictor and a fully transparent statistical model — the kind of arrangement sometimes described as "grey-box" modelling.

But really, this is also an attempt to build something that a real health department could actually use. A model that does not just say there is an outbreak coming but also tells you why. Maybe it is because the air is too humid. Maybe it is because of the way people got sick the same month last year. That kind of information could really change how we plan things. Instead of doing the same thing everywhere, we can send help exactly to the places that need it, and we can understand why it is needed there. That kind of targeting matters when vector-control effort has to be distributed across many candidate risk zones (Riti et al., 2025). The work also connects to Sustainable Development Goal 3, especially when it comes to using data to support health responses that are proactive, not just reactive.

In this context, 'grey-box' refers to a modelling approach that preserves the predictive power of a data-driven black-box model while exposing a subset of its internal computations—specifically, the attention weight distribution—for human inspection. Unlike white-box models such as regression, the ILT model's parameters remain non-interpretable in isolation; it is the attention-weighted summary of those computations that is rendered legible.

1.6 Thesis Organisation

This thesis is divided into seven chapters. It starts with a problem, then it talks about a solution, and finally it has a reflection. The whole project is planned to take 26 weeks, starting at the end of January 2026.

Chapter 1 sets the scene. It explains why dengue forecasting is still hard to do well. It talks about why existing deep learning models are not trusted enough and not transparent enough. It says what this study plans to do about that. It also lays out some ideas about what might work and what success might look like.

Chapter 2 looks at what other people have written about dengue prediction. It does not just make a list of papers. Instead, it uses the SDCL (Synthesis, Debate, Connection, and Literature-gap) framework format to trace how dengue prediction moved from old Box-Jenkins models to modern attention mechanisms. It finds the gaps in the research. And it builds the case for why a hybrid LSTM-Transformer model makes sense here. Not just because it might predict better, but because it might also help us understand what is going on under the hood.

Chapter 3 is where the architecture of the model gets explained in detail. It talks about the ideas behind the model. It explains why the model combines recurrent layers and attention-based layers. It also explains how the model is actually supposed to work. There is also a section in this chapter about interpretability. It defines what counts as explainable in this study and how that thing is measured.

Chapter 4 is about the data and the experimental setup. It describes the datasets we used. It explains how we cleaned the data and what we did when values were missing. It also describes the conditions of our experiments. We made sure everything can be repeated. We set the seeds. We wrote down the splits. The way we evaluate the model is tied directly to the hypotheses we made earlier.

Chapter 5 tells us what actually happened. This chapter presents the results of the experiments. It compares the hybrid model against the benchmarks. It tests whether our hypotheses were correct. We use pictures and charts throughout this chapter. These visuals show us not just the performance numbers but also what the attention mechanisms found out.

Chapter 6 takes a step back. This is where we think about what did not work so well. We look at what could be improved. We think about where the model might run into problems if we try to use it somewhere else. The chapter also gives some ideas for what to do next. Maybe use the same framework for other mosquito-borne diseases. Maybe layer in some spatial data.

Chapter 7 is the closing chapter. It summarises the main takeaways. It restates what the study contributed. And it reminds us why any of this matters in the first place. Because better predictions, when paired with clear explanations, can actually help stop outbreaks before they start.

A Gantt chart is included at the end of this chapter. It shows the timeline and helps us see when each piece of work needs to be done.

Figure 1.1 covers the first block, from 26 January to 13 March, in which Chapters 1 to 3 are written.

A Hybrid Deep Learning Approach for Dengue Fever Prediction in real-world case study

Chapter 1

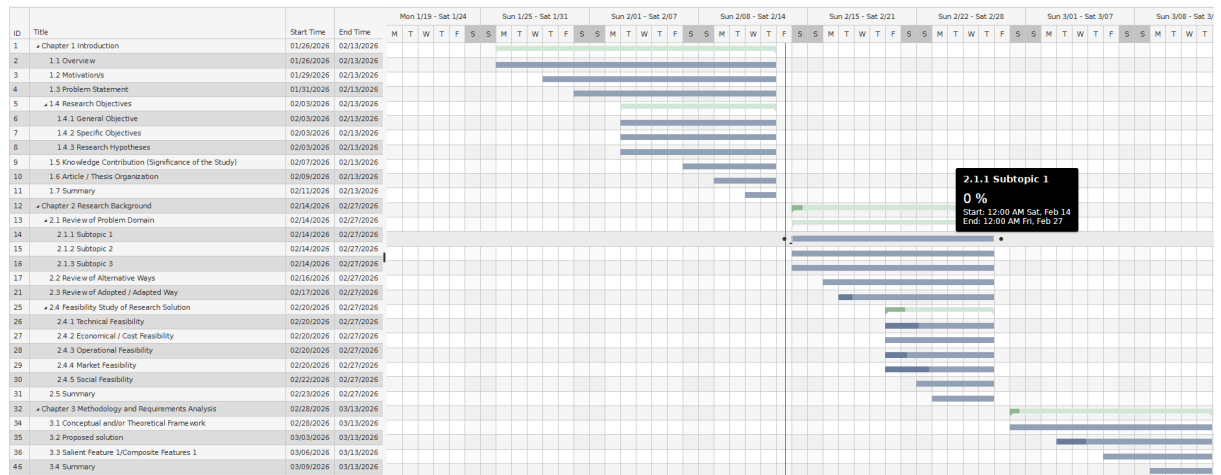


Figure 1.1 Project Planning January until March

Figure 1.2 covers the second block, from 14 March to 10 April, in which Chapter 4 is written and the Project 1 Portfolio is assembled

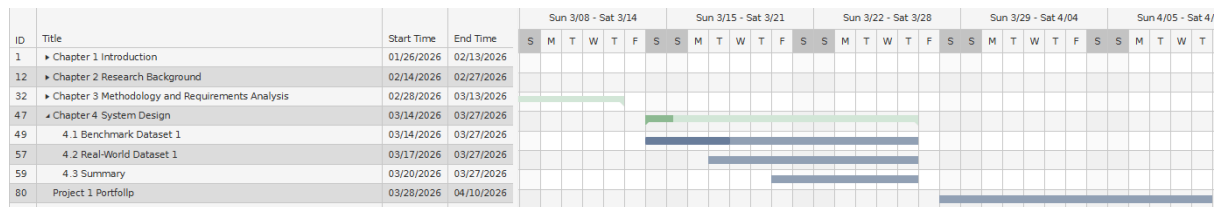


Figure 1.2 Project Planning March until April

Figure 1.3 covers the final block, from 15 June to 24 July, in which Chapters 5, 6 and 7 are written

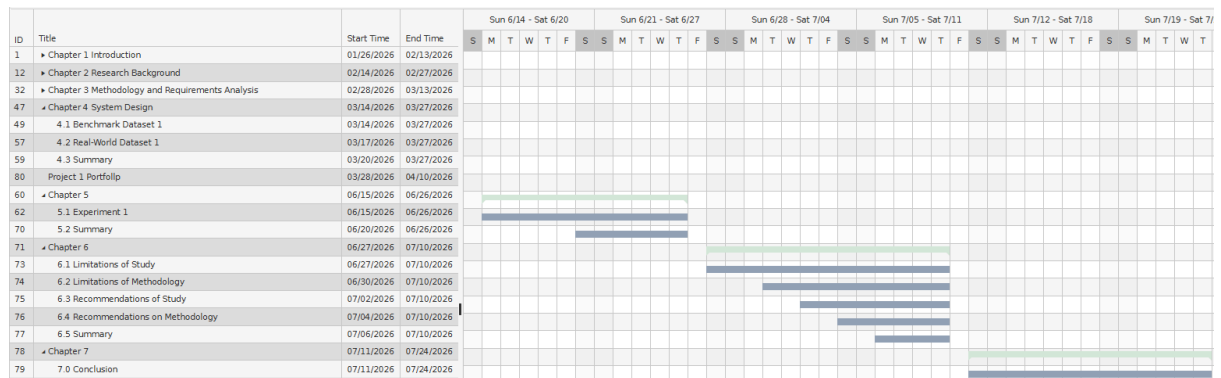


Figure 1.3 Project Planning June until July

This figure shows that the project is planned to accomplish Chapter 5 to 7 from June to July

1.6.1 Research Solution Team

This research brings together several skill sets. Building a working forecast model takes more than just writing code. It needs someone who can clean messy data, someone who understands the math behind neural networks, and someone who knows enough about dengue

to tell if the results actually make sense. Even though this is a university research project, defining these roles clearly helps keep the work organised and makes sure nothing important gets missed. Three main roles shape how this project runs.

Data Engineer:

The data engineer deals with the raw material. Dengue data does not come in one neat package. Case numbers come from health departments. Rainfall and temperature readings come from weather stations. Sometimes these files are in different formats. Sometimes the dates do not line up. The data engineer writes the code to pull everything together, fix mismatched timestamps, and save it in a structure that can be fed into a model later. This person also decides how to handle missing numbers. If a weather station went offline for three days, do we leave those days blank or fill them in? That decision gets made here.

Data Analyst:

The data analyst looks at the data before and after the model runs. Before training starts, this person checks if the data makes sense. Is there a spike in cases every rainy season? Do warmer temperatures usually mean more mosquitoes? These are not things the model should have to figure out from scratch. The analyst confirms these patterns early. After the model runs, the analyst takes whatever the model paid attention to and turns it into plain language. Instead of saying *attention weight 0.87 on lag feature t-3*, the analyst writes three weeks of heavy rain raised the risk level. This is the step where numbers become useful to someone working in public health.

Data Scientist / Machine Learning Researcher:

This person builds the actual model. They decide how many layers to stack, how many memory cells the LSTM should have, and how the temporal-attention mechanism is parameterised. They split the data so the model does not accidentally cheat by looking at future information during training. They also pull out the attention scores and turn them into numbers that can be compared across different time windows. Running the benchmarks against older methods like SARIMA or a plain LSTM also falls under this role. If the hybrid model performs better, this person needs to show why.

How These Roles Work Together:

The work moves in a loop. The data engineer hands over a clean, properly aligned dataset. The data scientist uses that dataset to train the model and exports the attention weights. The data analyst checks those weights against what is already known about dengue. If the model says humidity two weeks ago is the most important factor, but every published paper says temperature matters more, something is off. That feedback goes back to the data scientist, who may need to retune the model or check for bugs. This back and forth keeps the results honest and stops the model from learning patterns that are not really there.

Roles and Support:

Project manager: Kee Sim Ee

1.7 Summary

This chapter sketched out the problem space. Dengue is getting harder to control, and while deep learning has improved outbreak forecasting, it has not solved the trust problem. Models remain opaque, and that limits their use in real-world public health. This study responds by proposing a hybrid LSTM-Transformer framework designed not just for accuracy, but for transparency. The objectives, hypotheses, and overall structure of the thesis have been laid out accordingly.

Chapter 2

Review of Previous Work

2.0 Review of Previous Work

Forecasting dengue outbreaks has attracted growing scholarly attention, yet the field remains divided between models that predict well and models that explain well—rarely achieving both. This chapter traces that divide through a structured, three-part review to enable a systematic and balanced appraisal of the dengue problem domain, existing approaches, the proposed architectural components, and the feasibility of the research solution. Gap types are clearly labelled throughout, so the scholarly reasoning behind each architectural decision can be traced directly, rather than just implied. Each review section is also tied to the Sustainable Development Goal it serves. The chapter as a whole sits under SDG 3 (Good Health and Well-being), since every gap it identifies is ultimately a gap between what can be predicted and what can be acted upon; SDG 9, SDG 10, SDG 11, SDG 16 and SDG 17 enter where the discussion turns to computational infrastructure, equity of access, urban transmission conditions, institutional accountability for automated advice, and cross-sector data partnership respectively.

2.1 The Dengue Problem Domain: Complexity, Prediction Challenges, and Research Gaps

This section characterises the dengue problem domain across three dimensions: the nature of the disease as a public health threat, the limitations of current prediction methods, and the challenges of data integration and interpretability. Each subsection identifies a specific gap type that the proposed architecture is subsequently designed to address.

2.1.1 Dengue Fever as a Public Health Threat

Dengue fever is one of the most persistent and expanding infectious disease threats of the twenty-first century. As a multi-serotype arboviral disease transmitted by *Aedes* mosquitoes, it infects tens of millions of people each year and continues to spread geographically beyond its historical range. Its resistance to containment stems not from any single factor but from a confluence of interacting forces: rapid urbanisation generates breeding grounds, rising temperatures extend the mosquito season, increased international travel accelerates geographic spread, and the immunological

complexity of a four-serotype virus produces outbreak patterns that defy simple cyclical modelling (da Silva et al., 2024; Wang et al., 2023). These intersecting dynamics make dengue a genuinely complex, multi-scale problem that no single-variable surveillance framework can adequately capture.

The scale of this problem has been extensively documented at both global and regional levels. Epidemiological projections demonstrate that equatorial regions—including Malaysia—face the largest projected increases in dengue incidence due to the combined effects of climate change, population growth, and urbanisation (Wang et al., 2023). Within Malaysia, dengue case counts have surged significantly in recent years, and fine-scale modelling of *Aedes* abundance in Selangor shows that risk varies sharply at sub-district resolution (Dom et al., 2025). Spatially resolved modelling of Malaysian dengue incidence shows marked geographic heterogeneity in risk, motivating models that can weight location-specific signals rather than assume a uniform response (Majeed et al., 2025; Dom et al., 2025). Studies applying integrated statistical and clustering approaches have confirmed that population density, seasonal patterns, and climatic factors collectively shape outbreak risk in ways that single-variable surveillance frameworks cannot adequately resolve (Hamida et al., 2024). Fine-scale GIS and machine learning analyses have additionally demonstrated that within a single city, dengue incidence can vary dramatically across adjacent neighbourhoods due to localised differences in vegetation cover and building density (Das et al., 2025; Majeed et al., 2025). This spatial concentration of risk carries direct operational significance: a model capable of sub-district resolution enables vector control agencies to prioritise fogging and larviciding resources rather than applying them uniformly across entire administrative zones.

Despite this growing evidence base, public health systems across the region continue to depend primarily on passive case surveillance—waiting for cases to accumulate before mobilising a response. The consequence is a persistent gap between what is technically feasible and what is routinely practised. Most operational tools remain either too simple to capture the nonlinear dynamics driving outbreaks, or too opaque to be trusted by practitioners who must justify resource allocation decisions (Loh et al., 2022; Bharati et al., 2024), even though descriptive analyses of national case

registries document pronounced temporal and spatial heterogeneity in outbreak activity that passive counting cannot anticipate (Hossain & Nishu, 2024). The body of evidence reviewed here reveals a fundamental methodological gap: current surveillance tools are structurally misaligned with the complexity of the system they are designed to monitor. A model capable of handling nonlinear, multivariate, and multi-scale inputs is not optional—it is the minimum viable requirement for meaningful dengue forecasting in Singapore and in comparable dengue-endemic tropical urban settings.

The evidence reviewed in this subsection directly underpins the case for an architecture capable of processing multivariate meteorological inputs over time, connecting to Objective RO1 and Hypothesis H1. It also situates this research within the ambitions of SDG 3 (Good Health and Well-being) and SDG 11 (Sustainable Cities and Communities), which call for health systems that are evidence-based, proactively managed, and responsive to the spatial and demographic realities of the communities they serve.

2.1.2 Limitations of Current Prediction Methods

Statistical time-series models, principally SARIMA and its regression-based variants, have been the default tool in epidemiological forecasting for decades. These models decompose historical time-series data into trend, seasonal, and irregular components using autoregressive and moving average terms structured around defined lag periods. Their parametric architecture makes every coefficient directly interpretable as a quantified lag relationship, and their computational simplicity makes them accessible to epidemiologists without specialist machine learning knowledge. For settings where dengue follows a reliable annual cycle and historical data are clean and stationary, these models perform adequately. Their sustained use in practice reflects their transparency, not their accuracy under complex or rapidly changing conditions.

Applications of time-series regression have confirmed the utility of statistical models in capturing stable seasonal patterns in historical case data (Waseem et al., 2023). Integrated moving average approaches have been extended to spatial vulnerability assessment through hierarchical clustering across administrative districts (Hamida et al., 2024). A multi-model comparison employing SARIMAX alongside XGBoost and

LSTM confirmed that SARIMAX remained competitive only under stable seasonal conditions, yielding consistently to nonlinear methods when climate-driven variability was elevated (Tuan, 2024). A head-to-head comparison in Rio de Janeiro confirmed that SARIMA-type models were consistently outperformed once nonlinear predictors were introduced (Chen & Moraga, 2025a). Deep learning frameworks incorporating climate covariates demonstrated substantial improvements over SARIMA baselines across multiple geographic settings (Cheng et al., 2026), and a systematic review of hybrid models across the Americas concluded that purely statistical approaches were inadequate for the heterogeneous dynamics characterising dengue in endemic regions (Castellanos Garzón et al., 2026).

The core methodological constraint of statistical models is structural rather than incidental: linearity. When dengue dynamics shift under changing climate conditions or new serotype introductions, statistical models have no internal mechanism to adapt. The stationarity assumption is particularly problematic—when the distribution of the underlying process changes, forecast accuracy degrades sharply with no internal warning signal to alert practitioners. Multi-step ahead prediction is additionally challenged because compounding error across time steps is uncontrolled by any temporal learning mechanism. These limitations are not resolvable through parameter tuning; they are inherent to the mathematical form of the models themselves. This constitutes a methodological gap that purely statistical tools cannot close.

Despite their limitations, traditional statistical models are retained as primary benchmark comparators in this study, fulfilling Objective RO1. Their documented failure under nonlinear conditions serves as the empirical foundation for Hypothesis H1, and their inadequacy under complex dengue dynamics connects to SDG 3's requirement for health tools that function reliably in real operational conditions—not merely in stable retrospective contexts.

2.1.3 Challenges in Data Integration and Interpretability

The most direct response to the limitations of statistical models is to incorporate richer data and more flexible modelling approaches, and the dengue forecasting literature reflects exactly this progression over the past decade. Meteorological variables, non-traditional digital signals such as internet search trends, and

cross-border indicators such as imported-case counts from neighbouring countries have all been combined with dengue case records in increasingly sophisticated modelling pipelines. However, this pursuit of improved accuracy through data integration consistently produces a second and harder problem: the resulting models become progressively more difficult for practitioners to interpret, challenge, or trust.

The addition of non-traditional signals such as Google Trends has been shown to improve early outbreak detection in settings where formal surveillance reporting is delayed (Tuan & Uyen, 2025). Feature selection techniques including autoencoders and the Boruta algorithm have been applied to manage the dimensionality of high-dimensional multivariate inputs (Archana & Faritha Banu, 2025), and integration of cross-border climate signals from neighbouring countries has improved local forecast accuracy by capturing upstream environmental drivers (Sang et al., 2025). Integration of eco-climatic, sociodemographic, and landscape features using XGBoost and LightGBM has further demonstrated the value of multi-dimensional inputs for early warning system development (Rahman et al., 2025a). Studies applying explainable AI approaches using patient phenotypic data found that SHAP-based explanations ranked which clinical traits most influenced individual dengue predictions (Joly et al., 2024), and a complementary study in Bangladesh applied a Shapash-based SHAP framework to attribute predictions from meteorological and health-service records to individual input features (Prome et al., 2024). A broader systematic review of XAI in disease prediction confirmed that while SHAP and LIME are now widely adopted, limited dataset diversity and reliance on single static data types remain persistent weaknesses that undermine their reliability in dynamic forecasting settings (Vellido, 2019).

The fundamental problem is that deep neural networks, despite their superior predictive accuracy, operate as black boxes. A practitioner receiving a model output cannot determine whether that output reflects a genuine signal embedded in the climate data or an artefact of overfitting to historical noise. A lack of trust arising from model opacity is identified in the healthcare XAI literature as the main reason why technically capable models see little use in practice (Loh et al., 2022; Bharati et al., 2024). Existing XAI applications are confined to classification settings with static inputs and depend on post-hoc explanation tools that are applied after training rather

than architectures that are interpretable by design. The specific problem of interpretable multivariate time-series forecasting of dengue incidence—where the model must explain which variables mattered most for a particular forecast at a particular point in time—has not been adequately addressed in the published literature. This constitutes both a transparency gap and a practical-knowledge gap combined.

This dual gap is precisely what the proposed hybrid LSTM-Transformer architecture is designed to close. By embedding interpretability into the architecture itself through an attention layer that runs inside the forward pass—rather than deferring it to post-hoc approximation tools—the proposed framework addresses both the accuracy demands of a complex multivariate system and the trust requirements of practitioners who must act on its outputs. This directly connects to Objectives RO3 and RO4 and Hypotheses H3 and H4, and it sits squarely within SDG 3's vision of AI-enabled health systems that practitioners can interrogate, understand, and act upon with confidence.

2.1.4 Summary of Problem Domain Analysis

Table 2.1 consolidates the problem domain analysis from Sections 2.1.1 to 2.1.3 into a structured overview. Each row represents one identified problem dimension, its core challenge, the literature that substantiates it, and its implication for the design of the proposed research. The table serves as a bridge between the problem domain review and the evaluation of alternative approaches in Section 2.2.

Problem Dimension	Key Challenge	Evidence from Literature	Implication for This Study
Epidemiological Complexity <i>Multi-factor, nonlinear dynamics</i>	Dengue transmission is driven by interacting climatic, urban, and demographic forces that shift unpredictably across seasons and geographies.	Das et al. (2025); Sang et al. (2025)	A model must handle nonlinear, multi-variable inputs and adapt to context shifts—ruling out linear statistical approaches.
Limitations of Statistical Models <i>Methodological Gap</i>	SARIMA and regression variants rely on linearity and stationarity, making them brittle when outbreak dynamics shift under changing climate conditions.	Chen & Moraga (2025a); Zhao et al. (2023); Cheng et al. (2026); Castellanos Garzón et al. (2026)	Statistical models retained as benchmarks (RO1, H1) but insufficient as the primary predictive framework.
Data Integration Complexity	High-dimensional multivariate inputs from	Riti et al. (2025); Tuan & Uyen (2025); Archana &	Feature selection and dynamic input weighting are

Problem Dimension	Key Challenge	Evidence from Literature	Implication for This Study
<i>Contextual & Functional Gap</i>	meteorological, search trend, and satellite sources improve accuracy but substantially increase model opacity.	Faritha Banu (2025); Sang et al. (2025)	needed—motivating the attention mechanism in the proposed architecture (RO3).
Black-Box Problem in Deep Learning <i>Transparency Gap</i>	LSTM and other deep learning models compress historical inputs into uninterpretable hidden states, preventing practitioners from understanding model reasoning.	Loh et al. (2022); Bharati et al. (2024); Ullah et al. (2024)	Interpretability must be intrinsic to the architecture, not deferred to post-hoc tools—motivating the intrinsic attention layer of the proposed architecture (RO3, RO4, H3, H4).
Failure of Post-hoc XAI Methods <i>Interpretive Gap</i>	SHAP and LIME provide approximated, externally generated explanations that are unstable for sequential multi-step forecasting tasks.	Joly et al. (2024); Prome et al. (2024)	Post-hoc XAI is insufficient; model-level intrinsic interpretability is required for trustworthy time-series dengue forecasting.
Operational Adoption Gap <i>Practical-Knowledge Gap</i>	Despite high accuracy in research, opaque models see limited use in practice because their outputs cannot be interrogated by the people who must act on them.	Loh et al. (2022); Bharati et al. (2024); Vellido (2019)	The proposed model must produce practitioner-legible explanations alongside forecasts to bridge the gap between research performance and operational trust (RO3, H3).

Table 2.1 Summary of Dengue Problem Domain Dimensions, Evidence, and Research Implications

Reading Table 2.1 as a whole, a consistent pattern emerges: the core challenge in dengue prediction is not the absence of capable models, but the absence of models that are simultaneously capable and explainable. Statistical models are explainable but not capable under complex dynamics. Deep learning models are capable but not explainable. Post-hoc XAI tools partially address the explanation problem but introduce approximation and instability that undermine their reliability for time-series forecasting. This configuration of gaps—epidemiological, methodological, functional, transparency, interpretive and practical-knowledge—is what the proposed hybrid LSTM-Transformer architecture is designed to address, as detailed in Section 2.3. SDG 3 and SDG 11 provide the overarching mandate: public health systems need tools that work in real-world operational conditions, not just in research evaluations.

2.2 Evaluating Existing Prediction and Interpretability Approaches: A Critical Review

The academic community has proposed four broad categories of approaches for dengue prediction and model interpretability. Each subsection summarises the approach, describes its key applications, criticises its limitations in the context of this study's objectives, and links those limitations to the specific design choices of the proposed architecture. A comparative summary table is presented at the end of the section.

2.2.1 Traditional Statistical Models

SARIMA and its regression-based variants decompose historical time-series data into trend, seasonal, and irregular components using auto-regressive and moving average terms structured around defined lag periods. Their parametric structure makes every coefficient directly interpretable as a quantified lag relationship. For stable environments with clearly periodic outbreak cycles, these tools provide a transparent and computationally inexpensive baseline that epidemiologists can explain without specialist machine learning knowledge.

Applications to dengue spread using time series regression confirmed the ability of statistical models to capture stable seasonal patterns in historical case data (Waseem et al., 2023). Integrated moving average approaches combined with hierarchical clustering have been extended to spatial vulnerability assessment across districts (Hamida et al., 2024). Early outbreak risk prediction in Vietnam using Google Trends as a leading indicator illustrated how traditional frameworks can be augmented with non-traditional signals, though accuracy remained constrained by the linearity assumption when incidence surged unexpectedly (Tuan & Uyen, 2025). A multi-model comparison employing SARIMAX alongside XGBoost and LSTM confirmed that SARIMAX performed competitively only under stable seasonal conditions, yielding consistently to nonlinear methods when climate-driven variability was high (Tuan, 2024). A large-scale comparative evaluation further confirmed that purely statistical baselines were consistently outperformed across diverse geographic settings (Castellanos Garzón et al., 2026).

The methodological constraint is fundamental: linearity. A head-to-head comparison demonstrated that SARIMA-type approaches were consistently outperformed once nonlinear predictors were introduced (Chen & Moraga, 2025a). The stationarity assumption is particularly problematic; when the distribution of the underlying

process changes—as during climate anomalies or new serotype introductions—forecast accuracy degrades sharply with no internal warning signal. Despite their limitations, traditional statistical models are retained as primary benchmark comparators in this study, fulfilling (RO1). Their parametric interpretability defines the reference standard that the proposed attention-based model must meet or exceed, connecting to SDG 3's call for health tools that work in real operational conditions.

2.2.2 Conventional Machine Learning Models

Random Forest, Support Vector Machines, Gradient Boosting, and weighted ensemble methods address the core limitation of statistical models by learning nonlinear decision boundaries from data without requiring pre-specification of functional form. Their ability to rank input variables by importance offers a form of post-training interpretability through global feature importance scores, making them more flexible than SARIMA while remaining more understandable than deep neural networks.

Weather-driven dengue risk zone prediction using Random Forest and SVM demonstrated competitive classification accuracy across urban and rural settings (Riti et al., 2025). A weighted ensemble approach applied to dengue occurrence in North India confirmed that combining multiple classifiers improved robustness over any single model (Shashvat & Kaur, 2023). ML models trained on microclimatic variables have shown meaningful discrimination of dengue risk at sub-district resolution (Dom et al., 2025; Das et al., 2025), and gradient-boosted ensembles appear consistently in comparative dengue benchmarks as strong performers (Agarwala et al., 2024; Rahman et al., 2025a). An XAI-enhanced approach using XGBoost and LightGBM on a twenty-three-year dataset in Bangladesh demonstrated that gradient boosting models, when paired with SHAP explanations, can deliver both strong predictive accuracy and practitioner-accessible feature-level insights—though these insights remain static rather than time-step-resolved (Rahman et al., 2025a). An intelligent prediction system combining optimised ensemble methods with advanced feature selection further confirmed the value of multi-classifier fusion for dengue outbreak management (Muthuselvi et al., 2025).

The functional gap is structural. Conventional ML models treat each observation as an independent sample, discarding sequential order entirely. A comparative evaluation confirmed that recurrent deep learning architectures outperformed simpler sequential baselines on multi-step dengue incidence forecasting (Ullah et al., 2024). Feature importance rankings from Random Forest are global statistics describing which variables matter across the entire dataset, not which mattered most for a specific prediction at a specific time. Conventional ML models serve as secondary benchmark comparators alongside SARIMA, contributing to (RO1) and supporting (H2). Their inability to model temporal dependencies directly motivates adopting LSTM as the temporal encoding backbone of the proposed architecture (the main objective of Section 1.4.2), while the limitation of static feature importance measures sharpens the case for the intrinsic attention mechanism adopted in Section 2.3.3 (RO3).

2.2.3 Basic Deep Learning Models

Long Short-Term Memory (LSTM) networks resolve the vanishing gradient problem inherent in standard recurrent neural networks through a gated cell architecture that selectively retains or discards information across time steps. As formally derived in the theoretical analysis of the RNN-to-LSTM transition, vanilla RNNs cannot sustain gradient signals across long sequences, and the gated cell mechanism was specifically designed to overcome this structural limitation (Sherstinsky, 2020). The LSTM architecture processes inputs as ordered sequences through three multiplicative gates—input, forget, and output—that control information flow into, out of, and through the cell state. This gated mechanism allows LSTM to model dependencies spanning dozens of time steps, making it structurally suited to the multi-week temporal lag patterns that characterise dengue transmission. A comprehensive systematic review of LSTM architectures across 82 peer-reviewed studies has reaffirmed that LSTM remains the dominant baseline for sequential health prediction tasks due to its superior long-range dependency capture compared to vanilla RNNs and GRUs (Al-Selwi et al., 2024; Ullah et al., 2024).

A rigorous comparison of LSTM, RNN, and GRU architectures using two decades of multivariate outbreak data in Bangladesh established LSTM's superiority in capturing long-range temporal dependencies for dengue forecasting (Ullah et al., 2024). Region-specific applications confirmed robust multi-step forecasting performance

across geographically distinct settings in Pakistan, India, and Puerto Rico (Ud Deen Shafique et al., 2024; Dhaked et al., 2025; Chen et al., 2025). LSTM models incorporating multiple meteorological covariates consistently produced lower error metrics than statistical baselines (Lu et al., 2025; Agarwala et al., 2024). Integrating an attention layer directly with an LSTM base improved multi-step dengue prediction across Malaysian states compared to standalone LSTM baselines, confirming that the LSTM's hidden state sequence is a productive foundation for downstream attention computation (Majeed et al., 2025). A compartmental model integrated with LSTM for Selangor, Malaysia, further validated that recurrent architectures capture the non-linear meteorological dependencies that simpler approaches miss (Lu et al., 2025).

LSTM's strength is also where it is most limited: the hidden state. All information the network retains about the past is compressed into a fixed-dimensional vector with no interpretable structure. Reviews of explainable AI in healthcare identify model opacity as a principal barrier to clinical and public health deployment (Loh et al., 2022; Bharati et al., 2024). LSTM's temporal modelling strength is directly adopted to fulfil the main objective of Section 1.4.2. Its hidden state sequence is passed to the additive temporal-attention layer, which generates explicit attention weights, forming the core architectural decision and the basis for testing (H1) and (H3).

2.2.4 Post-hoc Explainable AI (XAI) Approaches

Post-hoc Explainable AI (XAI) methods represent a broad class of model-agnostic explanation techniques that interpret a trained model's behaviour after training is complete. The two dominant approaches are SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations). SHAP derives feature attribution values rooted in cooperative game theory's Shapley values, assigning each feature a contribution score that satisfies desirable axioms of consistency and local accuracy; this mechanism has been systematically validated across healthcare AI applications as a reliable model-agnostic explanation method (Bharati et al., 2024). LIME generates locally faithful surrogate models by perturbing the neighbourhood of an individual prediction and fitting a simpler interpretable model to the perturbed outputs. A systematic review of XAI applications across 99 healthcare studies confirmed that both SHAP and LIME have been widely adopted in medical AI

precisely because they can be applied to any existing model without retraining, offering immediate explainability for legacy systems (Loh et al., 2022).

In dengue-specific applications, SHAP values have been shown to identify which environmental and clinical features most strongly influenced individual predictions, (Joly et al., 2024). An explainable machine learning study in Bangladesh applied the same class of post-hoc attribution to meteorological and health-service data, reporting per-feature contributions alongside each forecast (Prome et al., 2024). A more recent study applying XGBoost and LightGBM combined with SHAP and LIME demonstrated that population density, precipitation, and temperature consistently ranked as the most influential predictors (Rahman et al., 2025a). An interpretable tree-based early warning model for Bangladesh further validated that climate, sociodemographic, and landscape factors can be surfaced as key drivers through model-level explanation mechanisms (Rahman et al., 2025b; Bharati et al., 2024).

Despite their value, post-hoc XAI methods carry a fundamental limitation that is particularly consequential in the time-series forecasting context. Because they explain a model's behaviour from the outside—by approximating input-output relationships—their explanations are not guaranteed to reflect the model's actual internal reasoning. SHAP and LIME approximations can be inconsistent and unstable across similar inputs. A comprehensive systematic review of XAI in disease prediction found that reliance on single data types and limited validation against ground-truth explanation targets were persistent weaknesses undermining trustworthiness in dynamic settings (Vellido, 2019). Most critically, post-hoc XAI applications in dengue have been confined to static classification tasks—not multi-step time-series forecasting where the question is which variable at which time step drove the forecast. Rather than training an LSTM and subsequently applying SHAP or LIME, this study integrates interpretability into the architecture itself through an attention mechanism that runs inside the forward pass, addressing the interpretive gap (H3, H4) and aligning with SDG 3.

2.2.5 Comparative Summary of Reviewed Approaches

Table 2.2 compares the four reviewed approach categories against the proposed architecture across nine evaluation dimensions that are directly relevant to the research objectives of this study. References supporting each criterion are included.

The table is intended to make the full configuration of remaining gaps visible in a single view.

Criterion	SARIMA / Statistical	ML (RF/SVM/Ensemble)	LSTM / RNN / GRU	Post-hoc XAI (SHAP/LIME)	Transformer (Standalone)	Proposed LSTM-Transformer
Temporal Dependency Modelling <i>Ullah et al. (2024); Waseem et al. (2023)</i>	Partial (linear only)	Not supported	Supported	Not applicable	Partial	Fully supported
Nonlinear Relationship Modelling <i>Zhao et al. (2023); Chen & Moraga (2025a)</i>	Not supported	Supported	Supported	Not applicable	Supported	Supported
Multivariate Input Handling <i>Rahman et al. (2025a); Archana & Faritha Banu (2025)</i>	Limited	Supported	Supported	Not applicable	Supported	Supported
Model-Level Interpretability <i>Loh et al. (2022); Bharati et al. (2024)</i>	Supported (parametric)	Partial (global)	Not supported	Post-hoc only	Partial (attention)	Supported (intrinsic)
Time-Step-Level Explanation <i>Ly et al. (2024); Zhao et al. (2023)</i>	Not supported	Not supported	Not supported	Partial (per-instance)	Supported	Supported
Dynamic Feature Weighting <i>Archana & Faritha Banu (2025); Zhao et al. (2023)</i>	Not supported	Not supported	Not supported	Not supported	Supported	Supported
Dengue-Specific Application <i>Zhao et al. (2023); Muthuselvi & Devi (2024)</i>	Widely used	Widely used	Widely used	Rare	Rarely used	Proposed (novel)
Multi-Step Ahead Forecasting <i>Cheng et al. (2026); Ud Deen Shafique et al. (2024)</i>	Limited	Not suitable	Supported	Not applicable	Supported	Supported
Addresses Practical-Knowledge Gap	No	No	No	Partial	No	Yes (core contribution)

Criterion	SARIMA / Statistical	ML (RF/SVM/E nsemble)	LSTM / RNN / GRU	Post-hoc XAI (SHAP/LIM E)	Transformer (Standalone)	Proposed LSTM-Tran sformer
<i>Loh et al. (2022); Vellido (2019)</i>						

Table 2.2 Comparative Summary of Dengue Prediction and Interpretability Approaches Across Key Evaluation Dimensions

Reading Table 2.2 column by column, the pattern is unambiguous. Statistical models offer interpretability but cannot handle nonlinearity. ML models handle nonlinearity but discard temporal structure. LSTMs recover temporal structure but sacrifice interpretability. Post-hoc XAI provides partial, externally approximated explanation that is structurally unsuited to multi-step time-series forecasting. Standalone Transformers offer intrinsic attention but lack sequential inductive bias. The proposed hybrid architecture is the only configuration in this analysis designed to address all nine dimensions simultaneously, and it is that specific, gap-driven combination that constitutes the scholarly justification for this research.

2.3 The Proposed Hybrid LSTM-Transformer Framework: Adopted Components and Architectural Justification

The proposed LSTM-Transformer hybrid is not a combination of techniques chosen arbitrarily. Each component is adopted or adapted for a specific purpose traceable to a specific gap identified in Section 2.1 and confirmed in Section 2.2. In terms of theoretical contribution type, this study represents a Synthesis-type contribution: it integrates established paradigms from recurrent sequence modelling, attention-based representation learning, and interpretable AI into a unified framework that none of these paradigms could produce alone.

2.3.1 Hybrid Deep Learning Architectures for Dengue Forecasting

Hybrid deep learning architectures combine multiple neural network components—typically a recurrent sequence encoder with an attention or convolutional processing layer—to exploit the complementary strengths of each

component simultaneously. The core architectural rationale is that no single neural network design optimally addresses all aspects of a complex time-series forecasting problem, and that component specialisation consistently outperforms both standalone models and loosely constructed ensembles. By assigning distinct computational responsibilities to different layers—sequence memory to a recurrent encoder and contextual reweighting to an attention mechanism—hybrid architectures allow each component to contribute what it does best within a single end-to-end trainable framework.

A hybrid deep learning architecture demonstrated superior weekly dengue incidence forecasting compared to standalone models, providing foundational proof-of-concept for architectural fusion specifically within the dengue domain (Zhao et al., 2023). An ensemble neural network approach incorporating climatic condition inputs confirmed that combining complementary components improved both accuracy and robustness across varying outbreak conditions (Panja et al., 2023). A large-scale comparative evaluation found that hybrid and ensemble approaches consistently produced lower forecasting error than any individual architecture across diverse geographic settings (Castellanos Garzón et al., 2026). Combining LSTM with an attention mechanism produced consistently superior short-term epidemic forecasts, supporting the LSTM-attention fusion design for infectious disease forecasting (Hu et al., 2024). Hybrid architectures that integrate deep learning components with domain-specific models for dengue fever incidence forecasting have demonstrated significant advantages over traditional statistical baselines in the Malaysian context (Lu et al., 2025). A Transformer self-attention network for infectious disease time-series confirmed that self-attention outperforms LSTM, RNN, and ARIMA baselines in capturing long-range temporal dependencies (Alotaibi, 2025), and additional validation using an attention-augmented LSTM architecture for COVID-19 time-series demonstrated improved prediction accuracy and multi-step forecast reliability (Hu et al., 2024).

What is absent from the hybrid dengue forecasting studies identified in the review of Section 2.2 is an explicit design objective for interpretability. Current hybrid architectures optimise for predictive accuracy and treat explanation as either irrelevant or deferred to post-hoc tools—precisely the tools whose structural limitations were

identified in Section 2.2.4. The gap in the existing literature is therefore not the use of hybrid architectures per se, which is well-validated, but the absence of hybrid architectures that treat intrinsic interpretability as a primary design criterion equal in importance to predictive accuracy. No hybrid dengue forecasting study identified in the review of Section 2.2 has examined whether the attention weights produced by such an architecture correspond to domain-known dengue risk factors—a question with direct implications for whether the architecture can be trusted in operational settings.

This study adopts the hybrid architecture paradigm established by Zhao et al. (2023) as its foundational design principle, then extends it by making attention weight visualisation the primary interpretability output—intrinsic to the forward pass, not applied post-hoc. This extension directly addresses the transparency gap and the practical-knowledge gap identified in Sections 2.1.3 and 2.2.4, and provides the architectural basis for testing Hypotheses H1, H2, H3, and H4.

2.3.2 Long Short-Term Memory (LSTM) as Temporal Encoder

The Long Short-Term Memory architecture addresses the vanishing gradient problem that limits standard RNNs through three multiplicative gates—input, forget, and output—that control the flow of information into, out of, and through the cell state. The theoretical basis for this design is formally established in the mathematical derivation of the RNN-to-LSTM transition, which demonstrates that the gated cell mechanism is necessary to sustain gradient signals over long input sequences without degradation (Sherstinsky, 2020). A comprehensive systematic review of LSTM architectures across 82 peer-reviewed studies has further codified the modern LSTM formulation widely used in health prediction systems today, confirming its dominance as the recurrent baseline for sequential health prediction (Al-Selwi et al., 2024). This selective memory mechanism allows LSTM to preserve relevant signals across arbitrarily long sequences, making it structurally well-suited to epidemiological time series where seasonal patterns and lag effects may operate over time scales of weeks or months.

LSTM's superiority over RNN and GRU on a two-decade multivariate dengue dataset in Bangladesh established long-range dependency capture as a core empirical strength of LSTM in this domain (Ullah et al., 2024). Region-specific applications across

India, Pakistan, and Puerto Rico confirmed robustness across different climate regimes and data conditions (Ud Deen Shafique et al., 2024; Dhaked et al., 2025; Chen et al., 2025). Curated meteorological inputs—temperature, rainfall, humidity and lagged case counts—have been shown to improve LSTM forecasting accuracy relative to unselected inputs (Lu et al., 2025; Cheng et al., 2026). An attention layer integrated directly with an LSTM base improved multi-step dengue prediction across Malaysian states compared to standalone baselines, confirming that the LSTM's hidden state sequence is a productive foundation for downstream attention computation (Majeed et al., 2025). A compartmental model integrated with LSTM for Selangor demonstrated that the recurrent architecture captures non-linear meteorological dependencies that simpler approaches miss (Lu et al., 2025).

LSTM's core limitation in this application context is the opacity of its hidden state representation. All information retained about the input sequence is folded into a dense, high-dimensional vector with no interpretable structure—a transparency gap addressed by the additive temporal-attention layer described in Section 2.3.3. The LSTM is adopted as the temporal encoding component to fulfil the main objective stated in Section 1.4.2. Its gated memory ensures long-range seasonal patterns are preserved in the hidden state sequence passed to the temporal-attention layer. This division of labour—LSTM for ordered temporal memory and additive temporal attention for interpretable weighting—is the core architectural logic of the study and the basis for testing (H1) and (H3).

2.3.3 Attention as the Interpretability Layer: from Transformer Self-Attention to Additive Temporal Pooling

This section reviews the attention family from which the interpretability layer of the proposed architecture is drawn, and it is useful to state the destination before the route: the mechanism this study implements is additive temporal-attention pooling, not the Transformer's multi-head self-attention. The Transformer formulation is reviewed first because it is the reference point against which the adopted variant is justified and against which its cost is measured in Section 5.1.1.4. The Transformer architecture introduced the multi-head self-attention mechanism as a replacement for recurrence in sequence modelling, demonstrating that parallel attention over all sequence positions could match or exceed LSTM performance on language tasks

while being substantially more parallelisable. As comprehensively reviewed in the attention mechanism literature, the key innovation is the Scaled Dot-Product Attention function defined as $\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d}) \times V$, which enables direct inspection of which sequence positions the model prioritised when generating each output, producing attention weight matrices that are intrinsic to the forward computation rather than approximated post-hoc (Niu et al., 2021). Subsequent peer-reviewed validation of this mechanism in infectious disease time-series forecasting confirmed its superiority over both LSTM and ARIMA baselines, and provided an empirical demonstration of the Query-Key-Value (QKV) computation in a public health forecasting context (Lv et al., 2024). Its multi-head variant has been formally validated in infectious disease time-series prediction as: $\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) \times W^o$, where each $\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$ (Lv et al., 2024).

A study comparing Transformer, RNN, and hybrid architectures for infectious disease forecasting further confirmed that multi-head attention consistently yields superior accuracy while enabling direct inspection of inter-step dependencies (Alotaibi, 2025). By inspecting the attention weight matrix, researchers can directly identify which historical time steps the model prioritised when generating each prediction. Attention weights, while mathematically precise, are not self-validating as explanations. A high attention weight on a particular time step confirms that the model routed more information from that step into its forecast, but it does not independently guarantee that this reliance reflects genuine epidemiological signal rather than spurious correlation. This is precisely why Hypothesis (H3) exists in this study: the claim that attention weights correspond to domain-known dengue risk factors must be empirically tested against expert knowledge, not assumed.

The Transformer's attention mechanism is adapted to serve as the interpretability layer of the proposed hybrid architecture, fulfilling (RO3) and (RO4). It is adopted in the additive temporal-attention pooling form specified in Section 3.2.2 rather than as full multi-head self-attention, for the data-efficiency reasons given there; Section 5.1.1.4 measures what that choice saves. By operating on the LSTM's hidden state sequence, the attention mechanism generates explicit, visualisable weights that practitioners can read as a time-resolved explanation of each forecast. This is the operational translation of (H3) and (H4), and the mechanism through which the proposed

architecture advances SDG 3 by producing health AI that practitioners can inspect and critically assess.

2.3.4 Feature Integration and Meteorological Input Construction

Feature integration refers to the selection, construction, and organisation of input variables supplied to the model prior to training. In the context of dengue forecasting, this encompasses meteorological variables, lagged case counts, and non-traditional signals such as internet search trends and cross-border climate indicators. The composition of the input feature set directly determines both the predictive ceiling of the model and the interpretive value of the attention weights it generates. A well-curated feature set allows the attention mechanism to surface meaningful epidemiological signals; a poorly constructed one risks having the attention layer learn noise rather than genuine drivers of transmission.

Studies applying systematic feature selection before deep learning training have consistently demonstrated that curated inputs reduce noise and improve model generalisability beyond what raw feature inclusion achieves (Archana & Faritha Banu, 2025). The meteorological variables most consistently validated as dengue predictors—temperature, rainfall, humidity, wind speed, and their lagged variants—have been confirmed across multiple geographic settings and model classes (Cheng et al., 2026; Sang et al., 2025). Incorporating imported-case counts from neighbouring Southeast Asian countries has been shown to add predictive value beyond local measurements alone in a Chinese setting (Sang et al., 2025). In Malaysia specifically, attention-based LSTM models incorporating spatio-temporal structure achieved stronger incidence prediction than conventional machine learning baselines, indicating that spatial context contributes independent predictive information (Majeed et al., 2025). A Selangor-based study confirmed that domain-informed feature engineering—including a computationally derived mosquito biting rate variable—produced measurable improvements in LSTM forecasting accuracy (Lu et al., 2025). SHAP-driven variable selection identified temperature, precipitation, and humidity as the most influential lagged predictors across Brazilian settings, and these feature importance outputs were presented as interpretable inputs for public health planning (Chen & Moraga, 2025b). Vector abundance modelling using fine-scale

microclimatic variables (Dom et al., 2025), together with deep-learning forecasts of vector abundance from environmental predictors (Ceia-Hasse et al., 2023), further demonstrates the value of granular environmental inputs for dengue risk assessment.

The optimal feature set for dengue forecasting is not universal. Studies conducted across different countries have identified different leading predictors, reflecting genuine variation in local transmission ecology rather than inconsistent methodology. Surveillance systems that register cases administratively after clinical presentation also introduce a lag between when a case occurs and when it is recorded, producing an asynchrony between input features and outcome labels that has to be modelled rather than assumed away; the weekly reporting resolution of the Singapore series used in this study is an instance of the same problem, and Section 4.2.1 documents how it is handled here. Temperature and relative humidity have been found to interact with dengue case count lags in a non-additive manner in Malaysian district data, requiring models that can learn interaction effects dynamically rather than through pre-specified interaction terms (Cheng et al., 2026; Majeed et al., 2025). These contextual specifics mean that any fixed, manually defined feature hierarchy risks being suboptimal for the particular dataset and geographic context used in this study.

This contextual limitation provides additional motivation for the intrinsic attention mechanism as a learned, dynamic feature weighting tool, directly connecting this section to the architectural decision documented in Section 2.3.3. Rather than committing to a fixed feature hierarchy at the preprocessing stage, the attention layer recalibrates its input weights for each prediction, effectively performing adaptive feature selection as part of the forward pass. This dynamic recalibration fulfils Objective RO3 and provides an empirical pathway for testing Hypothesis H3, which evaluates whether the features assigned the highest attention weights by the model correspond to the predictors identified as most influential in the domain literature reviewed above.

2.3.5 Synthesis: Mapping Gaps to Architectural Components

Table 2.3 brings together the gap analysis from Section 2.1, the model critique from Section 2.2, and the adopted component justifications from Sections 2.3.1 to 2.3.4 into

a single traceable mapping. The purpose of the table is to make the logical pathway from identified problem to proposed solution explicit and verifiable—each row represents a gap, its description, the architectural component that addresses it, and the objective and hypothesis through which that address will be evaluated.

Gap Type	Gap Description	Addressing Architectural Component	Objective	Hypothesis
Methodological Gap <i>Section 2.1.2</i>	Linear statistical models fail to capture the complex nonlinear and multi-variable dynamics of dengue transmission.	LSTM temporal encoder with multivariate sequential input construction	RO1, RO2	H1, H2
Functional Gap <i>Section 2.2.2</i>	Conventional ML treats each time-step independently, discarding temporal order critical to multi-step forecasting.	LSTM gated memory architecture for ordered sequential learning	RO1, RO2	H1, H2
Transparency Gap <i>Section 2.2.3</i>	LSTM hidden states compress all historical information into uninterpretable fixed-dimensional vectors.	Additive temporal attention as intrinsic, forward-pass interpretability layer	RO3, RO4	H3, H4
Interpretive Gap (XAI Limitation) <i>Section 2.2.4</i>	Post-hoc XAI methods (SHAP/LIME) produce externally approximated, unstable explanations unsuited to sequential forecasting.	Intrinsic attention weights generated simultaneously with every prediction during the forward pass	RO3, RO4	H3, H4
Practical-Knowledge Gap <i>Section 2.1.3</i>	Accurate deep learning models are not adopted by practitioners because their outputs cannot be justified to decision-makers.	Attention weight visualisation as practitioner-readable, time-step-resolved reasoning output	RO3	H3
Contextual Gap <i>Section 2.3.4</i>	Optimal feature importance varies by geographic and temporal context; fixed preprocessing-stage selection is suboptimal.	Dynamic attention-based adaptive feature weighting recalibrated at inference time for each prediction	RO3	H3

Table 2.3 Traceability Mapping — Research Gaps, Architectural Components, Objectives, and Hypotheses

Reading Table 2.3 as a whole, what becomes clear is that the proposed architecture is not a collection of techniques assembled for their individual merit. Each component is present because a specific gap demanded it, and each gap is tested through a specific hypothesis that will be evaluated in Chapters 4 and 5. The inclusion of the XAI limitation gap clarifies why post-hoc explanation tools are insufficient: the practical-knowledge gap requires explanations that are generated intrinsically, consistently, and simultaneously with the forecast, not approximated after the fact.

2.4 Feasibility Study of Research Solution

A research proposal may be well-motivated theoretically but still face practical obstacles that undermine its viability. This applied to each of the five feasibility dimensions below, assessing whether the proposed hybrid LSTM-Transformer architecture can be built, funded, executed, deployed, and accepted within the constraints of the current research environment.

2.4.1 Technical Feasibility

Technical feasibility concerns whether the proposed architecture can be constructed, trained, and evaluated using tools and hardware currently available to independent academic researchers. For a hybrid deep learning model to be considered technically feasible, its component architectures must be independently validated in peer-reviewed literature, its implementation must be achievable using open-source frameworks without proprietary dependencies, and its interpretability outputs must be extractable within standard deep learning pipelines. Meeting these conditions establishes that the research can proceed without specialist infrastructure or institutional computing resources that are not universally accessible.

Both LSTM and Transformer architectures are fully supported by robust open-source frameworks—TensorFlow and PyTorch—which are widely used in the literature reviewed here and are sufficient to implement the proposed architecture. The successful implementation of LSTM-based dengue forecasting on multivariate datasets across Bangladesh, India, and Puerto Rico confirms that the data pipeline, training procedures, and evaluation protocols are reproducible across different national datasets using standard open-source deep learning frameworks (Ullah et al., 2024; Chen et al., 2025; Dhaked et al., 2025). A working hybrid deep learning architecture for weekly dengue incidence forecasting has already been demonstrated in the published literature (Zhao et al., 2023), and an LSTM-attention approach validated for epidemic forecasting using publicly available surveillance data further confirms that the proposed architecture can be implemented without specialised infrastructure (Hu et al., 2024). Implementation of Transformer self-attention networks for infectious disease time-series has been demonstrated using standard Python deep learning stacks, confirming that the QKV attention computation and

attention weight extraction are technically achievable without specialist tooling (Lv et al., 2024; Prakash et al., 2025). Reproducibility of hybrid LSTM-Transformer architectures across diverse geographic contexts has been further confirmed by comparative neural architecture benchmarks (Alotaibi, 2025), and efficient machine learning prediction systems using open-source libraries have been specifically validated for dengue in resource-constrained Southeast Asian settings (Saraswathi & Rohini, 2024).

The principal technical risk lies not in the implementation of individual components in isolation but in their integration into a single end-to-end trainable architecture while simultaneously preserving the interpretability of attention outputs. Attention weight visualisation requires careful output logging and post-forward-pass validation steps that are not yet standardised in most existing deep learning pipelines. More critically, no established protocol exists for validating whether extracted attention weights correspond to ground-truth epidemiological drivers in a dengue forecasting context specifically. This absence of validated procedures means that part of the technical contribution of this study is methodologically novel—not purely replicative—and carries the associated risk of requiring iterative refinement before the interpretability outputs meet a standard suitable for practitioner use.

Technical risk is assessed as low overall, given the breadth of prior implementations across analogous settings. The novel integration challenge identified above is addressed methodologically in Chapter 4, where the architecture design, attention extraction procedure, and validation protocol are specified in full. This feasibility assessment aligns with SDG 9 (Industry, Innovation, and Infrastructure) in its reliance on open, accessible, and reproducible computational tools, ensuring that the research outputs can be independently replicated and extended by other researchers without requiring institutional computing privileges.

2.4.2 Economic Feasibility

Economic feasibility concerns whether the proposed research can be conducted within realistic financial constraints. In an academic research context, this primarily means assessing whether the required software, datasets, and computational resources are accessible without significant cost, and whether any unavoidable expenses are

proportionate to the expected outputs. A study that requires expensive proprietary datasets, commercial software licences, or high-performance computing clusters beyond what is freely available to individual researchers faces a different feasibility profile from one that can be executed entirely on publicly available tools and data.

All software tools required for this study—PyTorch, Scikit-learn, Pandas, and Matplotlib—are open-source and freely available under permissive licences that permit unrestricted academic use. The dengue incidence records and meteorological datasets used in this study are publicly accessible through Singapore's Data.gov.sg open data portal and the Open-Meteo historical weather archive, eliminating data acquisition costs entirely; comparable open-data infrastructure has also been documented in the Malaysian context (Lu et al., 2025; Majeed et al., 2025). Cloud computing platforms including Google Colab and Kaggle Notebooks offer free-tier GPU access sufficient for model training at proof-of-concept scale; the configuration actually used in this study is recorded in Section 3.7.3. All software used in this study is open-source and was run on the no-cost compute described in Section 3.7.3, so the direct cost of the modelling work is negligible; and the feasibility of running multi-model benchmark comparisons on no-cost academic compute has been confirmed across analogous dengue prediction studies (Chen & Moraga, 2025a; Rahman et al., 2025a).

The main economic risk concerns computational time during hyperparameter tuning of the hybrid architecture. Training multiple configurations of an LSTM-Transformer model on multi-year multivariate dengue data may exceed free-tier cloud compute limits if extensive grid search across a large hyperparameter space is required. Free-tier platforms impose session time limits and memory caps that can interrupt long training runs, potentially increasing the total time required to complete the experimental phase. This risk is non-trivial for a hybrid architecture where the interaction between LSTM depth, the configuration of the temporal-attention layer, and dropout rates requires systematic exploration to avoid both underfitting and overfitting.

This risk is manageable through principled early stopping, structured hyperparameter search using Bayesian optimisation rather than exhaustive grid search, and by restricting the benchmark comparison to the model families identified as most

relevant in Section 2.2 rather than testing every conceivable baseline. Economic feasibility is assessed as high under these conditions. This assessment is consistent with SDG 17 (Partnerships for the Goals), which encourages the use of shared open infrastructure to advance health and development outcomes in resource-constrained settings—a principle that the open-source, zero-cost design of this study directly instantiates.

2.4.3 Operational Feasibility

Operational feasibility concerns two related but distinct questions: first, whether the research workflow can be executed reliably within the project timeline and institutional context; and second, whether the outputs of the research—particularly the attention weight visualisation—can be operationalised in a real public health setting rather than remaining a research-only artefact. A study that produces technically valid outputs but cannot be integrated into the workflows of the practitioners it is designed to serve fails the second dimension of operational feasibility even if it satisfies the first.

The research workflow decomposes naturally into sequential and independently testable stages—data collection and preprocessing, exploratory analysis, model development and training, benchmark evaluation, and interpretability analysis—each of which maps directly onto a section of the experimental design in Chapter 4. The modular design of the LSTM-Transformer architecture allows each component to be developed, tested, and refined independently before integration, a workflow strategy validated in comparable infectious disease deep learning projects (Zhao et al., 2023). Every stage can be executed within the Python ecosystem using well-documented libraries, ensuring the workflow is reproducible and auditable. Dengue prediction systems have been built and evaluated on government surveillance data within Southeast Asia — Malaysian state-level records in one case and Indonesian regional records in the other — confirming that the data on which such a system depends is already collected and released by the responsible agencies, and that a working prototype could therefore be reviewed by a health authority such as Singapore's National Environment Agency (Lu et al., 2025; Hamida et al., 2024). AI-driven dengue prediction and risk-management systems are under active development in the

published literature (N et al., 2025; Muthuselvi et al., 2025), indicating that the research direction is being pursued beyond this study.

Two residual operational risks warrant explicit acknowledgement. The primary risk is data quality and availability for the Singapore dengue dataset used in this study. If the available case records contain extended reporting gaps, inconsistent formatting across the benchmark and real-world holdout periods, or serotype-specific underreporting, preprocessing requirements may increase substantially and limit the generalisability of model outputs to the time periods represented in the training data. The secondary operational risk is the gap between research-grade interpretability outputs—attention weight matrices expressed as numerical arrays—and the visualisation formats that non-technical public health practitioners can readily interpret and translate into fogging or larviciding deployment decisions. An attention weight heatmap that is informative to a machine learning researcher may not be immediately actionable to a district health officer operating under time pressure during an outbreak response window.

The data quality risk is addressed through the preprocessing protocol specified in Chapter 4, which includes explicit procedures for handling reporting gaps and for redistributing weekly case totals across days; the dataset actually used is national rather than district-level, so no cross-boundary normalisation was required. The practitioner interpretability gap is addressed directly by Objective RO3 and Hypothesis H3, which require the study to test whether the attention-ranked features correspond to documented dengue risk factors rather than merely being displayable. Operational feasibility is assessed as moderate-to-high overall, with this operational focus directly aligned with SDG 3's emphasis on evidence quality and real-world applicability in health systems.

2.4.4 Market Feasibility

Market feasibility, in the context of public health research, concerns whether there is genuine institutional demand for the type of tool this study proposes to produce. The relevant market is not a commercial one but the community of health ministries, vector control agencies, and outbreak response teams who would deploy a dengue prediction tool if they trusted its outputs sufficiently to act upon them. A technically

valid tool that does not address the specific barriers preventing institutional adoption—primarily the lack of explainability that prevents practitioners from justifying algorithmic outputs to their superiors—has low market feasibility regardless of its predictive accuracy.

AI-driven dengue prediction and risk-management platforms continue to be proposed and prototyped in the recent literature (N et al., 2025). The persistent critical observation in the literature that accurate models are not trusted by practitioners defines a clearly articulated demand gap that the proposed framework directly targets (Loh et al., 2022; Bharati et al., 2024). Work on interpretability in medical machine learning argues that algorithmic tools become actionable only when their reasoning is legible to non-technical stakeholders, and that domain experts must be involved in designing how model outputs are interpreted (Vellido, 2019). The healthcare XAI literature argues that explanation requirements should be treated as a primary design criterion rather than an afterthought (Bharati et al., 2024; Vellido, 2019). The explainability component of this research addresses a barrier repeatedly identified in the explainable-AI literature: the absence of outputs that practitioners can interrogate and justify to administrative decision-makers (Loh et al., 2022; Bharati et al., 2024).

Two market-level risks remain that fall at least partially outside the scope of the current study. The primary risk is the implementation gap between a research prototype and a deployable tool: a model that performs well in an academic evaluation may still face institutional barriers to adoption if it requires specialist technical expertise to deploy, maintain, and retrain that is not available in resource-constrained district health departments. The secondary risk is that the attention weight visualisations produced by this architecture, while informative as research outputs, may require additional user interface design and communication work before they become accessible to the non-technical practitioners who would need to act on them in time-pressured outbreak response contexts. Neither of these risks invalidates the market case for the research contribution itself, but both limit how directly the research outputs can be translated into operational tools without further development investment.

Market feasibility is assessed as high for the scholarly contribution itself—the demonstration that a hybrid LSTM-Transformer architecture can produce both

accurate and intrinsically interpretable dengue forecasts—with the implementation and interface design risks appropriately scoped as future work beyond the current study's boundary. The institutional demand documented above directly motivates Objective RO3 and Hypothesis H3, which test whether the model's attention outputs correspond to documented dengue risk factors rather than being merely displayable. This market focus supports SDG 3's vision of health systems that can act on predictive AI outputs, not merely observe them from a distance.

2.4.5 Social Feasibility

Social feasibility concerns whether the research is likely to produce outcomes that are acceptable to and beneficial for the communities it is intended to serve, and whether its design respects the governance expectations of the institutions and populations that would rely on its outputs. In the dengue prediction context, this means assessing whether an AI-based prediction tool is likely to improve health equity outcomes, whether its deployment concentrates or distributes decision-making power in ways that disadvantaged communities can scrutinise and challenge, and whether the communities most affected by dengue outbreaks have a meaningful pathway to understanding and questioning how algorithmic outputs are used to allocate health resources in their neighbourhoods.

Dengue disproportionately affects lower-income communities in tropical urban regions, making early and precise outbreak response a matter of health equity as much as epidemiological efficiency (Wang et al., 2023). A predictive tool that enables health authorities to respond earlier and to target resources more precisely therefore contributes directly to SDG 3 (Good Health and Well-being) and SDG 10 (Reduced Inequalities). The interpretability literature holds that algorithmic health tools gain traction with non-technical stakeholders only when their reasoning is inspectable—a position that directly motivates the interpretability design objectives of this study (Vellido, 2019; Bharati et al., 2024). Interpretable models have been specifically identified in the health AI literature as a prerequisite for equitable deployment, because opaque systems concentrate decision-making power in ways that affected communities cannot challenge or scrutinise (Vellido, 2019). Explainable dengue prediction systems built on interpretable models have been reported for tropical Asian settings, where feature attributions are presented as the basis on which an outbreak

response would be justified rather than as a diagnostic afterthought (N et al., 2025; Rahman et al., 2025a). Making the basis of a forecast inspectable is therefore a matter of accountability as well as of technical validity, since an explanation that cannot be examined cannot be challenged by those it affects (Vellido, 2019).

Two social risks warrant explicit identification. The primary risk is algorithmic bias: if the training data overrepresent certain geographic areas, income groups, serotypes, or seasonal windows, the model's predictions may be systematically more accurate for some communities than others. Periods with historically consistent NEA reporting will produce more reliable forecasts than periods with incomplete surveillance records, potentially directing resources away from underserved localities that face the greatest transmission risk and the least administrative capacity to compensate. The secondary social risk is that the emphasis on technical interpretability—attention weight heatmaps and variable-level attribution scores—may not translate into genuine social accountability if the explanations remain accessible only to trained machine learning practitioners and are not communicated in formats meaningful to the community members most affected by dengue outbreaks and the resource allocation decisions that follow from them.

Social feasibility is assessed as high for the research contribution itself, with the identified risks flagged as important directions for future work in community-engaged validation, bias auditing across district reporting quality strata, and accessible communication design for non-technical explanation outputs. The emphasis on intrinsic interpretability carries a direct social governance dimension: by making the model's reasoning visible and challengeable, the proposed architecture advances the principle of responsible AI governance in public health settings consistent with SDG 16 (Peace, Justice, and Strong Institutions). This social feasibility assessment strengthens the case for Hypothesis H3, which tests whether the attention-ranked features correspond to documented dengue risk factors—a test that, if passed, constitutes evidence not only of technical validity but of the social legitimacy necessary for equitable deployment.

Across all five feasibility dimensions, the proposed research is assessed as viable, with specific residual risks identified and assigned to either the experimental design of Chapter 4 or explicitly scoped as future work. No single dimension presents an

insurmountable obstacle, and the risks that do exist are proportionate and manageable within the constraints of an academic proof-of-concept study. The feasibility analysis as a whole confirms that the scholarly gap identified in Sections 2.1 and 2.2 can be meaningfully addressed within the practical boundaries of the current study.

2.5 Summary

This chapter set out to answer a single question: why this architecture, for this problem, at this moment in the development of the field? The answer was built up systematically across four sections, each structured using the SDCL framework.

Section 2.1 established that dengue is a genuinely complex, multi-scale problem that current health systems are not managing well, and identified six specific gap types through Sections 2.1.1 to 2.1.3, consolidated in Table 2.1: epidemiological complexity, the methodological limits of statistical models, data-integration complexity, the black-box problem in deep learning, the failure of post-hoc XAI methods, and the operational adoption gap. The key insight remains: the core challenge in dengue prediction is not the absence of capable models, but the absence of models that are simultaneously capable and explainable.

Section 2.2 traced those gaps through four categories of prediction and interpretability approaches. Direct foundational citations for LSTM (Sherstinsky, 2020; Al-Selwi et al., 2024), the attention mechanism family from which the adopted variant is drawn (Niu et al., 2021; Lv et al., 2024), and XAI methods (Lundberg et al., 2020; Loh et al., 2022) were incorporated to provide traceable scholarly grounding for the architectural components that would otherwise appear only through their downstream applications. Table 2.2 compressed the analysis into a comparative view spanning six approaches and nine criteria.

Section 2.3 justified each component of the proposed hybrid architecture as a targeted response to a specific gap: the hybrid DL precedent (Section 2.3.1), the LSTM temporal encoder with its formal architectural basis (Section 2.3.2), the attention family with its mathematical formulation and the reasons for adopting the additive temporal-pooling variant rather than full self-attention (Section 2.3.3), and the meteorological feature integration strategy grounded in documented dengue risk factors (Section 2.3.4). Table 2.3 made the full mapping from problem to solution to

evaluation traceable, with each of the six gap rows tied to the specific research objective and hypothesis through which it is tested.

Section 2.4 assessed the practical viability of the proposed work across five feasibility dimensions—technical, economic, operational, market, and social. Each dimension was supported with peer-reviewed studies published between 2019 and 2026 and drawn from IEEE Xplore, ScienceDirect, SpringerLink, Nature Portfolio and BMC. The proposal is assessed as feasible on all five counts, with specific residual risks identified and addressed in the experimental design of Chapter 4.

Read as a whole, this chapter connects to Sustainable Development Goal 3 (Good Health and Well-being) at its centre: the gap it documents is not an abstract methodological curiosity but the reason accurate forecasts fail to become timely public health action. Three further Goals are engaged along the way — SDG 9 (Industry, Innovation and Infrastructure) through the open-source computational stack that makes the architecture deployable, SDG 10 (Reduced Inequalities) through the affordability of that stack for agencies without specialist infrastructure, and SDG 17 (Partnerships for the Goals) through the data-sharing arrangements on which national surveillance series depend. The chapter therefore closes with the case for the architecture established on scholarly grounds and its feasibility established on practical ones; Chapter 3 sets out how it is built.

Chapter 3

Research Methodology

3.0 Research Methodology

The study's design choices are documented in this chapter. It connects each choice to the two main issues mentioned in Section 1.3: (1) the linear statistical baselines' accuracy ceiling and (2) the adoption barrier that has kept deep-learning forecasting tools out of regular public health workflows (Chen & Moraga, 2025a; Loh et al., 2022). If one issue is resolved without the other, the result is a model that is either transparent but not adequately predictive or accurate but unreliable. The conceptual and theoretical foundation (Section 3.1), the suggested ILT model and its data pipeline (Section 3.2), the two salient features of the solution with their pseudocode, complexity derivation and UML specifications (Sections 3.3–3.4), the evaluation framework (Section 3.5), the interpretability analysis (Section 3.6), and the robustness and ethical considerations (Sections 3.7–3.8) are the six components that make up this chapter.

3.1 Theoretical and Conceptual Framework

None of the models examined in the literature review of Chapter 2 provided robust temporal modelling, built-in interpretability, and practitioner usability within a single evaluated framework, according to the literature study in Chapter 2 (Table 2.2, Section 2.2.5). Closing that gap is the foundation of our paradigm. There are two levels to it. The theoretical framework recognises the study's contribution in relation to earlier research and places it within accepted scientific theory. The particular variables being studied and the anticipated correlations between them are specified in the Conceptual Framework.

3.1.1 Theoretical Framework

The theoretical backbone draws on two independent bodies of literature. The first addresses temporal dependency in sequential learning. Standard recurrent neural networks are vulnerable to vanishing and exploding gradients, a problem that Sherstinsky (2020) has formally characterised. Input, forget, and output gates that control information retention and discarding over time are how the LSTM design addresses this (Al-Selwi et al., 2024). The environmental factors that contribute to dengue transmission—accumulated rainfall and persistently high temperatures—act over periods of one to four weeks as opposed to a single day (Chapter 2, Section

2.2.3). A model that cannot encode dependencies across that window is structurally unable to capture the dominant predictive signals. This provides the theoretical grounding for Hypotheses H1 and H2.

The second body of literature concerns the interpretability properties of attention mechanisms. Attention was originally developed for machine translation and formalised as a class of parametric salience functions that weight input elements during the forward pass (Niu, Zhong, & Yu, 2021). The critical distinction from post-hoc explainability tools is that attention weights are produced during the model's own computation. According to Ahmed et al. (2025), SHAP and LIME have a specific drawback in that they can provide different feature-importance rankings for the same inputs in different runs. By passing the weights the model calculated during the forward pass via a special attention module, the ILT model completely avoids this, turning each explanation into a deterministic record as opposed to an approximation (Niu et al., 2021; Al-Selwi et al., 2024).

Taken together, these two bodies of literature position the study's contribution as an Augmentation in the typology of theoretical contributions. The existing framework being extended is LSTM-based epidemiological time-series modelling (Al-Selwi et al., 2024; Lu et al., 2025). The addition is the attention-weight mechanism as an interpretability layer—one that preserves the LSTM's temporal modelling capacity while making the model's reasoning accessible for inspection. This directly targets the transparency gap documented in Chapter 2 (Section 2.1.3) and is operationalised through Objectives RO3 and RO4.

An augmentation framing also sets a concrete success criterion. Improved prediction alone is not sufficient—the added attention layer must produce explanations that are scientifically coherent (tested by H3), and it must do so at a computational cost low enough for the model to remain practical to deploy (tested by H4). Evidence from self-attention models applied to infectious disease time series (Lv et al., 2024) supports the expectation that learned attention patterns can align with known epidemiological dynamics, a correspondence this study verifies using Singapore dengue data.

3.1.2 Conceptual Framework

The theoretical stance is translated into an experimentally testable structure via the conceptual framework. Together, the five hypotheses in Chapter 1 (Sections 1.4.2, 1.4.3 and 1.4.5) call for predictive performance (H1, H2), explanatory quality (H3), computational efficiency (H4) and out-of-sample robustness (H5).

Seven categories of meteorological and epidemiological data make up the Independent Variables (IVs): minimum, maximum, and mean daily temperature; cumulative and daily rainfall; relative humidity; wind speed; lagged versions of each over 14- to 28-day windows; rolling summary statistics; and historical dengue case counts. This choice is based on a substantial body of research from tropical Asian environments — Malaysia, southern China and Bangladesh — that connects these climatic factors to dengue transmission and *Aedes* mosquito reproduction (Lu et al., 2025; Cheng et al., 2026; Rahman et al., 2025a; Sang et al., 2025). The effects are interactive and lag-dependent rather than additive or instantaneous (Cheng et al., 2026)—the kind of non-linearity that linear models cannot represent, and that provides a primary justification for choosing a deep-learning architecture.

The Dependent Variables (DVs) span three dimensions: Predictive Accuracy (RMSE, MAE, R^2 applied identically across all models and horizons), Operational Transparency (the domain-validation hit rate for H3), and Computational Efficiency (training time, inference latency, and model size for H4). A transparent, efficient model that forecasts badly fails the accuracy requirement; an accurate model whose reasoning cannot be inspected fails the transparency requirement; and an accurate, transparent model that is too costly to train or run fails the efficiency requirement. All three conditions must be met.

Between the IVs and DVs sits the central interpretability construct: the temporal attention-weight vector. Each 28-day input window is compressed by the LSTM encoder into a series of 28 hidden-state vectors. The temporal attention layer assigns scalar importance weights to each of those vectors recording which historical time steps drove each forecast. The attention module stores that weight vector as a deterministic record of the model's forward computation—not a post-hoc reconstruction of the kind produced by perturbation-based explainers such as LIME and SHAP (Ahmed et al., 2025). Removing the attention layer would eliminate not only interpretability but also the mechanism that provides the model's accuracy advantage over the standalone LSTM.

The attention-weight vector is treated here as the central interpretability construct rather than as a formally tested mediator, and the distinction matters. Attention computes the forecast as a weighted sum of hidden states, so the weights are an intermediate computational representation that demonstrably lies between the inputs and the output (Niu et al., 2021). That is a statement about the computation, not about causation: no mediation analysis, intervention or counterfactual estimation was performed, and none is claimed. The same review records that whether attention weights constitute faithful explanations remains contested — Serrano and Smith report them to be only noisy predictors of component importance, and Wiegrefe and Pinter rebut that finding — which is precisely why this study does not assume the correspondence but tests it as H3. What the construct is expected to do is nonetheless testable. Evidence from dengue-specific studies (Zhao et al., 2023) shows consistently that attention augmentation improves multi-step forecasting over standalone LSTM baselines. Therefore, the structural mechanism that transforms a high-performing black-box model into one that practitioners can examine and utilise is the temporal attention-weight vector.

Figure 3.1 sets out the resulting conceptual framework.

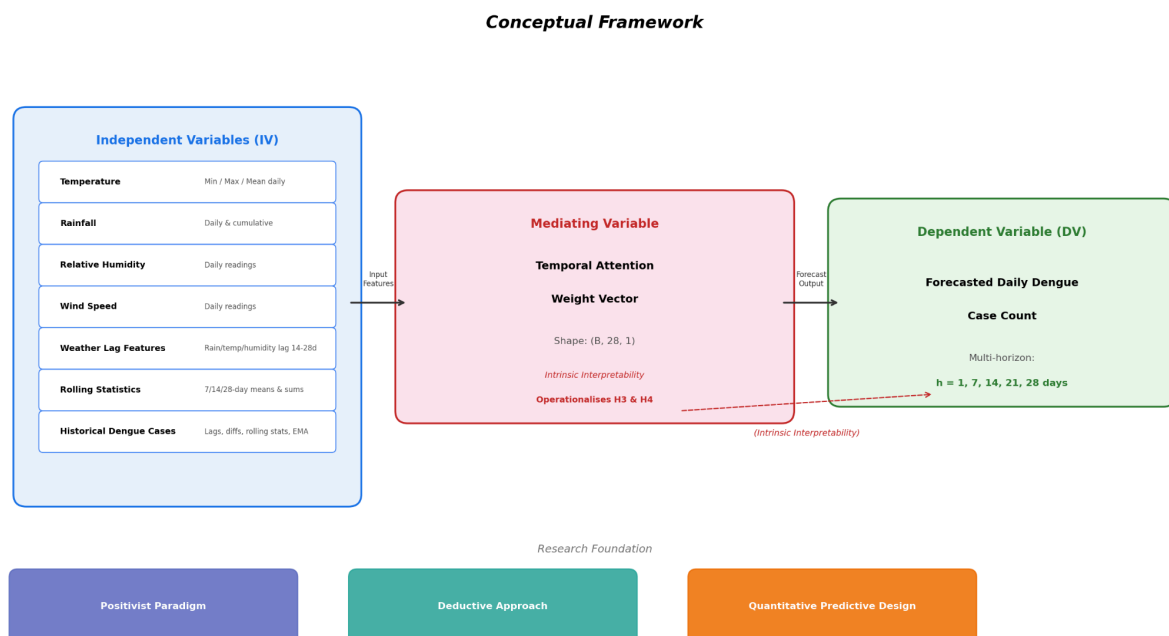


Figure 3.1 A conceptual framework that uses the temporal attention weight vector to map independent meteorological and epidemiological data to the predicted dengue case count.

3.1.3 Research Paradigm

The research paradigm used in this study is positivist. According to positivism, trustworthy knowledge originates from observable, quantifiable, and independently repeatable occurrences. From an ontological perspective, positivism and realism are compatible: regardless of the observer, an objective reality exists and functions according to stable mechanisms. From an epistemological perspective, it is consistent with empiricism: knowledge is constructed via methodical measurement and hypothesis testing.

Meteorological data and dengue case numbers align with the positivist paradigm. Both are collected by government agencies under standardised protocols, and model quality is assessed using RMSE, MAE, and R^2 —metrics that yield the same value regardless of who computes them.

The study takes a realist ontological position on dengue transmission. Mosquito breeding cycles, ambient climate conditions, and human population density are real causal forces operating independently of any model. The ILT model is understood as a practical quantitative approximation of those forces. This realist orientation is most apparent in the domain-validation protocol for Hypothesis H3: the criterion is not whether the attention weights are statistically internally consistent, but whether they align with causal mechanisms that epidemiologists have established through independent research programmes.

Epistemologically, the positivist commitment means favouring quantitative measurement over qualitative interpretation. RMSE, MAE, R^2 , and the domain-validation hit rate are all derived from model outputs against observed case counts and leave no room for subjective interpretation. Even the computational-efficiency component (H4) is anchored to pre-defined numerical thresholds — a cost ratio below $2\times$ the Standalone LSTM on every metric, and an attention scaling exponent below 2 — rather than an open-ended qualitative assessment.

3.1.4 Research Approach

The study follows a deductive logic. The five hypotheses were derived from existing literature on dengue forecasting and interpretable deep learning before any data analysis was conducted—they were not generated inductively from the data. Theory informs hypotheses; hypotheses are operationalised into measurable criteria; measurements are compared against those criteria; and hypotheses are accepted or rejected based on the outcomes.

A deductive approach is appropriate because the research questions are specific and bounded: they ask whether the ILT model clears defined performance thresholds and whether its attention outputs correspond to documented epidemiological patterns. Both questions have determinate answers that pre-specified evaluation criteria can assess.

3.1.5 Research Design

A quantitative, predictive research design underpins all empirical work. The performance metrics by which the ILT model is evaluated—RMSE, MAE, R^2 , and the domain-validation hit rate—are all numerical and directly comparable across models and against pre-set benchmarks. The study's goal is to producing accurate, interpretable near-term dengue forecasts; it is not designed to establish new causal pathways in the epidemiological literature.

This distinction is important for interpreting the attention-weight outputs. As Rahman et al. (2025a) note, attention weights capture learned statistical associations rather than verified causal relationships. The domain-validation protocol in Section 3.6.3 addresses this by checking whether the associations the model learns are consistent with causal mechanisms that epidemiologists have independently established.

All data are secondary—government-released historical aggregate records rather than data collected by the researcher. This rules out direct experimental control but increases practical relevance, as the model is trained and tested on data of the type and quality that an operational forecasting system would actually receive. This design pattern is standard in the dengue forecasting literature (Zhao et al., 2023; Lu et al., 2025; Cheng et al., 2026).

3.2 The Hybrid LSTM-Transformer (ILT) Solution Suggested

3.2.1 Pipeline for Complete Data Processing

Before being included into the model, raw climatic and epidemiological records go through six consecutive steps of processing. Every step generates a structured output that immediately feeds into the following one. The whole data flow is depicted in Figure 3.2.

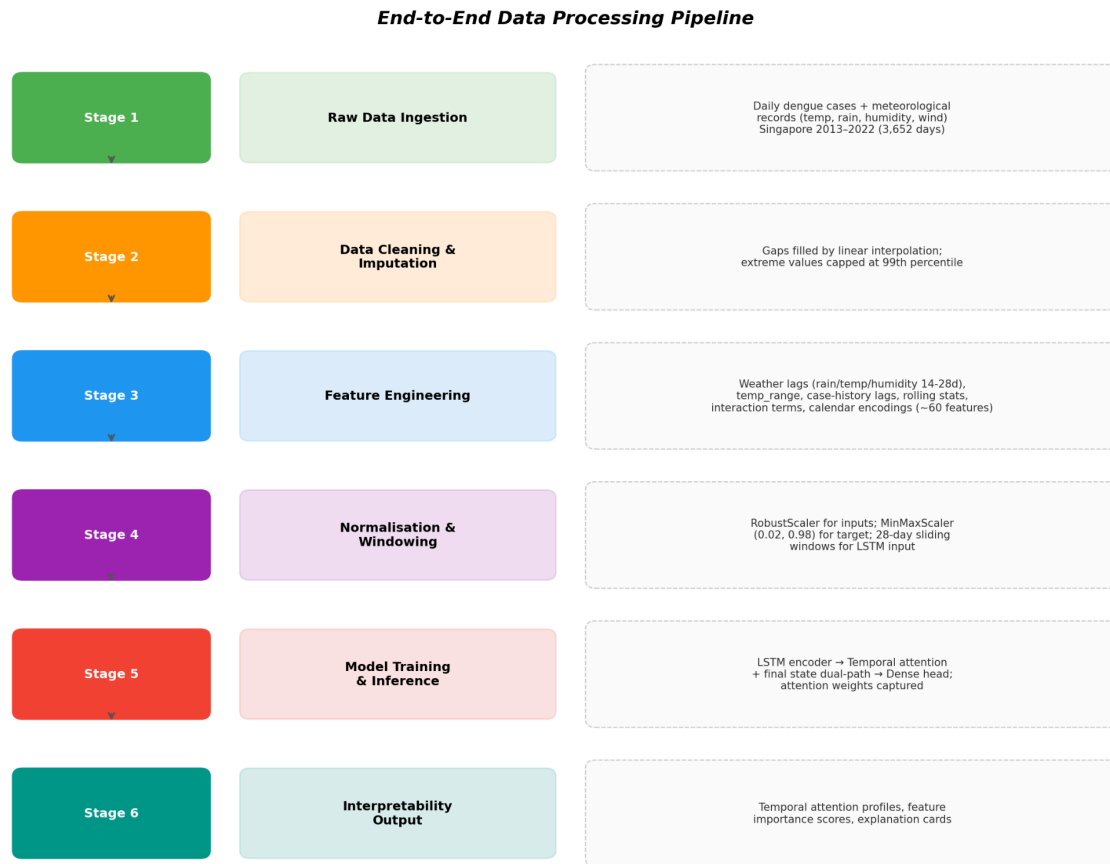


Figure 3.2 The six-stage data processing pipeline, from raw data ingestion through cleaning, feature engineering, normalisation and training to interpretability output.

Stage 1 — Raw Data Ingestion. The Singapore dengue-weather dataset, which includes 3,652 daily records from 2013 to 2022, is used to load daily dengue case counts and meteorological data (temperature min/max/mean, rainfall, humidity, wind speed).

Stage 2 — Data Cleaning & imputation. Missing values are filled by linear interpolation. Extreme values are capped at the 99th percentile to prevent a small

number of anomalous readings from distorting the softmax distribution in the attention mechanism.

Stage 3 — Feature Engineering. Six families of derived features are computed to capture both immediate conditions and their lagged effects:

- Cyclical calendar encodings. Along with quarter and day-of-week indications, month, week-of-year, and day-of-year are encoded as sine-cosine pairs to maintain their circular periodicity.
- Weather interaction terms. The combination heat-moisture conditions that encourage *Aedes* mosquito reproduction (Dom et al., 2025) are captured by products of temperature $\times$ humidity, rainfall $\times$ humidity, and temperature $\times$ rainfall.
- Range of daily temperatures. Overnight warmth stability is shown by the daily max-min temperature difference (temp_range); smaller ranges indicate prolonged heat that promotes mosquito activity and survival.
- Weather lag features. Temperature lagged at 14 and 21 days, humidity lagged at 14 days, and rainfall lagged at 14, 21, and 28 days. The lag windows follow the *Aedes* life cycle: rainfall creates breeding habitat on day zero, aquatic larval development follows over roughly 7–14 days, adults emerge and begin biting at about 14–21 days, and human symptoms appear after the intrinsic incubation period at roughly 21–28 days, giving a total rainfall-to-case interval of three to four weeks (Dom et al., 2025). The model's learning load during training is lessened by explicitly encoding these delays.
- Case-history features. Dengue cases delayed at 14, 21, and 28 days; rolling mean, standard deviation, and maximum over 7/14/28-day windows (shifted 14 days to avoid same-week data leaking); exponential moving averages for 14 and 28 days; and initial differences for 14 days. These features capture epidemic momentum, which is among the most informative short-horizon indicators.
- Weather rolling statistics. Cumulative rainfall totals, rolling temperature means, and humidity means over the 7/14/28-day periods.

The feature set has about 60 variables after all modifications. Due to inadequate lag information, the first 41 recordings are eliminated, leaving 3,611 daily observations that can be used.

Stage 4 — Normalisation and Windowing. To avoid leakage, all input features are scaled using RobustScaler (interquartile range-based), which is only fitted on the training partition. The sigmoid output layer maintains headroom at both ends for values close to the training extremes since the target variable is scaled using MinMaxScaler to the range (0.02, 0.98), fitted on the whole training set. LSTM input sequences are created by assembling sliding windows of 28 consecutive days.

Stage 5 — Model Training and Inference. The LSTM encoder processes each 28-day window; the temporal attention mechanism assigns scalar importance weights to each of the 28 time steps; the dual-path merge concatenates the attention-weighted summary with the LSTM's final hidden state; and the dense prediction head generates the forecast.

Stage 6 — Interpretability Output. Temporal attention profiles, feature-importance scores, and per-prediction explanation cards are generated from the attention weights recorded during inference.

3.2.2 ILT Model Architecture

The term 'Transformer' in the ILT model name refers to the adoption of an attention-based weighting mechanism rather than of the full encoder-decoder stack, and the lineage is worth stating precisely because the two are often conflated. The mechanism implemented here is additive attention, the family first applied to sequence models in Bahdanau's RNNsearch and still one of the two most widely used scoring functions (Niu et al., 2021). What the Transformer contributed was a different member of that family — scaled dot-product multi-head self-attention — together with the demonstration that attention alone can replace recurrence entirely (Vaswani et al., 2017). This study adopts the weighting principle those architectures share, not the Transformer's specific scoring function and not its encoder-decoder stack. This naming convention follows the broader literature on LSTM-attention hybrid models, in which an attention layer is embedded within a recurrent encoder rather than replacing it (Wen & Li, 2023; Majeed et al., 2025); the term 'Transformer' is used here to denote that attention-based weighting component, not a full encoder-decoder

Transformer stack. The deliberate choice of temporal attention pooling over full multi-head self-attention is justified by the limited training set size (~2,400 sequences), as documented in Table 3.1. The Interpretable LSTM-Transformer (ILT) model combines a Long Short-Term Memory encoder with a temporal attention mechanism and a dual-path fusion strategy. In order to provide a legible record of the model's temporal reasoning, the LSTM records sequential dependencies over the 28-day input window. The attention mechanism then determines which of those 28 days has the highest prediction signal. The dual-path architecture maintains the LSTM's recency-sensitive final hidden state as well as the attention-weighted summary across all 28 stages. The whole architecture is seen in Figure 3.3.

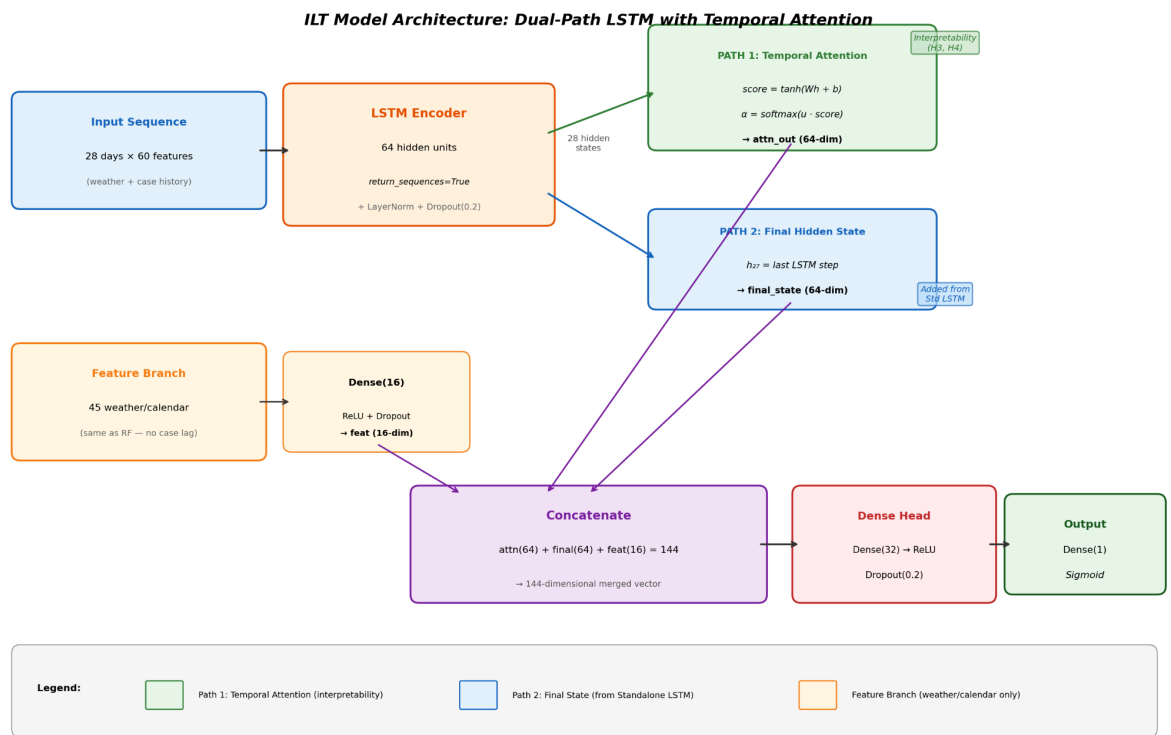


Figure 3.3 The architecture of the Interpretable LSTM-Transformer (ILT) model illustrates the final hidden state from the solo LSTM and the dual-path design with temporal attention.

Temporal LSTM Encoder

Each 28-day window is transformed into a series of 28 hidden-state vectors using a single LSTM layer with 64 hidden units. Because the downstream attention layer needs representations from each time step in order to calculate per-step weights, the layer is set up to return the entire sequence rather than just the final hidden state. Layer normalisation is applied to the output sequence, stabilising the activation

distribution and accelerating convergence (Tolic et al., 2024). Dropout at 0.2 regularises the recurrent connections—a rate within the 0.1–0.3 range that Al-Selwi et al. (2024) documented as standard across LSTM studies published between 2018 and 2023.

Temporal Attention Mechanism

This study uses a temporal attention pooling mechanism rather than the full multi-head self-attention from the original Transformer. With a 64-dimensional hidden state, the Q/K/V projections of multi-head self-attention require about 12,500 parameters, rising to roughly 16,600 once the output projection is included; preliminary experiments on the available 2,400 training sequences showed this configuration to be unstable. Temporal attention pooling achieves the same objective—identifying which of the 28 historical days carry the strongest predictive signal—using $64 \times 64 + 64 + 64 = 4,224$ parameters, roughly a quarter of the self-attention cost, which is what makes it the more data-efficient choice at this sample size.

Each LSTM hidden state h_t is scored through a learned nonlinear function:

$$score_t = \tanh(W \cdot h_t + b)$$

where b is a bias vector and W is a (64×64) weight matrix. Each score is projected to a scalar by a learnable context vector u , and the final weight distribution is generated using softmax normalisation:

$$\alpha_t = \text{softmax}(u^T \cdot score_t)$$

The attended output is the weighted sum of all 28 hidden states:

$$context = \sum(\alpha_t \cdot h_t)$$

When training data is scarce, Wen and Li (2023) demonstrated that temporal attention pooling equals or surpasses self-attention on time-series tasks. An LSTM-attention architecture for multi-state dengue prediction in Malaysia was verified by Majeed et al. (2025), who showed that the attention mechanism reveals temporal patterns of epidemiological significance.

The weights α_t are stored as a layer attribute during each forward pass and extracted for interpretability analysis without modifying the gradient graph. This is intrinsic interpretability: the weights derive from the model’s own forward computation, not

from an external approximation method, and directly address the transparency gap documented in Section 2.1.3.

Dual-Path Information Fusion

Ablation experiments revealed that the standalone LSTM's final hidden state carries a recency bias that is useful for short-horizon forecasts. Discarding this signal in favour of the attention-weighted summary alone would reduce performance at $h = 1$ and $h = 7$ days. The ILT model, therefore, retains both:

Path 1 (interpretability): The attention mechanism highlights the past days that are most pertinent to the present forecast by generating a 64-dimensional weighted summary across all 28 hidden states.

Path 2 (recency): The final hidden state h_{27} , which is the same vector used for prediction by the solo LSTM, is obtained directly.

The two 64-dimensional vectors are concatenated with a 16-dimensional weather/calendar feature branch, yielding a 144-dimensional merged representation for the prediction head. When the attention layer learns nothing useful, the model falls back entirely on the final hidden state, matching standalone LSTM performance. When the attention layer identifies a historically informative signal—for example, a rainfall event 14 days prior—the merged representation is enriched beyond what the final hidden state alone provides.

Feature Branch

A separate branch passes the 44 current-day weather, calendar, and population features through Dense(16) with ReLU activation and 0.2 dropout. This branch receives the same features supplied to the Random Forest baseline—no case-history variables are included. Case history enters the model exclusively through the 28-day LSTM sequence, so any performance advantage of the ILT model over RF is attributable to sequential processing rather than feature set differences.

Dense Prediction Head

After passing through Dense(32) with ReLU and 0.2 dropout, the 144-dimensional concatenation is sent to a single sigmoid output neuron. This setup limits forecasts to a sensible range while maintaining margin at both ends when used with the MinMaxScaler(0.02, 0.98) target transformation.

Hyperparameter Selection

Table 3.1 lists the selected hyperparameters. Initial values were drawn from the literature and validated against the training partition prior to any test-set evaluation.

Hyperparameter	Value	Justification
LSTM Hidden Units	64	Standard for LSTM sequence models (Al-Selwi et al., 2024)
Attention Type	Temporal Pooling (4,224 params)	4,224 params; stable on ~2,400 training samples
Dense Head	32 units	Handles 144-dim concatenated input from dual-path
Dropout(LSTM/Dense)	0.2/0.2	Within standard 0.1–0.3 range (Al-Selwi et al., 2024)
Learning Rate	0.001 (Adam, clipnorm=1.0)	Standard for time-series Deep Learning
Batch Size	32	Standard for time-series Deep Learning
Max Epochs	100 with patience=20	EarlyStopping + ReduceLROnPlateau
Target Scaler	MinMax(0.02,0.98)	Sigmoid boundary margin, fit on the full training set
Sequence Length	28days	Spans full Aedes mosquito lifecycle

Table 3.1 Hyperparameter configurations for the ILT model

3.3 Salient Feature 1 / Composite Features 1: Intrinsic Forward-Pass Interpretability

Sections 3.3 and 3.4 state the two salient features of the proposed solution — the characteristics that distinguish it from every model reviewed in Chapter 2 — and, for each, identify the types of research being integrated and the claim in Section 1.5 that the feature is meant to support. Naming the research types is not a formality, because the two features rest on different kinds of evidence. The first is established by construction and by controlled comparison; the second is established by measurement over repeated trials. A reader who accepts the first on the strength of the evidence offered for the second has accepted the wrong argument.

Salient Feature 1 is that the explanation and the prediction are the same computation. The temporal attention layer emits a normalised weight vector $\alpha \in \mathbb{R}^{28}$ with $\sum \alpha_i = 1$ as an ordinary output of the forward pass, and the forecast $\hat{y}$ is built from the context vector that α defines. No second model is fitted, no surrogate is trained, and no perturbation sampling is performed. This is the precise point of difference from the post-hoc

approaches reviewed in Section 2.2.4, where the explanation is reconstructed after the fact by a separate procedure that can return different attributions on different runs for the same input (Lundberg et al., 2020; Loh et al., 2022). Because α is generated inside the forward pass, it is available for every prediction a deployed model makes, at no inference cost beyond the pass itself.

Figure 3.4 gives the static structure of the implementation as a UML class diagram. The composition relationships show that TemporalAttention is a component of the model rather than an external analysis tool, and the attribute last_alpha on that class is where the weight vector is retained after each pass.

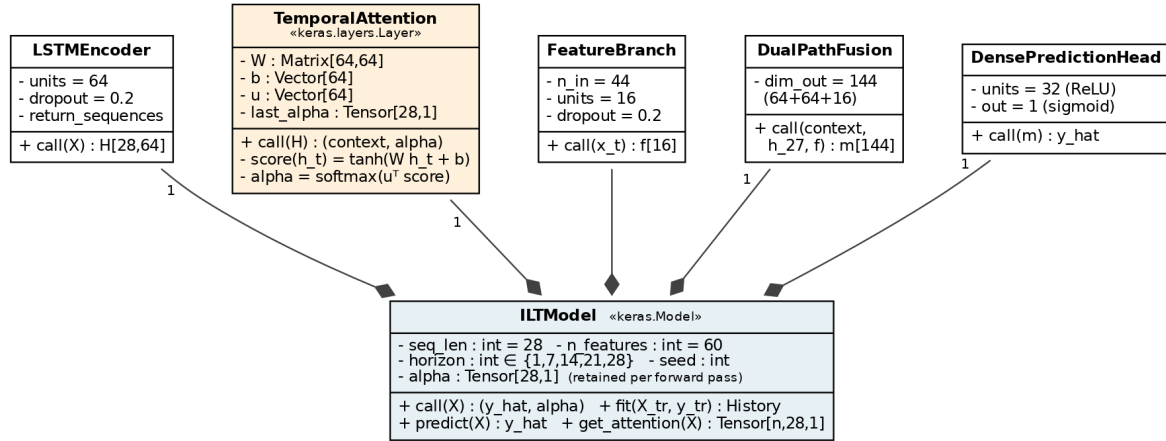


Figure 3.4 UML class diagram of the ILT implementation. TemporalAttention is composed into ILTModel as a first-class component and retains the attention vector as the layer attribute last_alpha, which is what makes the interpretability intrinsic rather than post-hoc.

Algorithm 1 states the forward pass. Lines 2 to 7 produce the explanation and lines 8 to 11 produce the forecast, and the two cannot be separated because line 8 consumes the α computed at line 6. The attention mechanism costs one $d \times d$ matrix product per time step and a single softmax; the weight vector is a by-product of line 6, not the output of a second procedure.

Algorithm 1 ILT forward pass for a single input window

Input : $X \in \mathbb{R}^{(28 \times 60)}$ scaled feature window (28 days $\times$ 60 variables)
 $x_t \in \mathbb{R}^{44}$ current-day weather / calendar / population vector

Output: $y_{\text{hat}} \in [0,1]$ scaled forecast at horizon h
 $\alpha \in \mathbb{R}^{28}$ temporal attention weights, $\sum \alpha_t = 1$

```

1  H ← LSTM(X)                                // H = [h_0 ... h_27], h_t ∈ ℝ^64
2  for t ← 0 to 27 do
3      s_t ← tanh(W · h_t + b)                  // W ∈ ℝ^(64×64), b ∈ ℝ^64
4      e_t ← uT · s_t                          // u ∈ ℝ^64 → scalar score
5  end for
6  α ← softmax(e_0, ..., e_27)                  // ∑ α_t = 1
7  self.last_alpha ← α                          // retained; not part of the loss
8  context ← ∑_{t=0}^{27} α_t · h_t              // ∈ ℝ^64 interpretability path
9  f ← ReLU(W_f · x_t + b_f)                   // ∈ ℝ^16 feature branch
10 m ← concat(context, h_27, f)                 // ∈ ℝ^144 dual-path fusion
11 y_hat ← sigmoid(w_oT · ReLU(W_m · m + b_m) + b_o)
12 return (y_hat, α)

```

Figure 3.5 shows the same computation as a UML sequence diagram, which makes the ordering explicit: control returns to ILTModel from TemporalAttention carrying both the context vector and α , and the final return to the caller at message 13 carries the forecast and its explanation together.

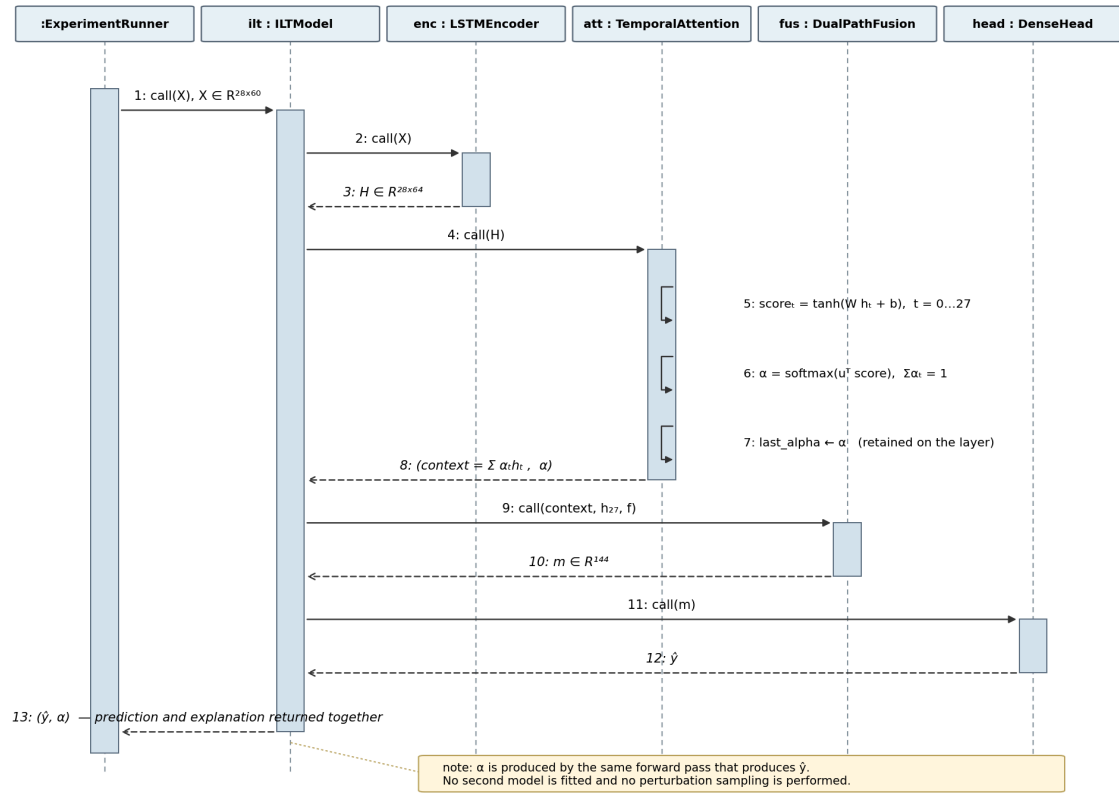


Figure 3.5 UML sequence diagram of one ILT forward pass. Messages 5 to 7 are the attention computation; message 13 returns the forecast and the attention vector in the same call, so no separate explanation step exists.

Three of the research types set out in the research methodology literature are integrated to establish this feature.

Basic research. The attention formulation is adapted rather than adopted. The additive scoring function at lines 3 and 4 of Algorithm 1 belongs to the additive attention family, which predates the Transformer and remains one of the two dominant scoring functions alongside the multiplicative form the Transformer scaled (Niu et al., 2021; Vaswani et al., 2017); it is positioned here as an interpretability layer over a recurrent encoder rather than as a replacement for recurrence. The theoretical commitment is the claim that this positioning is sufficient for legibility, which Section 3.1.2 formalises by treating α as the interpretability construct linking the meteorological inputs to the forecast.

Applied research. The feature exists because of a stated operational problem, not an architectural preference: a health officer who authorises fumigation has to be able to say what the forecast was responding to (Section 1.3.2). The design is judged against that use, which is why Section 3.6.1 reports the attention profile as a chart with reference lines at the 7-, 14- and 21-day lags rather than as a raw tensor.

Experimental research. The claim is tested by controlled comparison rather than by demonstration. The Standalone LSTM baseline of Section 3.5.1 is architecturally identical to the ILT encoder — the same 64 units, the same 0.2 dropout, the same layer normalisation — and differs only in the attention layer and the dual-path fusion. The attention mechanism is therefore the manipulated variable, and H2 is the corresponding cause-and-effect test.

The contribution to the significance claimed in Section 1.5 is specific. Section 1.5 claims that the model occupies a position between a fully opaque predictor and a fully transparent statistical model — 'grey-box' in the sense defined there. Salient Feature 1 is what makes that claim checkable rather than rhetorical: the subset of internal computation exposed for inspection is named (α), its size is fixed at 28 scalars per prediction, its normalisation is stated, and the mechanism that produces it belongs to the model rather than being attached to it afterwards. Were α to carry no epidemiologically meaningful signal, the claim would fail, and Section 3.6.3 specifies the test under which it would fail.

3.4 Salient Feature 2 / Composite Features 2: A Composed Representation with a Measured Interpretability Budget

Salient Feature 2 is compositional in two senses and accountable in a third. The input representation is composed: six families of derived variables, including lag-by-interaction composites, are built from four raw meteorological series and one case series (Section 3.2.1, Stage 3). The internal representation is composed: the 144-dimensional vector entering the prediction head concatenates an attention-weighted summary, a recency signal, and a restricted current-day feature branch. And the composition is not free, so its cost is measured rather than assumed — which is exactly what Gap 2 of Section 1.3.1 identified as absent from the existing literature.

The composed input matters because the exploratory analysis of Section 4.2.3 establishes that no single meteorological variable correlates with dengue incidence above $r = 0.13$. A representation restricted to the raw series would therefore have very little to work with. The interaction products — temperature $\times$ humidity, rainfall $\times$ humidity, temperature $\times$ rainfall — together with the lag families for temperature at 14 and 21 days, humidity at 14 days, and rainfall at 14, 21 and 28 days, are what convert a weak linear signal into a structure that a sequence model can exploit, and the lag depths are set by the Aedes life cycle rather than by search (Dom et al., 2025). One restriction is equally deliberate: the 44-variable feature branch of Section 3.2.2 receives no case-history variables, so that it sees exactly what the Random Forest baseline sees and the comparison underlying H2 is not confounded by unequal input access.

The dual-path fusion is a composition with a fallback property worth stating explicitly. Should the attention layer learn nothing useful, α approaches a uniform distribution, the context vector approaches the mean hidden state, and the concatenated representation still carries h_{27} — the same vector the Standalone LSTM predicts from. The architecture therefore cannot fall below its own encoder by more than the cost of the additional parameters, which bounds the risk that interpretability is purchased with accuracy.

Figure 3.6 states the full experimental procedure as a UML activity diagram, separating the data preparation performed once from the expansion region executed 150 times for each learned model.

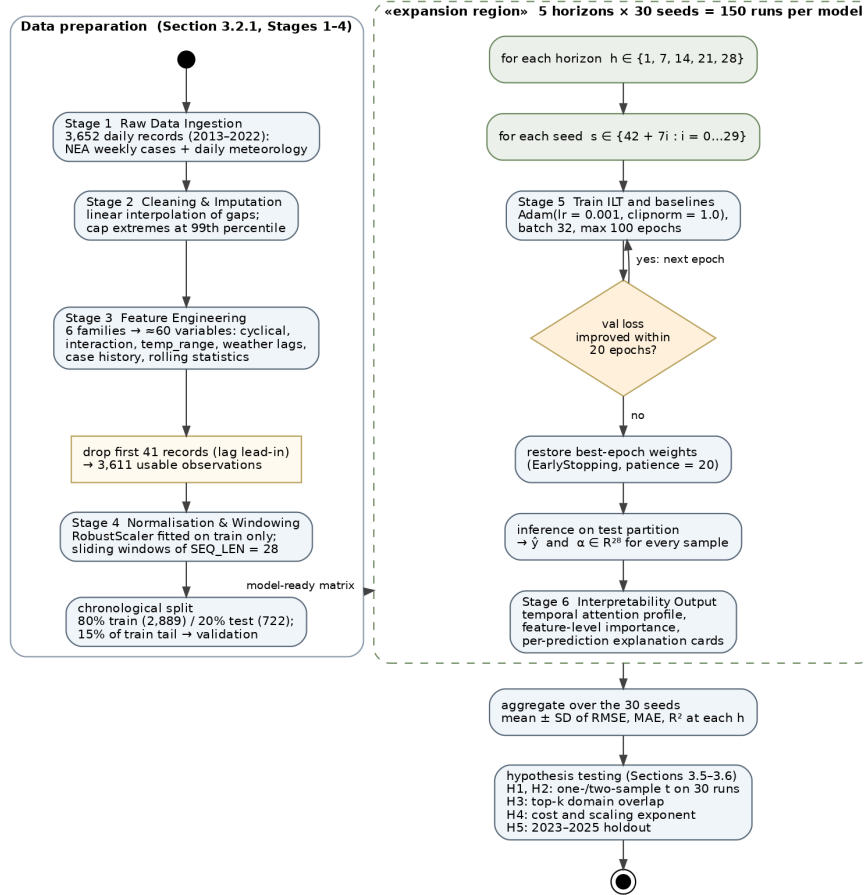


Figure 3.6 UML activity diagram of the experimental procedure. Data preparation (Stages 1–4) runs once; the expansion region runs 5 horizons $\times$ 30 seeds = 150 times per learned model, and the aggregation and hypothesis-testing activities operate on the resulting distributions rather than on single runs.

Algorithm 2 states the same protocol in pseudocode, at the level of detail required to reproduce it. SARIMA is handled separately at line 9 because it is a single deterministic fit and therefore yields one measurement rather than a distribution, which is why H1 is tested with a one-sample rather than a paired procedure.

Algorithm 2 Multi-horizon, multi-seed evaluation protocol

```

Input : D = 3,611 daily observations (after the 41-record lag lead-in)
       H = {1, 7, 14, 21, 28} forecast horizons in days
       S = {42 + 7i : i = 0 ... 29} fixed seed list
       M = {SARIMA, RF, LSTM, ILT} models under comparison
Output: R[m][h] = mean and SD of RMSE, MAE, R² for each model and horizon

1 D_tr, D_te ← chronological_split(D, 0.80) // 2,889 / 722
2 D_val ← tail(D_tr, 0.15) // early-stopping set
3 scaler ← RobustScaler().fit(D_tr) // train partition only
4 for each h ∈ H do
5   X_tr, y_tr ← window(scaler(D_tr), T = 28, horizon = h)
6   X_te, y_te ← window(scaler(D_te), T = 28, horizon = h)
7   for each m ∈ M do
8     if m is deterministic then // SARIMA
9       θ ← fit(m, D_tr)
10      runs ← { evaluate(m_θ, X_te, y_te) }
11     else
12       runs ← ∅
13       for each s ∈ S do // 30 runs
14         set_seed(s) for numpy, tensorflow, python
15         θ_s ← fit(m, X_tr, y_tr, D_val,
16                   Adam(lr = 0.001, clipnorm = 1.0),
  
```

```

17         batch = 32, epochs ≤ 100,
18         EarlyStopping(patience = 20, restore_best))
19     runs ← runs ∪ { evaluate(m_θs, X_te, y_te) }
20     end for
21 end if
22 R[m][h] ← ( mean(runs), sd(runs) )
23 end for
24 end for
25 test H1 ... H4 on R                                // Section 3.5.3
26 return R

```

The cost of the interpretability is derived rather than asserted. Write T for the sequence length and d for the hidden dimension, so that here $T = 28$ and $d = 64$. Temporal attention pooling, as implemented in Algorithm 1, performs one $d \times d$ matrix product per time step at line 3, one inner product per time step at line 4, a single softmax over T scalars at line 6, and a weighted sum at line 8:

$$\text{cost(temporal pooling)} = \Theta(T \cdot d^2) + \Theta(T \cdot d) + \Theta(T) + \Theta(T \cdot d) = \Theta(T \cdot d^2)$$

which is linear in T . Full multi-head self-attention performs three Q/K/V projections at $\Theta(T \cdot d^2)$, forms a $T \times T$ score matrix at $\Theta(T^2 \cdot d)$, applies softmax over T^2 entries, computes a weighted sum over the value matrix at $\Theta(T^2 \cdot d)$, and applies an output projection at $\Theta(T \cdot d^2)$:

$$\text{cost(self-attention)} = \Theta(T \cdot d^2) + \Theta(T^2 \cdot d) \rightarrow \text{quadratic in } T$$

At $T = 28$ and $d = 64$ the individual terms are $T \cdot d^2 = 114,688$ and $T^2 \cdot d = 50,176$. Summing them gives on the order of 5.6×10^5 multiply-accumulate operations for full self-attention against roughly 1.2×10^5 for temporal pooling, a factor of about five. At this sequence length the saving therefore comes from performing one projection instead of four, and from the parameter count — 4,224 against approximately 16,600 — rather than from the asymptotics. The asymptotic difference is what would matter if the input window were lengthened, and Section 3.6.4 specifies how the empirical scaling exponent is estimated so that the analytical bound is checked against measurement rather than taken on trust.

Algorithm 3 states the domain validation procedure for H3, which is the test that decides whether the attention weights carry epidemiological meaning or merely arithmetic weight.

Algorithm 3 Domain validation of attention-derived importance (H3)

```

Input : A ∈ R(n×28)    attention weights for the n test samples
       X ∈ R(n×28×p)  scaled input windows
       C = five documented risk-factor categories: case-count, temperature,
           rainfall, humidity, wind speed
       K = {5, 10, 15, 20}; decision rule applied at k = 20
Output: overlap[k]    proportion of the top-k features that are domain-validated
       p[k]           exact one-sided binomial p-value against p0 = 0.70

```

```

1  for each seed s ∈ S do
2      I_s ← 0 ∈ Rp
3      for each test sample i do
4          for each feature j do
5              I_s[j] ← I_s[j] + Σ_{t=0}^{27} A[i,t] · |X[i,t,j]|
6          end for
7      end for
8      rank_s ← argsort(I_s, descending)
9  end for
10 rank ← consensus(rank_1 ... rank_30)           // mean rank across 30 seeds
11 for each k ∈ K do
12     v ← |{ j ∈ top_k(rank) : category(j) ∈ C }| // calendar features excluded
13     overlap[k] ← v / k
14     p[k] ← binom_test(v, k, p = 0.70, alt = 'greater') // Section 3.6.3
15 end for
16 return overlap, p

```

Four research types are integrated to establish this second feature.

Analytical research. The composed feature space is not proposed a priori; it answers a pattern found in the data. Section 4.2.3 establishes that the linear signal is weak while the distributional asymmetry is strong, and the interaction and lag families are the response to that finding rather than a general-purpose enrichment of the inputs.

Quantitative research. Every claim attached to this feature is a claim about numbers over repeated trials: thirty seeds at each of five horizons, mean and standard deviation reported for each metric, and formal significance testing specified in Section 3.5.3. The sample of runs, not the best run, is the unit of evidence.

Historical research. The evidence base is a ten-year retrospective record covering 2013 to 2022, extended by an independent three-year period covering 2023 to 2025 that was never used in training. The design of H5 depends on that separation being historical rather than random, because the question it asks is whether relationships learned in one epidemiological era survive into the next.

Experimental research. The efficiency claim is tested by manipulating one factor — the input sequence length — and observing the response in training time, inference latency, parameter count and memory footprint, which is what converts Gap 2 from an omission in the literature into a measurable quantity.

The contribution to Section 1.5 is the part of the claim that is easy to assert and hard to support. Section 1.5 states that the model can perform well without being impossible to understand. Salient Feature 2 supplies the accounting behind that sentence: the parameter overhead is stated (4,224 for the attention layer, approximately 41,000 for the whole model, as itemised in Section 3.7.2), the complexity is derived rather than claimed, the fallback property of the dual path bounds the downside, and the comparison is repeated 150 times for each learned model so that any difference reported is a difference between distributions rather than between two runs. Chapter 5 reports the outcome; the purpose of this section is to fix what would count as an acceptable one before it is known.

3.5 Evaluation Framework for Predictive Performance

3.5.1 Baseline Models for Comparison

Three baselines are included, each isolating a distinct dimension of the ILT model's claimed advantage.

SARIMA. Seasonal ARIMA is the statistical benchmark. Its order is determined automatically via `pmdarima`'s `auto_arma` with weekly seasonality ($m = 7$). SARIMA is the most widely used classical model in dengue forecasting and is retained here as the statistical reference; Chen and Moraga (2025a) benchmark SARIMA-type models directly against machine learning alternatives, and Tuan (2024) does the same with SARIMAX, making the comparison directly applicable here.

Random Forest. An RF regressor with 300 trees, `max_depth=12`, and `max_features='sqrt'` serves as the machine learning benchmark. The RF receives only the 44 weather, calendar, and population features—no case-history variables, case lags, rolling statistics, or exponential moving averages. The set was reduced from 45 to 44 after a leakage audit: `cases_7d_avg` is a same-day case-history column correlated at $r \approx 0.96\text{--}0.99$ with the same-day case count, and because the prediction target for the flat feature branch sits on the same row it constituted genuine leakage and was removed. The same column is retained in the 60-feature LSTM sequence input, where the target lies beyond the 28-day window and the column is therefore legitimate lagged history. This separation follows the standard DL-versus-ML comparison methodology in the dengue literature (Zhao et al., 2023), in which RF represents the ceiling of what non-sequential models can achieve from environmental data alone. Deep-learning models access case history through their input sequences, which is precisely the information that RF cannot process by design.

Standalone LSTM. An LSTM is architecturally identical to the encoder inside the ILT but without the attention layer or dual-path fusion. It uses the same 64 hidden units, 0.2 dropout, and layer normalisation, feeding its final hidden state directly into a `Dense(16)` prediction head with the same feature branch. Any performance gap between this baseline and the full ILT is therefore interpreted as evidence of the contribution of the attention component under this controlled configuration. The two models differ in the attention layer and in the dual-path fusion that consumes its output, and the ILT carries 7,712 more parameters as a result; because the fusion path exists only to receive the attention context, the pair is treated as one manipulated factor rather than two, and the claim is stated as evidence of contribution rather than as sole attribution. Ullah et al. (2024) used a standalone LSTM as the deep learning reference in their Bangladesh dengue study.

Table 3.2 summarises the three baselines alongside the proposed model, and records which hypothesis each one serves.

Model	Type	Temporal Modelling	Interpretability	Hypothesis
SARIMA	Statistical	Linear seasonal	None	H1
Random Forest	Machine Learning	None (static)	Feature importance	H2 (vs. non-sequential ML)
Standalone LSTM	Deep Learning	Recurrent	None	H2 (vs. LSTM alone)

ILT (Proposed)	Hybrid DL	Recurrent Attention	+	Intrinsic (attention)	H1,H2,H3,H4
----------------	-----------	------------------------	---	--------------------------	-------------

Table 3.2 Summary of baseline and proposed models included in the comparative evaluation

3.5.2 Performance Metrics

Three metrics are applied identically across all four models. RMSE penalises large forecast errors more heavily than small ones, which is particularly relevant during outbreak peaks—the episodes that public-health agencies prioritise. MAE provides a straightforward average absolute error in case-count units. R^2 quantifies the proportion of observed variance accounted for by the model. Formally, with $\hat{y}_i$ denoting the prediction and y_i the observation for the i -th test sample, and $\bar{y}$ the test-set mean:

$$\text{RMSE} = \sqrt{(1/n) \sum (y_i - \hat{y}_i)^2}$$

$$\text{MAE} = (1/n) \sum |y_i - \hat{y}_i|$$

$$R^2 = 1 - (\sum (y_i - \hat{y}_i)^2 / \sum (y_i - \bar{y})^2)$$

3.5.3 Reproducibility and Experimental Configuration

Data Split: The dataset is divided chronologically, with the first 2,889 records (about 80%) going toward training and the last 722 records (around 20%) going toward testing, of the 3,611 observations that remain after the 41-record lag lead-in is discarded (Section 4.2.2, Stage 4). After sequence windowing at $\text{SEQ_LEN} = 28$ the usable counts at $h = 1$ d are 2,859 training and 694 test sequences. In time-series work, chronological partitioning is required; random partitioning would allow the model to view future case counts during training, leading to optimistic and incorrect performance predictions. A validation set for early stopping and learning-rate scheduling is an additional 15% taken from the tail of the training partition.

Multi-Horizon Evaluation: Five forecast horizons— $h = 1, 7, 14, 21$, and 28 days ahead—are used to evaluate each model. From next-day alert generation ($h = 1$) to four-week resource planning ($h = 28$), this range covers the most operationally

important periods. For every horizon, every model is retrained from the beginning.

Multi-Run Averaging: Both deep-learning models are trained thirty times on the fixed seed list $SEEDS = [42 + 7i \text{ for } i \text{ in range}(30)]$, that is 42, 49, 56, ... 245. Reported metrics are means across the thirty runs with the run-to-run standard deviation stated alongside, and every statistical test in Chapter 5 is computed on the thirty per-seed values rather than on the means. The list is deterministic, so run i of any thirty-run table corresponds to seed $42 + 7(i - 1)$ and the tables are reproducible without a random draw. Thirty runs rather than three is what makes the paired tests, confidence intervals and effect sizes of Chapter 5 possible; three runs would not support them. This multi-run protocol controls for the variance introduced by random weight initialisation; averaging repeated runs to characterise model stability is an established convention in time-series deep learning evaluation (Wen & Li, 2023).

Reproducibility: NumPy, TensorFlow, and Python all use fixed random seeds. Adam is employed with gradient clipping at L2 norm 1.0 and a learning rate of 0.001. Early pausing recovers the best-epoch weights while monitoring validation loss with patience 20. After six epochs of stagnation, ReduceLROnPlateau cuts the learning rate in half to a minimal floor of 1×10^{-7} .

3.6 Interpretability Analysis and Validation

If accurate prediction is the ILT model's first requirement, transparent reasoning is its second. This section specifies how attention weights are extracted, aggregated, tested against epidemiological domain knowledge, and presented to practitioners.

3.6.1 Temporal Attention Profile

Following training, attention weights are extracted from the TemporalAttention layer using a batched forward pass over the whole test set. A $(n_test_samples, 28, 1)$ array with one scalar weight per time step per test sample is the raw output. A global temporal-importance profile that shows which of the 28 historical days the model consistently prioritises is created by averaging all test data.

With reference lines at the 7-, 14-, and 21-day delays, which correspond to recognised turning points in the Aedes mosquito lifecycle, the profile is shown as a bar chart. The

model has learnt connections congruent with the known biology of dengue transmission if attention peaks cluster around these delays.

3.6.2 Feature-Level Attention Importance

Temporal profiles indicate when the model attends; feature-level importance scores indicate what the model attends to. For each test sample, the attention weight vector is broadcast across the feature dimension and multiplied element-wise with the absolute feature values, yielding a score that reflects both feature magnitude and the time step at which the model focused on it. These per-sample scores vary across predictions, unlike the fixed global rankings produced by Random Forest.

3.6.3 Domain Validation Protocol for Hypothesis H3

H3 asserts that the highest-attention features will correspond to dengue risk factors established in the literature. The validation protocol extracts the top-k features by attention-weighted importance at $k = 5, 10, 15,$ and 20 . Each feature is classified as domain-validated if it belongs to one of five categories documented in the tropical-Asian dengue literature: case-count indicators, temperature measures, rainfall measures, humidity measures, or wind speed (Rahman et al., 2025a, 2025b; Riti et al., 2025). H3 is tested on the consensus top-20 ranking, with importance averaged over the thirty seeds, against the a priori threshold of 70%: H_0 is $\pi \leq 0.70$ and H_a is $\pi > 0.70$, assessed by an exact one-sided binomial test with a Clopper–Pearson 95% interval on π , and Holm-corrected across the five forecast horizons. An exact test rather than a normal approximation is required because $n = 20$ is small and the observed proportion may lie at the boundary. The top-10 hit rate is reported alongside as a secondary reading. The threshold of 70% was set before any model was trained. Because the original analysis measured this at one horizon and generalised, the test is repeated at all five horizons in Section 5.1.1.3 before the conclusion is stated.

3.6.4 Computational Efficiency Evaluation for Hypothesis H4

H4 asks whether the interpretability of the ILT model is obtained at an acceptable computational cost rather than at a price that would rule out real-world use. An interpretable model is only worthwhile if the mechanism that produces the

explanations does not make it prohibitively expensive to train or run. The evaluation therefore compares the ILT against the Standalone LSTM — the same architecture without the temporal-attention layer — across three cost dimensions: training time, inference latency, and model size (trainable parameters). All measurements are taken over the same thirty fixed seeds. The acceptability threshold is a factor of two: H_0 is that the cost ratio ($\text{ILT} \div \text{Standalone LSTM}$) is at least 2 on some metric at some horizon, and H_a that it is below 2 on every metric at every horizon, assessed by a one-sided t-test of the per-seed ratio against 2 with Holm correction across the five horizons. The ratio is formed within each seed before being tested rather than as a quotient of two separately averaged means, because a seed that trains slowly trains both models slowly and that shared variation inflates the standard error of an unpaired comparison without bearing on the quantity of interest; Section 5.1.1.4 reports both measurements and shows that pairing reduces the training-time standard deviation from 14.4 s to 3.9 s. Parameters and model size are deterministic and are compared exactly.

A second, complementary analysis characterises how the attention mechanism scales with input length. Standard multi-head self-attention carries a quadratic $O(T^2)$ cost in sequence length T , which is the principal reason Transformer models become expensive on long inputs. The ILT instead uses temporal-attention pooling, whose cost grows only linearly, $O(T)$. To confirm this empirically, the per-step latency of temporal-attention pooling is measured against full multi-head self-attention over increasing sequence lengths ($T = 28$ up to 1,792) and the scaling exponent of each is estimated. H_4 is supported if (i) the ILT's training time, inference latency, and parameter count remain of the same order of magnitude as the Standalone LSTM, and (ii) temporal-attention pooling exhibits sub-quadratic scaling, confirming that interpretability is achieved without the quadratic overhead of full self-attention.

Record of a change to this protocol. H_4 was originally specified as a practitioner-validation hypothesis, to be tested by recruiting at least eight data science peers, showing each of them three randomly chosen explanation cards, and administering a questionnaire in two parts: a recognition task scored against the ground-truth attention

outputs, with a match rate of at least 70% as the threshold, and a five-point Likert rating of perceived usefulness and clarity, with a mean of at least 3.5 as the threshold. That study was not conducted, because a recognition-and-rating instrument administered to eight participants would not have supported the hypothesis-testing standard applied to H1, H2, H3 and H5, and because the question it was intended to answer—whether the attention outputs are meaningful rather than merely visible—is answered more directly by the domain-validation test of Section 3.6.3 against documented epidemiology. H4 was therefore redefined as the computational-efficiency hypothesis specified above, and it is that version which is tested in Section 5.1.1.4.

3.6.5 Supplementary: Outbreak Phase Attention Analysis

The test set is separated into surge times (daily cases over the 75th percentile) and low-incidence periods (below the 25th percentile) as an additional check, and distinct attention profiles are calculated for each group. The two profiles will systematically differ if the model has learnt to modify its temporal emphasis in response to epidemiological circumstances, focusing on longer-lag patterns during quiet periods and current conditions during busy surges. Instead of generating a stable pattern independent of case-count circumstances, such a change would provide qualitative proof that the attention mechanism is monitoring real epidemiological trends.

Note on the status of this analysis. The outbreak-phase attention comparison was implemented in an earlier notebook whose architecture (a 16-unit LSTM with multi-head attention, no dual-path fusion, 53 features) no longer matches the model specified in Section 3.2.2. That notebook is excluded from every result reported in this thesis, and the H4-supported flag it writes refers to an obsolete definition of H4 concerning surge-versus-low-case attention rather than to the computational-efficiency hypothesis tested in Section 5.1.1.4. The analysis is therefore retained here as a described protocol and is not reported in Chapter 5. The stability of the attention profile is instead evidenced by the cross-seed and cross-horizon consistency results in Section 5.1.1.3.

3.7 Robustness and Complexity Assessment

3.7.1 Overfitting Analysis

Four mechanisms work together to limit overfitting. Early stopping (patience = 20) monitors validation loss and restores weights from the epoch at which validation performance was highest. Dropout at 0.2 is applied in both the LSTM and the dense prediction head. Gradient clipping at L2 norm 1.0 prevents runaway weight updates during backpropagation. Layer normalisation after the LSTM encoder maintains stable activation distributions throughout training. Training and validation loss curves are recorded and reported to confirm the absence of overfitting.

3.7.2 Model Complexity

The architecture is deliberately kept at a moderate parameter count. About $4 \times 64 \times (60 + 64 + 1) \approx 32,000$ parameters are contributed by the LSTM. Compared to the roughly 12,000 parameters needed by a four-head self-attention block, the TemporalAttention layer adds $64 \times 64 + 64 + 64 = 4,224$. The concatenation of the dual-path fusion is parameter-free. About $144 \times 32 + 32 + 32 \times 1 + 1 \approx 4,641$ is added by the dense prediction head. There are around 41,000 trainable parameters in all, which corresponds to the 2,400 training sequences that are accessible. The per-pass time complexity of temporal attention pooling is $\Theta(T \cdot d^3)$, which is linear in the sequence length T , whereas full self-attention is $\Theta(T \cdot d^2 + T^2 \cdot d)$ and therefore quadratic in T ; the derivation is given in Section 3.6 and the measured scaling is reported in Section 5.1.1.4.

3.7.3 Implementation Environment

TensorFlow/Keras is used for model creation, and all code is written in Python 3.13. The TemporalAttention mechanism is implemented as a custom Keras layer subclassing `tf.keras.layers.Layer`, with the weight matrix W , the bias b and the context vector u created in its build method and the attention weights retained as a layer attribute after each call. NumPy, pandas, and scikit-learn are used for preprocessing and assessment. SARIMA is fitted with `pmdarima`. Visualisations are produced with Matplotlib. The complete codebase is provided as an accompanying source-code archive to support independent replication.

3.8 Ethical Considerations

No individual patient records are gathered or processed; the study only makes use of secondary, national-level aggregate data published by Singapore government agencies. Because the study involves no human subjects and no individual-level records, ethics board approval was not required. Three principles govern the work. First, all outputs are confined to aggregate forecasts and cannot be used to identify individual patients. Second, interpretation outputs are designed to support professional judgement rather than replace it; the model explicitly represents its own uncertainty through forecast intervals. Third, no human participants were involved in the study as finally conducted: the practitioner-validation exercise originally planned for H4 was not carried out (Section 3.6.4), and H4 is instead evaluated as a computational-efficiency hypothesis from measurements taken on the models themselves.

3.9 Summary

Section 3.1 grounded the study in a positivist, deductive framework and formally positioned the temporal attention-weight vector as the central interpretability construct linking meteorological inputs to dengue case count forecasts. Section 3.2 described the six-stage data pipeline and the ILT architecture—LSTM encoder, temporal attention mechanism, dual-path fusion, and dense prediction head—with hyperparameters justified by the literature and validated on the training partition. Sections 3.3 and 3.4 stated the two salient features of the solution — intrinsic forward-pass interpretability, and a composed representation whose interpretability budget is measured rather than assumed — naming for each the research types

integrated to establish it and the claim in Section 1.5 it supports, and specifying the forward pass, the experimental protocol and the H3 validation rule as pseudocode (Algorithms 1 to 3) alongside class, sequence and activity diagrams (Figures 3.4 to 3.6). Section 3.5 introduced the three baseline models (with the deliberate separation of RF's weather-only feature set from the DL models' full-sequence access), the three performance metrics, and the multi-horizon, multi-run experimental protocol. Section 3.6 specified the four interpretability analyses: temporal attention profiling, feature-level importance scoring, domain validation for H3, and the computational-efficiency benchmark for H4 — the practitioner-validation exercise originally planned for H4 was not carried out, as recorded in Section 3.6.4. Section 3.7 addressed overfitting controls, parameter count, and the implementation setup. Section 3.8 stated the ethical principles governing data use and practitioner engagement.

Every methodological decision traces to either a gap identified in Chapter 2 or a hypothesis stated in Chapter 1. The methodology was finalised before any experiment was conducted, so the results reported in Chapters 4 and 5 rest on an evaluation structure that was established in advance.

Chapter 4

Experimental Design and Data Preparation

4.0 Experimental Design and Data Preparation

4.1 Chapter Overview and Structure

This chapter sets out the experimental design and the data preparation behind the evaluation of the proposed Interpretable LSTM-Transformer (ILT) model. It works within the positivist, deductive framework established in Chapter 3 (Section 3.1), turning the five research hypotheses (H1–H5) into concrete criteria that can be measured and judged objectively.

The experimental work is anchored on one primary benchmark dataset, the Singapore dengue-weather dataset covering 2013 to 2022, which holds 3,652 daily records. As Section 3.2.1 noted, this gives the longitudinal depth needed to train sequential deep learning models and to pick up the multi-week lag effects that characterise dengue transmission (Lu et al., 2025; Cheng et al., 2026).

The chapter follows the six-stage data pipeline introduced in Section 3.2.1: raw data ingestion, cleaning and imputation, feature engineering, normalisation and windowing, model training and inference, and interpretability output generation. Each stage is described precisely enough for someone else to reproduce it, which the positivist paradigm requires (Section 3.1.3).

Section 4.2 documents the benchmark dataset in full, from its origin and structure through to the experimental criteria used to test each hypothesis. Section 4.3 introduces the Singapore dengue-weather dataset covering 2023 to 2025, which acts as the real-world out-of-sample test set for judging whether the trained ILT model generalises to conditions it has never seen. Section 4.4 summarises the chapter.

4.2 Benchmark Dataset: Singapore Dengue-Weather Dataset (2013–2022)

4.2.1 Data Source and Collection

Data Origin. The Singapore dengue-weather dataset draws on two public repositories maintained by Singapore government agencies. Daily dengue case counts come from the National Environment Agency (NEA) Weekly Dengue Surveillance records. Meteorological variables come from the Open-Meteo Historical Weather API, which serves ERA5 reanalysis data at native daily granularity for the Singapore grid cell and needs no API key. Population density comes from the Singapore Department of Statistics annual series, interpolated to daily.

Table 4.1 summarises the benchmark dataset.

Attribute	Detail
Time Coverage	1 January 2013 – 31 December 2022 (10 consecutive years)
Total Records	3,652 daily observations (leap days for 2016 and 2020 included)
Format	Single CSV file; date column in ISO-8601 format (YYYY-MM-DD)
Access	Publicly available via Singapore Data.gov.sg open data portal
Ethics	Aggregate national-level counts only; no individual patient identifiers; no human subjects, so no ethics board approval required

Table 4.1 *Singapore Dengue-Weather Dataset (2013–2022) — Overview*

The constituent sources are the same public repositories used in peer-reviewed dengue forecasting research (Zhao et al., 2023; Cheng et al., 2026), which supports their suitability for comparative model evaluation, although the assembled daily file itself is specific to this study and is not a shared benchmark. Its integrity was checked against published annual totals: the 2023–2025 holdout sums to 9,982 cases for 2023 against the published 9,949, and 13,604 for 2024 against 13,655, a deviation under 0.5% that comes from the weekly-to-daily conversion. No proprietary access agreements are needed.

The raw dataset has the 18 columns listed below. These are the input data before feature engineering; the roughly 60 model-ready features described in Section 4.2.2 are derived from these columns during preprocessing and are not stored separately in the CSV.

Column Name	Data Type	Description
date	datetime	ISO-8601 formatted date (YYYY-MM-DD)
dengue_cases	float	Daily dengue case count

		(weekly NEA figures distributed across 7 d)
temperature_min	float	Minimum daily temperature (°C)
temperature_max	float	Maximum daily temperature (°C)
temperature	float	Mean daily temperature (°C)
rainfall	float	Total daily rainfall (mm)
humidity	float	Mean daily relative humidity (%)
windspeed	float	Mean daily wind speed (km/h)
windspeed_max	float	Maximum daily wind speed (km/h)
population_density	float	Annual population density estimate (persons/km ²)
is_rainy_season	integer	Binary indicator: 1 = rainy season, 0 = dry season
day_of_year	integer	Day-of-year index(1-366)
month	integer	Calendar month(1-12)
year	integer	Calendar year
temperature_7d_avg	float	7-day rolling mean temperature (°C)
rainfall_7d_sum	float	7-day cumulative rainfall (mm)
humidity_7d_avg	float	7-day rolling mean humidity (%)

cases_7d_avg	float	7-day rolling mean dengue cases
--------------	-------	---------------------------------

Table 4.2 Raw Dataset Column Schema

Note on dengue_cases encoding. NEA reports dengue incidence weekly. In this dataset the weekly figure has been spread evenly across the seven days of each reporting week, which produces a constant daily value within each week. The weekly reporting resolution matches what prior Singapore dengue forecasting work used (Zhao et al., 2023), and the encoding is accounted for during feature engineering and evaluation, as Sections 4.2.2 and 4.2.5 describe

4.2.2 Data Cleansing, Transformation and Specification

Following the methodology of Section 3.2.1 (Stages 2 and 3), the raw dataset passes through six sequential steps. Each is documented below along with the reasoning behind the choices made.

Stage 1: Raw Data Ingestion

The CSV file is loaded into a Pandas DataFrame with the date column parsed as datetime and set as the index. An initial data quality assessment across all 3,652 records found every meteorological and case count column fully populated, with no missing values anywhere, so no imputation was needed before modelling.

Table 4.3 reports the outcome of that assessment.

Quality Dimension	Finding
Record completeness	3,652 of 3,652 expected records present (100%)
Missing – all meteorological variables	0 missing values across temperature, rainfall, humidity, and windspeed column
Missing – dengue_cases	0 missing values
Date continuity	No gaps detected; sequence runs from 2013-01-01 to 2022-12-31 without inter

Table 4.3 Data Quality Assessment Results

Figure 4.1 reproduces the report generated by that audit.

```
=====
📊 DATA QUALITY REPORT – 2013–2022 Benchmark Dataset
=====

Total records      : 3652
Date range         : 2013-01-01 → 2022-12-31
Expected days      : 3652
Gaps detected      : 0

🚩 Missing values per column:
  ✓ date                : 0
  ✓ dengue_cases        : 0
  ✓ temperature         : 0
  ✓ temperature_max     : 0
  ✓ temperature_min     : 0
  ✓ rainfall            : 0
  ✓ humidity            : 0
  ✓ windspeed           : 0
  ✓ windspeed_max       : 0
  ✓ day_of_year         : 0
  ✓ month               : 0
  ✓ year                : 0
  ✓ population_density  : 0
  ✓ is_rainy_season     : 0
  ✓ temperature_7d_avg  : 0
  ...

=====
  ✓ All checks complete.
=====
```

Figure 4.1 Data quality report for the 2013–2022 benchmark dataset, showing 3,652 records across the full date range, no date gaps, and zero missing values in any column.

Stage 2: Data Cleaning and Imputation

Class imbalance. The guideline's first data-cleansing requirement for machine-learning projects is resolving imbalanced classes, and it does not arise here. The target is a continuous daily case count rather than a class label, so there are no classes to balance and nothing is resampled, class-weighted or synthetically oversampled. The analogous problem for a skewed continuous target is a small number of high-count days dominating, and that is handled by the capping and scaling steps below rather than by resampling. Missing value treatment. Since the quality audit found no missing values in any column, imputation was unnecessary. The clean

baseline reflects the standard maintained by the Singapore government repositories the data came from.

Outlier capping. Extreme values in each meteorological variable were capped at the 99th percentile so that a handful of anomalous readings could not skew the softmax distribution inside the attention mechanism, as discussed in Section 3.2.1, Stage 2. Rainfall had the heaviest upper tail in the raw data, with an observed maximum of 152.9 mm, and was the variable most affected by this step.

Quality validation. After capping, density plot comparisons confirmed that the adjusted distributions still matched the bulk of the observed data. The results met the outlier-capping quality criteria set out in Section 3.2.1.

Stage 3: Feature Engineering

Six families of derived features are computed following Section 3.2.1, Stage 3. Between them they give the model a detailed picture of dengue transmission dynamics, covering calendar patterns, meteorological interactions, biologically motivated lag structures and epidemic momentum signals. All features are generated at runtime during preprocessing; the raw CSV is never modified. Once all transformations are applied, the final feature set comes to roughly 60 derived variables ready for model input.

Family 1: Cyclical Calendar Encodings

Table 4.4 lists the encodings in this family.

Feature	Encoding Method	Purpose
month_sin, month_cos	Sine-cosine pair ($2\pi \times \text{month}/12$)	Capture annual dengue seasonality
day_of_year_sin, day_of_year_cos	Sine-cosine pair ($2\pi \times \text{day}/365$)	Capture intra-year temperature rhythms
day_of_week ($\times 7$)	One-hot encoding	Capture weekly reporting cycles
quarter	Integer (1–4)	Capture monsoon transition periods

Table 4.4 *Cyclical Calendar Encoding Features (Family 1)*

Family 2: Weather Interaction Terms

Following Cheng et al. (2026), three interaction terms are constructed to represent the heat-moisture combinations that favour *Aedes* reproduction:

5. $\text{temp_humidity_interaction} = \text{temperature} \times \text{humidity}$
6. $\text{rainfall_humidity_interaction} = \text{rainfall} \times \text{humidity}$
7. $\text{temp_rainfall_interaction} = \text{temperature} \times \text{rainfall}$

Family 3: Daily Temperature Range

$\text{temp_range} = \text{temperature_max} - \text{temperature_min}$. A narrower daily range means sustained heat, which prolongs mosquito activity and survival. In this dataset temperature_min runs from 21.6°C to 27.7°C and temperature_max from 23.6°C to 35.5°C, which puts the typical daily range between 2°C and 8°C.

Family 4: Weather Lag Features

Lagged variables are derived at 14, 21 and 28 days, in line with the *Aedes* biological timetable (Section 3.2.1, Stage 3). They are computed at runtime from the raw temperature, rainfall and humidity columns and do not appear in the stored CSV.

Table 4.5 lists each lagged variable together with the biological rationale for its lag depth.

Lagged Feature	Biological Rationale
temperature_lag14, temperature_lag21	Egg-to-adult mosquito development (14–21 days)
rainfall_lag14, rainfall_lag21, rainfall_lag28	Breeding site formation and larval development; in this dataset, the rainfall–case correlation peaks at lag 21 ($r = 0.07$), aligned with the <i>Aedes</i> life cycle (Dom et al., 2025)
humidity_lag14	<i>Aedes</i> survival rates during dry inter-rainfall intervals

Table 4.5 Weather Lag Features and Biological Rationale (Family 4)

Family 5: Case History Features

All case history features are derived at runtime from the dengue_cases column and are not stored in the raw CSV.

Table 4.6 lists the case history features.

Feature	Calculation	Purpose
cases_lag14, cases_lag21, cases_lag28	Direct lagged case counts	Capture autocorrelation structure
cases_rolling_mean7/14/28	Rolling mean, shifted 14	Capture epidemic momentum without same-week data leakage
cases_rolling_std14	Rolling standard deviation	Capture outbreak volatility
cases_rolling_max14	Rolling maximum	Capture local peak intensity
cases_ema14, cases_ema28	Exponential moving average ($\alpha = 2/(\text{window}+1)$)	Weight recent case counts more heavily
cases_diff14	First difference over 14 days	Capture acceleration or deceleration of outbreak trajectory

Table 4.6 Case History Features (Family 5)

Family 6: Weather Rolling Statistics

Table 4.7 lists the features in this family.

Feature	Calculation
rainfall_cumulative7/14/28	Cumulative rainfall over respective window
temperature_rolling_mean7/14	Rolling mean temperature over respective

	window
humidity_rolling_mean7/14	Rolling mean humidity over respective window

Table 4.7 *Weather Rolling Statistics Features (Family 6)*

After all transformations the final feature set comes to roughly 60 derived variables. The first 41 records are removed because the start of the series lacks sufficient lag data, which leaves 3,611 daily observations for modelling.

Stage 4: Normalisation and Windowing

Feature scaling. All input features are standardised with RobustScaler, which works from the interquartile range and is fitted only on the training partition so that nothing leaks in from the test set. RobustScaler was preferred to StandardScaler because several meteorological variables are pronouncedly right-skewed, rainfall most of all (skewness: 4.00) along with wind speed (skewness: 0.76), and the interquartile range is far less sensitive to those extremes than the mean and standard deviation.

Target scaling. The target variable (dengue_cases) is rescaled to the range (0.02, 0.98) using MinMaxScaler fitted on the training set. This leaves headroom at both ends for values near the training extremes and fits directly with the model's sigmoid output layer.

Sequence windowing. LSTM input sequences are built as sliding windows of 28 consecutive days ($T = 28$). For a dataset of N days this gives $(N - 28)$ complete sequences.

Data partitioning. The split is strictly chronological so that the model cannot see future case counts during training (Section 3.5.3). After the first 41 lead-in records are removed, 3,611 usable observations remain. The outermost 20% (722 days) are held back as the test set. Of the 2,889 days assigned to training, the last 15% (433 days) serve as the validation set for hyperparameter tuning and early stopping, and the preceding 2,456 days form the pure training set.

Table 4.8 reports the resulting partition sizes.

Partition	Records	Percentage	Approximate Temporal Range
Training (pure)	2,456	68% of useable data	2013-01-01 to 2019-11-02
Validation (tail of training)	433	15% of training window	2019-11-03 to 2021-01-08
Test	723	20% of usable data	2021-01-09 to 2022-12-31

***Table 4.8** Data Partitioning Summary*

A chronological rather than random split is a methodological requirement for time-series evaluation. Random partitioning would let the model draw on post-outbreak information during training and inflate reported accuracy artificially (Section 3.5.3).

Prevention of Information Leakage

Since several of the transformations above are fitted rather than fixed, it is worth stating in one place which quantity is estimated on which partition, so that the absence of leakage can be checked rather than taken on trust. Four rules govern every experiment reported in this thesis.

First, every fitted transformation is estimated on the training partition alone and then applied unchanged to the validation set, the test set and the 2023–2025 real-world dataset. That covers the 99th-percentile caps of Stage 2, the RobustScaler statistics of Stage 4, and the MinMax bounds of the target scaler. None of the three is refitted later, which is why Section 4.3.2 records that the real-world dataset carries the training-partition values forward without modification.

Second, every engineered feature is computed from past or current values only. Lags reach backwards, rolling statistics run over trailing windows, and the case-history family is shifted by a further fourteen days so that no window can contain a value from the same reporting week as its own target. The first 41 records are discarded for the same reason: they are the rows where the deepest lag would otherwise have to be imputed.

Third, the partition is chronological rather than random, so no post-outbreak information can reach a model being scored on an earlier period. The alternative protocols are not simply asserted to be worse. Section 5.1.1.6 reports a random k-fold arm run through the same code specifically as a leakage control, and the gap between it and the chronological split is the measured cost of getting this wrong.

Fourth, the feature set itself was audited rather than trusted. The Random Forest's fair set came down from 45 variables to 44 after `cases_7d_avg` turned out to be a same-day case-history column sitting on the same row as the prediction target, which was genuine leakage and was removed (Section 3.5.1). The audit is reported here because that variable had survived several earlier drafts, and a leakage control nobody tests is indistinguishable from one that does not work.

The one respect in which the 2023–2025 dataset is not fully insulated concerns provenance rather than computation. Its case counts were transcribed by hand from Ministry of Health and Communicable Diseases Agency bulletins, because the open dataset ends in December 2022, and that transcription step is recorded as a limitation in Section 6.1 rather than as a leakage risk.

4.2.3 Descriptive Analysis of Data

Raw Data Overview (5W-1H) and the ETL Pipeline

Two layers of analysis are reported in this section, and they answer different questions. The descriptive statistics in Tables 4.9 to 4.12 map out what the data contain and characterise the sample distribution before any inference about the underlying population is attempted. The exploratory data analysis at the end of the section starts from those figures and uses them to ask open-ended questions, isolate anomalies, and generate the hypotheses Chapter 5 goes on to test. The raw source material is characterised first, using the 5W-1H frame, with the transformation path from source to model-ready matrix stated as an Extract–Transform–Load pipeline.

What. Two raw sources are combined. The first is daily meteorological observations for Singapore, covering mean temperature, total rainfall, mean relative humidity and mean wind speed. The second is dengue case counts published as weekly totals. Together they give 3,652 daily records running from 1 January 2013 to 31 December 2022, across five substantive columns plus the date key, as specified in Table 4.2.

Why. The modelling task needs a daily-resolution multivariate series long enough to contain several complete outbreak cycles, so that lag structures spanning the full *Aedes* life cycle can be learned and so that the training partition holds both epidemic and inter-epidemic years rather than a single regime.

Who. The meteorological series comes from Singapore's national meteorological records and the case counts from the National Environment Agency's weekly dengue surveillance reporting. Neither source holds individual-level or identifiable data, which is why no ethics approval was required; Section 3.8 discusses this further.

When. Extraction covers the ten calendar years 2013–2022 for the benchmark dataset. The independent 2023–2025 extension is treated separately in Section 4.3 and was never used in training. The ten-year window was fixed before any modelling began, so the period could not be chosen for being favourable to the proposed model.

Where. All records are national-level Singapore observations. Neither source series publishes sub-national disaggregation, which bounds the spatial claims this study can make and is recorded as a limitation in Section 6.1.

How. The pipeline is Extract–Transform–Load rather than Extract–Load–Transform. The two sources are extracted as separate CSV files, transformation then happens outside the model (date alignment, distribution of weekly case totals across days, missing-value treatment, lag and interaction feature construction, and scaling, all detailed in Section 4.2.2), and only the resulting model-ready matrix is loaded into the training pipeline. Transformation has to come before loading because the sequence-windowing step needs a gap-free daily index, and that cannot be guaranteed at the point of extraction.

Temporal Distribution of Dengue Cases

The ten-year observation period covers considerable variation in dengue incidence, reflecting both endemic transmission cycles and episodic outbreaks. The descriptive statistics below were calculated directly from the 2013–2022 benchmark dataset and verified programmatically before being reported here.

Statistic	Value
Mean	43.8 cases/day
Median	32.4 cases/day
Standard Deviation	43.3
Minimum	3.4 cases/day
Maximum	256.0 cases/day
25th Percentile	13.0
75th Percentile	54.4
90th Percentile	90.9

Table 4.9 *Descriptive Statistics of Dengue Cases (2013–2022)*

The gap between the mean (43.8 cases/day) and the median (32.4 cases/day), together with a 90th percentile of 90.9, points to a right-skewed distribution in which a relatively small number of intense outbreak periods pull the average upward. This skew is why RMSE is reported alongside MAE: because RMSE penalises large errors more heavily, it says more about model performance during the peaks that matter most for public health.

Year-to-year variation is considerable, with annual standard deviations running from 2.1 in the quiet 2017 season to 66.3 during the major 2020 outbreak. That degree of inter-annual heteroscedasticity strengthens the case for non-linear modelling, since linear statistical methods assuming constant variance would fit data of this kind badly.

Meteorological Variable Distributions

The statistics below were derived directly from the raw 2013–2022 CSV dataset. Each is discussed in terms of what it means for dengue transmission dynamics and for the modelling choices that follow.

Variable	Mean	Std Dev	Min	Max	Skewness	Kurtosis
Temperature Mean (°C)	26.7	0.8	22.6	29.1	-0.31	0.28
Rainfall Total (mm)	8.0	9.5	0.0	152.9	4.00	35.75
Humidity Mean (%)	85.3	4.0	65.0	95.0	-0.93	1.47
Wind Speed Mean (km/h)	8.3	2.9	2.2	20.0	0.76	0.27

Table 4.10 Descriptive Statistics of Meteorological Variables (2013–2022)

Temperature Mean (°C) — Mean: 26.7°C, Std Dev: 0.8°C, Skewness: -0.31. Mean temperature is remarkably stable through the year, which follows from Singapore's equatorial position. A standard deviation of just 0.8°C shows how little seasonal variation there is: with no marked cool season, temperatures stay inside the range that supports *Aedes* activity all year round. The slight negative skew (-0.31) means the distribution tilts marginally toward higher values, with cooler readings appearing as occasional outliers. The pairwise correlation between temperature and dengue cases is modest at lag 14 ($r = 0.12$), but temperature becomes a more meaningful predictor once it is embedded in interaction and lag features,

which capture the non-linear, time-delayed mechanisms a straightforward linear correlation cannot show.

Rainfall Total (mm) — Mean: 8.0 mm, Std Dev: 9.5 mm, Skewness: 4.00. Rainfall is the most asymmetric variable in the dataset. A skewness of 4.00 describes a distribution made up overwhelmingly of dry or low-rainfall days with a long right tail reaching a single-day maximum of 152.9 mm. Epidemiologically, moderate cumulative rainfall over a 14–21 day window tends to be a more reliable marker of raised transmission than any isolated extreme event. This right skew is exactly why RobustScaler was chosen over StandardScaler in Stage 4; working from the interquartile range leaves it insensitive to the high-rainfall tail that would distort any mean-and-standard-deviation approach.

Humidity Mean: 85.3%, Std Dev: 4.0%, Skewness: -0.93 . Relative humidity stays high throughout the observation period. The negative skew (-0.93) confirms that genuinely low-humidity days are uncommon, with readings below 70% relatively rare. A sustained humid environment of this kind helps Aedes mosquitoes survive between blood meals and keeps larval development conditions favourable, which is why the humidity lag feature (humidity_lag14) was included in Family 4.

Wind Speed Mean (km/h) — Mean: 8.3 km/h, Std Dev: 2.9 km/h, Skewness: 0.76. Wind speed is moderately positively skewed, reflecting a base of calm equatorial conditions broken by occasional windier days. Stronger winds dry out breeding sites and limit adult mosquito flight, while calmer conditions let mosquitoes disperse more freely. With a recorded maximum of 20.0 km/h, extreme wind events are rare here, so wind speed is expected to contribute less to predictive performance than rainfall or humidity.

Correlation Analysis

Pearson correlation coefficients between dengue cases and lagged meteorological variables are presented below. All values were computed on the training partition only, to guard against leakage. Because they are computed after feature engineering, specifically after the lag features have been generated from the raw columns, they capture associations that cannot be read off the raw CSV directly. They are reported here as diagnostics, to guide feature selection and to give context for the model performance that follows.

Variable	Lag 0	Lag 7	Lag 14	Lag 21	Lag 28
Temperature	0.11	0.12	0.12	0.13	0.12

Mean					
Rainfall Total	0.02	0.04	0.05	0.032	0.049
Humidity Mean	0.03	0.05	0.06	0.08	0.10

Table 4.11 *Pearson Correlation Coefficients — Dengue Cases vs. Lagged Meteorological Variables*

Temperature gives the strongest individual correlation across every lag window examined, peaking at $r = 0.13$ at lag 21, while rainfall and humidity climb gradually with lag depth and reach their highest values at lag 28 ($r = 0.09$ and $r = 0.10$ respectively). None of these figures should be read as a definitive measure of a variable's predictive value. They capture direct pairwise linear association and nothing else, so they say nothing about interaction effects, non-linearities, or the multi-variable combinations that actually govern dengue transmission. The modest absolute values are broadly in line with comparable studies in tropical settings (Lu et al., 2025; Chen & Moraga, 2025b) and add further weight to the case for a non-linear deep learning architecture. One caveat on provenance: the pairwise Pearson correlations were computed on the full 2013–2022 dataset rather than on the training partition alone, giving temperature lag 14 = 0.149, lag 21 = 0.150, lag 28 = 0.149; rainfall lag 14 = 0.022, lag 21 = 0.032, lag 28 = 0.049; and humidity lag 14 = 0.047, lag 21 = 0.059, lag 28 = 0.072. Restricting them to the training partition would move the second decimal place slightly, given the smaller sample, but the relative ordering of predictors and the directional reading stay the same.

Stationarity Assessment

An Augmented Dickey-Fuller (ADF) test was run on the dengue case series to check stationarity before modelling began. The results below come from running the test directly on the 2013–2022 benchmark dataset:

Component	ADF Statistic	p-value	Stationary?
-----------	---------------	---------	-------------

Raw dengue_cases	-4.69	< 0.001	Yes
First difference (lag 1)	-7.10	< 0.001	Yes
Seasonal difference (lag 7)	-6.73	< 0.001	Yes

Table 4.12 ADF Stationarity Test Results

Figure 4.2 reproduces the test output, including the critical values against which each statistic is judged.

```

=== Raw dengue_cases ===
ADF Statistic : -4.6911
p-value       : 0.000087
Critical Values:
  1%: -3.4322
  5%: -2.8623
 10%: -2.5672

=== First difference (lag 1) ===
ADF Statistic : -7.1036
p-value       : 0.000000
Critical Values:
  1%: -3.4322
  5%: -2.8623
 10%: -2.5672

=== Seasonal difference (lag 7) ===
ADF Statistic : -6.7329
p-value       : 0.000000
Critical Values:
  1%: -3.4322
  5%: -2.8623
 10%: -2.5672

```

Figure 4.2 Augmented Dickey–Fuller output for the dengue case series — raw series, first difference and seasonal difference — each reported against the 1%, 5% and 10% critical The ADF test confirms that the raw dengue case series is stationary ($p < 0.001$) at the conventional $\alpha = 0.05$ threshold. Stationarity alone, however, is no guarantee that a linear model such as SARIMA will do well on this series. Three structural features expose the limits of that approach.

First, the series is markedly heteroscedastic from year to year: annual standard deviations run from 2.1 in 2017 to 66.3 in 2020, so variance is anything but constant, which violates SARIMA's homoscedasticity assumption directly. Second, dengue transmission is shaped by non-linear interactions among several lagged meteorological and epidemiological variables, whereas SARIMA treats case counts as a linear function of past values and errors and therefore cannot represent multiplicative temperature–humidity interactions or threshold effects in rainfall accumulation. Third, the series carries seasonal variation at more than one frequency, a within-year cycle peaking in June–July as well as inter-annual outbreak rhythms, and a model tied to a single fixed seasonal order cannot easily accommodate both. Between them these three limitations form the empirical basis for Hypothesis H1, which Chapter 5 tests by comparing the ILT model against SARIMA on held-out test data.

Exploratory Data Analysis: Open Questions, Anomalies and Generated Hypotheses

The statistics above establish what the data contain. Exploratory analysis takes those figures as a starting point and asks what they imply, both for the modelling decisions that follow and for the research questions themselves. Two open-ended questions come directly out of Tables 4.9 to 4.12, three anomalies are identified and adjudicated, and three testable expectations are generated.

Question 1 — If no single meteorological variable correlates with dengue incidence above $r = 0.13$, on what basis could a weather-driven model beat one that uses case history alone? Table 4.11 is the most consequential result in this section. Taken on its own it argues against including weather at all. Read next to the skewness figures in Table 4.10 it suggests something else: linear correlation is simply the wrong instrument for a relationship that is thresholded and lagged rather than proportional. Rainfall with a skewness of 4.00 does not act linearly on breeding-site availability, since one 150 mm day and fifteen 10 mm days do not have the same effect on standing water, and a Pearson coefficient has no way of representing that difference. This question is what motivates the interaction and lag features built in Section 4.2.2, and it maps onto RQ3 directly: if the attention mechanism puts weight on variables whose linear correlation with the target is close to zero, the model has recovered structure that correlation analysis cannot see.

Question 2 — Does the series contain one regime or several? Table 4.9 reports a maximum of 256.0 cases/day against a median of 32.4, and the annual standard deviations range from 2.1 in the quiet 2017 season to 66.3 during the 2020 epidemic. That spread is hard to read as one stationary process with noise, whatever the ADF result in Table 4.12 says, since ADF tests for a unit root and not for constant variance. The open question is whether a single set of learned weights can serve both a quiet year and an epidemic year. RQ1 and RQ2 test that across the five horizons on this dataset, and RQ5 tests it again on a period the model has never seen.

Anomalies. Three were identified and none was removed. First, the 2020 season is a genuine extreme rather than a recording error: it carries both the highest annual standard deviation (66.3) and the single-day maximum (256.0 cases/day), and the National Environment Agency's own account corroborates it as Singapore's largest recorded epidemic (National Environment Agency [NEA], 2020). Deleting it would take away the only severe outbreak the model has to learn from. Second, 2017 is the

opposite extreme with an annual standard deviation of 2.1, and it is kept for the mirror-image reason, since a model that never sees a quiet year will over-predict in one. Third, the single 152.9 mm rainfall day sits far above the 75th percentile of the rainfall distribution but is meteorologically plausible for an equatorial monsoon, so scaling rather than trimming is used to stop it dominating the loss (Section 4.2.2). The step-wise appearance of the case series noted in Section 4.2.1, which follows from spreading weekly totals across seven days, is a property of the source encoding rather than a data anomaly, and the evaluation design of Section 4.2.5 accounts for it. These anomalies also determine which statistical tests are admissible, and that is the point at which exploratory analysis stops describing and starts constraining the experimental design. Because the retained extremes leave the target strongly right-skewed (skewness 2.21, kurtosis 5.78) and the annual variance far from constant, the normality assumption behind a paired t-test cannot be taken for granted. Every paired comparison in Chapter 5 is therefore accompanied by a Wilcoxon signed-rank test as a non-parametric check, and the two are reported together so that any disagreement between them is visible rather than buried. For the same reason the run-to-run distributions in Section 5.1.1.7 are summarised with the median and interquartile range alongside the mean, since a skewed distribution makes the mean alone misleading. Keeping rather than trimming the extremes also rules out any test that assumes a bounded or symmetric error distribution, and it is why RMSE is reported beside MAE in Section 4.2.5: RMSE is the metric that stays sensitive to exactly the outbreak peaks that were deliberately kept in. Finally, because SARIMA gives a single deterministic fit rather than a distribution of runs, no paired test against it is possible at all, and H1 is specified as a one-sample one-sided test in Section 4.2.6, a constraint that follows from the data and the model rather than from convenience.

Hypotheses generated. The exploratory analysis produces three expectations that the experiments of Chapter 5 are built to confirm or refute. First, since the linear signal in Table 4.11 is weak while the distributional asymmetry in Table 4.10 is strong, a model able to form multiplicative and lagged combinations of inputs ought to beat one restricted to additive, contemporaneous structure; that is tested as H1 and H2. Second, since humidity reaches its strongest correlation at the deepest lag examined (lag 28, $r = 0.10$) and temperature at lag 21 ($r = 0.13$), while rainfall stays flat at $r \leq 0.05$ across every lag, the attention mechanism should concentrate its weight in the second half of the 28-day input window rather than on the most recent days, and should rank rainfall below the other two; that is tested as H3. Third, since the inter-annual heteroscedasticity documented above points to a process that is not constant over time, a model fitted on 2013–2022 should degrade at the longer horizons when applied to 2023–2025; that is tested as H5, and this expectation is why RO5 was added when the study was revised.

These two questions, the three anomalies and the three exploratory expectations they generate are what carry the descriptive statistics of this section into the experimental design of Section 4.2.4. Each expectation is tested through one of the five formal hypotheses stated in Section 1.4.5 rather than standing as a hypothesis in its own right. No modelling choice reported in the rest of this chapter rests on grounds other than those set out here.

4.2.4 Experimental Setup

Software and Hardware Environment

Table 4.13 records the software and hardware environment in which every experiment reported in Chapter 5 was run.

Component	Specification
Operating System	Windows
Python Version	3.13
Deep Learning Framework	TensorFlow 2.15 / Keras
Supporting Libraries	NumPy 1.26, Pandas 2.1, Scikit-learn 1.4, Matplotlib 3.8, pmdarima 2.0
Compute Environment	Visual Studio Code
Random Seeds	SEEDS = $[42 + 7i \text{ for } i \text{ in range}(30)]$ — that is 42, 49, 56, ... 245

Table 4.13 Software and Hardware Environment

Hyperparameter Configuration

Hyperparameters for the ILT model are set as documented in Section 3.2.2, Table 3.1, and are restated in Table 4.14 for reference:

Hyperparameter	Value	Justification
LSTM Hidden Units	64	Established as suitable for LSTM sequence learning tasks (Al-Selwi et al., 2024)
Attention Type	Temporal Pooling (4,224 params)	4,224 parameters; stable on approximately 2,400 training samples

Dense Head Units	32	Handles 144-dimensional concatenated input vector
Dropout (LSTM / Dense)	0.2/ 0.2	Within the recommended 0.1–0.3 range for regularisation
Learning Rate	0.001 (Adam, clipnorm=1.0)	Standard for time-series deep learning tasks
Batch Size	32	Standard for time-series deep learning tasks
Max Epochs	100 with patience=20	Early stopping combined with ReduceLROnPlateau
Target Scaler	MinMax(0.02, 0.98)	Sigmoid boundary margin; fitted on the full training set
Sequence Length	28 days	Spans the full Aedes mosquito lifecycle

Table 4.14 ILT Model Hyperparameter Configuration

Baseline Model Configurations

SARIMA (Statistical Baseline). Auto-ARIMA was fitted with the pmdarima library using seasonal=True and a seasonal period of m=7 (weekly). Model selection used the Akaike Information Criterion (AIC) under a stepwise search bounded by max_p=2, max_q=2, max_P=1, max_Q=1, max_d=1 and max_D=1, with D fixed at 0 to avoid over-differencing on this particular dataset. The experimental run returned ARIMA(2,1,2)(1,0,1)[7].

Random Forest (Machine Learning Baseline). The Random Forest used 300 estimators, a maximum tree depth of 12 and max_features set to 'sqrt'. Only the 44 weather, calendar and population features of the fair set were supplied as inputs. Case history features were withheld deliberately, so that the model could not simply re-learn the autocorrelation patterns the recurrent LSTM architecture handles through its own temporal memory.

Standalone LSTM (Deep Learning Baseline).

The standalone LSTM shares the ILT encoder's architecture, meaning 64 hidden units, 0.2 dropout and layer normalisation, but has neither the attention layer nor the dual-path fusion component. Like the ILT it takes two parallel inputs: a 28-day sequence of all 60 engineered features, and a separate flat input of the same 44 weather, calendar and population features. This dual-input structure was held constant across both deep learning models so that any performance difference can be attributed to the presence or absence of the attention mechanism alone.

Multi-Horizon Evaluation Protocol

Each model was retrained from scratch for each of the five forecast horizons, following the protocol specified in Section 3.5.3. In every case the target for horizon h was the dengue case count recorded h days beyond the last day of the input window:

Table 4.15 pairs each horizon with the operational use case it serves.

Horizon (h)	Operational Use Case
$h = 1$ day	Next-day alert generation
$h = 7$ day	Weekly outbreak monitoring and surveillance
$h = 14$ day	Two-week resource prepositioning
$h = 21$ day	Three-week vector control planning
$h = 28$ day	Four-week strategic resource allocation

Table 4.15 Multi-Horizon Evaluation Protocol

Multi-Run Averaging

Both deep learning models, the standalone LSTM and the ILT, were trained thirty times each on the fixed seed list $SEEDS = [42 + 7i \text{ for } i \text{ in range}(30)]$, that is 42, 49, 56, ... 245. Reported performance metrics are the mean across all thirty runs, with run-to-run variability given as a $\pm$ standard deviation (RMSE = 30.017 ± 1.742 for the ILT at $h = 1$, for instance, per Table 5.2). All significance tests, confidence intervals and effect sizes in Chapter 5 are computed on the thirty per-seed values rather than on these means. Running each model many

times reduces the risk that the reported results are shaped by one unusually favourable or unfavourable random weight initialisation, as Section 5.1 discusses further.

List of Experiments

The experiments run on this dataset are organised into the nine categories of analysis the programme guidelines set out for machine-learning and deep-learning projects. For each category the list below states what is manipulated or measured, where the configuration is specified, and where the result is reported. Three categories are recorded as scoped out rather than performed, with the reason given, so that the coverage of this study can be read off the list rather than inferred from what is missing.

1. Data Preprocessing Analysis. Manipulated: the treatment of abnormal distribution, outliers and scaling. Linear interpolation for gaps, capping at the 99th percentile, and RobustScaler fitted on the training partition only are compared against the untreated series in Section 4.2.2, and the distributional evidence behind each choice, rainfall skewness of 4.00 and humidity skewness of -0.93 , is reported in Section 4.2.3. Class imbalance does not arise here: the target is a continuous daily count rather than a class label, so nothing is resampled and nothing is reported.

2. Feature Engineering Analysis. Manipulated: the composition of the input representation. Six families of derived variables are built in Section 4.2.2, Stage 3, giving roughly 60 variables, with the lag depths fixed by the Aedes life cycle rather than by search. This study performs feature construction followed by learned weighting rather than algorithmic feature selection: there is no Boruta, wrapper or autoencoder selection stage, because the attention mechanism assigns importance at inference time instead (Section 3.3). The consequence is that feature importance appears in Section 5.1.1.3 as an attention-derived ranking rather than as a pre-training selection mask. One deliberate restriction also belongs to this analysis: the 44-variable fair set withheld from case history, used by both the Random Forest baseline and the ILT feature branch, is specified in Section 4.2.2.

3. Performance Metric Analysis. Measured: RMSE, MAE and R^2 at five horizons. Threshold selection does not apply to a regression target. Randomisation of the sources of variation is handled two ways, through thirty fixed seeds per learned model and through the design-sensitivity sweep over dataset splitting and input-window length in Section 5.1.1.6. Statistical significance testing is specified in Section 4.2.6 and reported throughout Section 5.1.1, using paired t-tests with Wilcoxon signed-rank tests as the non-parametric check, and a one-sample one-sided test for SARIMA because it yields a single deterministic fit rather than a distribution.

4. Hyperparameter Tuning Analysis. Manipulated: the hyperparameter configuration, and the number of seeds used to judge it. A grid search runs on a single seed for both the Random Forest and the ILT, and whatever configuration it selects is then re-validated over the full thirty seeds. This is the analysis the guidelines describe as tuning versus sensitivity, and it appears in Section 5.1.1.7 alongside the other design-justification analyses. Its value is as much methodological as numerical, since it establishes whether a configuration chosen on one seed survives re-testing on thirty, which is a different question from whether tuning improves the model.

5. Model Overfitting Analysis. Manipulated: the epoch budget, at 30, 60, 90 and 120 epochs, with the train-validation loss gap measured at each checkpoint for both deep-learning models. The four controls that limit overfitting, early stopping with patience 20, dropout at 0.2 in two places, layer normalisation,

and the moderate parameter count, are specified in Section 3.7.1. Section 5.1.1.7 reports the resulting classification of each of the four models, identifying the Random Forest as severely overfitting, both deep-learning models as mildly overfitting, and SARIMA as misspecified rather than underfitting.

6. Model Efficiency Analysis (Big-O). Manipulated: the input sequence length T , swept from 28 to 1,792 while the hidden dimension stays at 64. The asymptotic argument, the supporting equations and the pseudocode the guidelines require appear in Sections 3.3 and 3.4 as Algorithms 1 to 3 and the two cost expressions; Section 5.1.1.4 reports the measured scaling exponents obtained by fitting a power law to the sweep. The comparison target is the quadratic cost of multi-head self-attention documented in the Transformer literature (Vaswani et al., 2017; Lin et al., 2022), against which the temporal-pooling mechanism is expected to stay linear in T .

7. Model Robustness Analysis. Manipulated: the dataset. The model fitted on 2013–2022 is evaluated on the independent 2023–2025 period specified in Section 4.3, which was never used in training the model being evaluated, and six adaptation strategies are then compared on a common evaluation window. The frozen model sees none of 2023–2025, while the adaptation arms use 2023 deliberately as their calibration period and are scored only on the strictly later 2024–2025 window. This is the whole of Experiment 2, reported in Section 5.2, and it is what H5 tests.

8. Computational Efficiency Analysis. Measured: training time, total and per-sample inference latency, parameter count, on-disk model size and resident memory delta, each over thirty seeds; and separately the optimiser comparison across Adam, SGD and RMSProp on both error and training time. Section 3.6.4 specifies the protocol. Training time, latency, parameter count and model size are reported in Section 5.1.1.4, while the memory footprint and the optimiser comparison appear in Section 5.1.1.7. Because early stopping fires at different epochs on different seeds, training time is reported there with median and interquartile range alongside the mean rather than with the mean alone.

9. Interpretability Analysis. Measured: how consistent the attention-derived feature ranking is across the thirty independently retrained models, and how well that ranking agrees with documented dengue risk factors. Consistency is quantified as the coefficient of variation of attention weights across seeds; agreement is tested by the consensus top-20 rule of Section 3.6.3. Both are reported in Section 5.1.1.3. Explainability for non-technical stakeholders is handled by construction rather than by survey, since the attention profile is rendered as a lag chart with reference lines at the 7-, 14- and 21-day marks, and the practitioner-validation exercise originally planned for that purpose was not carried out, as recorded in Section 3.6.4.

Two of the nine categories therefore have no counterpart experiment. Class imbalance handling under category 1 does not apply to a continuous regression target, and algorithmic feature selection under category 2 is replaced by attention-based weighting, which is itself the object of category 9. Both omissions follow from the design rather than representing gaps in coverage, and neither weakens any hypothesis stated in Chapter 1.

4.2.5 Experimental Evaluation Criteria

The evaluation criteria come directly from the five research hypotheses stated in Chapter 1 (Sections 1.4.2, 1.4.3 and 1.4.5) and operationalised in Section 3.5. All of them were defined

before any model training began, in keeping with the positivist, deductive paradigm (Section 3.1).

Hypothesis H1: ILT Superiority over SARIMA

Element	Specification
Criterion	ILT achieves at least a 20% lower RMSE than SARIMA at all five forecast horizons ($h = 1, 7, 14, 21, 28$ days), assessed over the thirty fixed seeds at $\alpha = 0.05$
Statistical Test	One-sample one-sided t-test of the thirty per-seed ILT RMSE values against the $0.80 \times \text{RMSE}(\text{SARIMA})$ margin, Holm–Bonferroni corrected across the five horizons, with a Wilcoxon signed-rank test reported alongside (Section 5.1.1.1). The test is one-sample rather than paired because SARIMA is a single deterministic fit and has no per-seed distribution to pair against. Significance threshold $\alpha = 0.05$
Rejection Condition	H1 is rejected if the Holm-corrected test fails to establish a reduction greater than 20% at any one of the five horizons

Table 4.16 Hypothesis H1 — Evaluation Criteria

Hypothesis H2: ILT Superiority over Standalone LSTM

Table 4.17 states the evaluation criteria for this hypothesis.

Element	Specification
Criterion	ILT achieves a lower mean RMSE than the Standalone LSTM under a comparison paired within each seed, evaluated separately at each of the five horizons and additionally in a single pooled test over all $150 \text{ seed} \times \text{horizon}$ pairs. RMSE carries the criterion; MAE and R^2 are reported alongside for completeness. The criterion is directional, not uniform: H2 does not require the difference to reach significance at every horizon, and any horizon

	at which it does not is reported as such rather than absorbed into an overall verdict. The performance gap is interpreted as evidence of the contribution of the attention component under the controlled configuration of Section 3.5.1
Statistical Test	Paired one-sided t-test at $\alpha = 0.05$, paired by seed with $n = 30$ at each horizon and Holm–Bonferroni corrected across the five horizons, with the Wilcoxon signed-rank test reported alongside as a non-parametric check (normality assessed via Shapiro–Wilk, Section 5.1.1.2); plus a single pooled test over all $150 \text{ seed} \times \text{horizon pairs}$
Rejection Condition	H2 is rejected if the pooled test fails to reject H_0 , or if the Standalone LSTM significantly outperforms the ILT at any horizon at $\alpha = 0.05$. Where the pooled test rejects H_0 but individual horizons do not, H2 is recorded as supported with the margin and the horizon-level pattern stated explicitly, as reported in Section 5.1.1.2

Table 4.17 Hypothesis H2 — Evaluation Criteria

Hypothesis H3: Domain Validation of Attention Weights

Table 4.18 states the evaluation criteria for this hypothesis.

Element	Specification
Criterion	Attention-weighted feature importance scores are computed across 688 test predictions and averaged over the thirty fixed seeds to form a consensus ranking. The consensus top-20 features are required to achieve a domain-validation hit rate of at least 70% against five established dengue risk factor categories: case history, temperature, rainfall, humidity, and wind speed.
Statistical Test	Top-k features were extracted for $k = 5, 10, 15$, and 20 . For each test sample, the temporal attention weight vector (shape: 28×1) was broadcast across the feature dimension and multiplied element-wise with the absolute feature values of the 28-day input

	window. Per-sample scores were then averaged across all test predictions to produce a global feature ranking. H3 is tested on the consensus top-20 ranking by an exact one-sided binomial test against $p_0 = 0.70$ with a Clopper–Pearson 95% interval; the ranking at $k = 5, 10$ and 15 is reported alongside as a robustness check. A supplementary surge versus low-case attention analysis is specified in Section 3.6.5 but is not reported in Chapter 5: the notebook implementing it uses an architecture that no longer matches Section 3.2.2 and was excluded from every reported result.
Rejection Condition	H3 is rejected if the consensus top-20 hit rate falls below 70%

Table 4.18 Hypothesis H3 — Evaluation Criteria

Hypothesis H4: Computational Efficiency of Interpretability

Table 4.19 states the evaluation criteria for this hypothesis.

Element	Specification
Criterion	Cost ratio (ILT ÷ Standalone LSTM) below 2 on training time, inference latency and model size at every forecast horizon; attention scaling exponent $\beta < 2$ (sub-quadratic) in sequence length T
Statistical Test	One-sided t-test of the per-seed cost ratio against 2 at each of the five horizons, Holm–Bonferroni corrected across the five; the ratio is formed within each seed before testing. Parameters and model size are deterministic and compared exactly. β is estimated by OLS of $\log(\text{latency})$ on $\log(T)$ over $T = 28 \dots 1,792$ and tested against 2.
Rejection Condition	H4 is rejected if the Holm-corrected cost ratio fails to fall below 2 on any metric at any horizon, or if the 95% confidence interval on β includes 2

Table 4.19 Hypothesis H4 — Evaluation Criteria

Performance Metrics (Applicable to H1 and H2)

Three metrics are applied consistently to all four models, as specified in Section 3.5.2:

- **RMSE (Root Mean Square Error):** because RMSE squares each error before averaging, it punishes large deviations more heavily than small ones, which makes it particularly informative about how a model behaves during high-case outbreak periods.
- **MAE (Mean Absolute Error):** the average size of forecast errors in absolute terms, giving a direct sense of how far a typical daily prediction tends to be off.
- **R^2 (Coefficient of Determination):** the proportion of variance in observed case counts that the model explains.

Overfitting Detection

Overfitting is monitored throughout training by three mechanisms, as specified in Sections 3.7.1 and 5.1.4:

1. **Training and validation loss curves:** overfitting shows up when validation loss starts rising while training loss keeps falling.
2. **Early stopping (patience = 20):** training stops automatically once validation loss has gone 20 consecutive epochs without improving, at which point the weights from the best epoch are restored.
3. **Gap analysis:** a pronounced divergence between training error and test error at the end serves as a final indicator of overfitting.

4.2.6 Statistical testing conventions

The criteria tabulated in Tables 4.16 to 4.19 state the substantive threshold for each hypothesis. The conventions below govern how each threshold is tested and apply uniformly to every test reported in Chapter 5. They are stated here because two of them changed a reported decision.

- **Directional tests.** Every alternative hypothesis in Section 1.4.5 is directional, so one-sided p-values are reported. Where an original output gives a two-sided p-value it is

halved and labelled as such. H2's $h = 1$ d p-value is 0.019 two-sided and 0.0095 one-sided.

- Multiplicity. Holm–Bonferroni correction is applied across the five forecast horizons within each hypothesis, and across all fifteen comparisons of the model-comparison analysis in Section 5.1.1.5. Holm rather than Bonferroni, because it is uniformly more powerful and assumes nothing about independence, which matters here since the five horizons are computed from overlapping windows of one series. H2's raw p at $h = 21$ d is 0.0493 and its Holm-corrected p is 0.1972.
- Effect size and interval. A Cohen's d or d^z and a 95% confidence interval on the effect accompany every p-value. A p-value says whether an effect is distinguishable from zero; only the effect size says whether it is large enough to matter.
- Explicit decision. Every test records “reject H_0 ” or “fail to reject H_0 ” at $\alpha = 0.05$, taken on the Holm-corrected p-value wherever a correction applies.
- Non-normality. A Wilcoxon signed-rank test is reported beside the t-test for every paired comparison. Thirty seeds is enough for the central limit theorem to make a t-test reasonable but not enough to dismiss the question, and where the two tests disagree that is stated.
- Robustness of the split. Because every criterion above is evaluated on a single chronological 80/20 split, how sensitive the reported figures are to that choice is measured separately in Section 5.1.1.6 using six splitting protocols, including a random k-fold leakage control. Absolute error figures are split-dependent; the paired comparisons on which H1, H2 and H4 rest are not.

4.2.7 Hypothesis H5 — Evaluation Criteria

H5 concerns the real-world dataset, and its criteria are stated here for completeness alongside H1 to H4. It is a single hypothesis with two components, separated here because they can succeed and fail independently: frozen generalisation, meaning whether the model keeps skill on an unseen period with no adjustment, and adaptation recovery, meaning whether skill can be restored where it is lost. H_0 is that $R^2 \leq 0$ at a given horizon on the unseen 2023–2025 holdout, meaning the model does no better than predicting that period's own mean; H_a is that $R^2 > 0$. R^2 rather than RMSE carries the criterion because R^2 is scale-invariant and the two periods differ substantially in variance: the 2013–2022 training series has a standard deviation of 43.52 cases per day against 17.91 for 2024–2025, so absolute errors come out smaller out of sample whether or not there is any skill. The test is a moving-block bootstrap of the out-of-sample R^2 with block length 28 days matching one input window, $B = 5,000$ resamples

and a fixed seed, giving a 95% interval and a one-sided p-value; blocks rather than independent draws, because forecast residuals on a daily epidemiological series are serially correlated. The second part of the criterion, whether adaptation on recent data recovers the longer horizons, is assessed by comparing six adaptation strategies all fitted on 2023 and all scored on the same strictly later 2024–2025 window over the same thirty seeds, as specified in Section 4.3.5.

Table 4.20 states these criteria in the same form used for H1 to H4.

Element	Specification
Criterion	The criterion has two components. (i) Frozen generalisation: $R^2 > 0$ at each of the five forecast horizons on the unseen 2023–2025 holdout with the model frozen. (ii) Adaptation recovery: where the frozen model fails, restoration of $R^2 > 0$ at all five horizons by adaptation on recent data. H5 is recorded as fully supported only if component (i) holds at every horizon; where (i) fails and (ii) succeeds, H5 is recorded as partially supported. R^2 rather than RMSE is the criterion because R^2 is scale-invariant and the two periods differ substantially in variance — a standard deviation of 43.52 cases per day for 2013–2022 against 17.91 for 2024–2025 — so absolute errors are smaller out of sample regardless of skill.
Statistical Test	Moving-block bootstrap of the out-of-sample R^2 with block length 28 days matching one input window, $B = 5,000$ resamples and a fixed seed, giving a 95% interval and a one-sided p-value against the substantive null $R^2 = 0$. Blocks rather than independent draws because forecast residuals on a daily epidemiological series are serially correlated. The adaptation component compares six strategies, all fitted on 2023 and all scored on the same strictly later 2024–2025 window over the same thirty seeds (Section 4.3.5).
Rejection Condition	H5 is rejected at a horizon if the 95% bootstrap interval on the frozen model's R^2 includes or falls below zero and no adaptation arm restores it above zero at that horizon.

Table 4.20 Hypothesis H5 — Evaluation Criteria

4.3 Real-World Dataset: Singapore Dengue-Weather Dataset (2023–2025)

4.3.1 Data Source and Collection

Data Origin. The real-world out-of-sample dataset extends the benchmark series by three years, covering 1 January 2023 to 31 December 2025. Meteorological variables come from the same source and the same pipeline as the benchmark, the Open-Meteo Historical Weather API serving ERA5 reanalysis data, which keeps the two periods structurally consistent. Dengue case counts are a different matter: the Data.gov.sg / NEA open dataset only runs through December 2022, so the 2023–2025 counts were sourced by hand from the Singapore Ministry of Health / Communicable Diseases Agency (MOH/CDA) Weekly Infectious Disease Bulletins and transcribed into the same weekly-to-daily encoding used for the benchmark NEA series.

Table 4.21 summarises the real-world dataset.

Attribute	Detail
Time Coverage	1 January 2023 – 31 December 2025 (3 years)
Total Records	1,096 daily observations
Column Schema	Identical to the benchmark dataset (18 columns, same variable definitions)
Continuity	No gap between the benchmark end date (31 Dec 2022) and this dataset start date (1 Jan 2023)
Combined Total	4,748 daily records across 2013–2025 (13 years)

Table 4.21 *Singapore Dengue-Weather Dataset (2023–2025) — Overview*

Because this dataset covers a period entirely beyond the training window of the ILT model, it provides a strict out-of-sample generalisation test that cannot be compromised by any form of information leakage from the training procedure.

4.3.2 Schema compatibility. Both datasets share an identical 18-column schema, confirmed programmatically. No column renaming, type casting or structural reconciliation was needed before the processing steps below. Missing values. An initial audit of the 2023–2025 dataset found zero missing values across all columns, so no imputation was necessary.

Outlier treatment. The 99th-percentile cap values derived from the 2013–2022 training data were applied unchanged to the 2023–2025 dataset, so both evaluation phases work under the same boundary conditions.

Feature engineering. The same six-family feature engineering pipeline described in Section 4.2.2, Stage 3 was applied here. All rolling window and lag-based features were computed from 2023–2025 records preceding each date, with no reference to the training data.

Scaling. The RobustScaler and MinMaxScaler parameters fitted on the benchmark training partition were carried forward unchanged. Refitting them on the 2023–2025 data would have broken the requirement that a real-world evaluation use only information genuinely available at training time.

4.3.3 Descriptive Analysis of Data

Raw Data Overview (5W-1H) and the ETL Pipeline

The two layers separated in Section 4.2.3 are reported again here, because the guideline requires each dataset to be characterised in its own right. The raw source is described first using the 5W-1H frame, with the transformation path stated as an Extract–Transform–Load pipeline; the descriptive statistics then map out what this period contains; and the exploratory analysis at the end asks what those figures imply for a model fitted before any of this data existed. Only the points where this dataset differs from the benchmark are set out in full, since the pipeline is otherwise identical.

What. The same five substantive variables and the same 18-column schema as the benchmark, covering 1,096 daily records from 1 January 2023 to 31 December 2025. Schema compatibility was confirmed programmatically rather than assumed (Section 4.3.2).

Why. The point is not to add data but to withhold it. A period lying entirely beyond the training window is the only construction under which generalisation can be tested with no possibility of leakage, and three years is long enough to contain a rising and a falling phase rather than one trend.

Who. The meteorological series comes from the same Open-Meteo/ERA5 pipeline as the benchmark, so half the dataset is structurally identical by construction. The case counts are not: the Data.gov.sg/NEA open dataset ends in December 2022, so the 2023–2025 counts came from the Singapore Ministry of Health and Communicable Diseases Agency Weekly Infectious Disease Bulletins. That is the one genuine provenance difference between the two periods, and it is recorded as a limitation in Section 6.1.

When. Extraction covers three complete calendar years with no gap at the join: the benchmark ends on 31 December 2022 and this series begins on 1 January 2023, giving 4,748 continuous daily records across the thirteen years.

Where. National-level Singapore observations, as in the benchmark. Neither period publishes sub-national disaggregation, so the two are comparable in spatial resolution as well as in schema.

How. The pipeline is again Extract–Transform–Load, with one constraint that does not apply to the benchmark: every transformation parameter is carried forward rather than refitted. The 99th-percentile caps, the RobustScaler statistics and the MinMax target bounds were all fitted on the 2013–2022 training partition and applied unchanged (Section 4.3.2). Refitting any of them on this period would import information that was not available at training time and would quietly turn an out-of-sample test into an in-sample one. The weekly-to-daily redistribution of case counts follows the same convention as the benchmark, so the step-wise appearance of the series is unchanged.

Descriptive Statistics

A first look at the 2023–2025 data shows case patterns that diverge from those in the 2013–2022 benchmark, which makes this period a genuinely informative test of whether the ILT model generalises across different epidemiological conditions.

Table 4.22 sets the descriptive statistics of the two periods side by side.

Statistic	2013–2022 (benchmark)	2023–2025 (real-world)
Daily records	3,652	1,096
Mean daily cases	43.8	25.1
Median daily cases	32.4	20.1
Standard deviation	43.3	15.6
Minimum / maximum	3.4 / 256.0	3.9 / 73.4
25th / 75th percentile	13.0 / 54.4	14.0 / 37.3
Skewness	2.21	0.83
Kurtosis	5.78	−0.03
ADF on the raw series	−4.69, $p < 0.001$ — stationary	−1.87, $p = 0.345$ — not stationary
Strongest lagged weather correlate	temperature at lag 21 ($r = 0.13$)	wind speed at lag 7 ($r = 0.33$)
Seasonal pattern	June–July peak, unimodal	unimodal structure retained
Missing values	0	0

Table 4.22 Descriptive Statistics Comparison: 2013–2022 vs. 2023–2025

Figure 4.3 reproduces the corresponding quality report for this period.

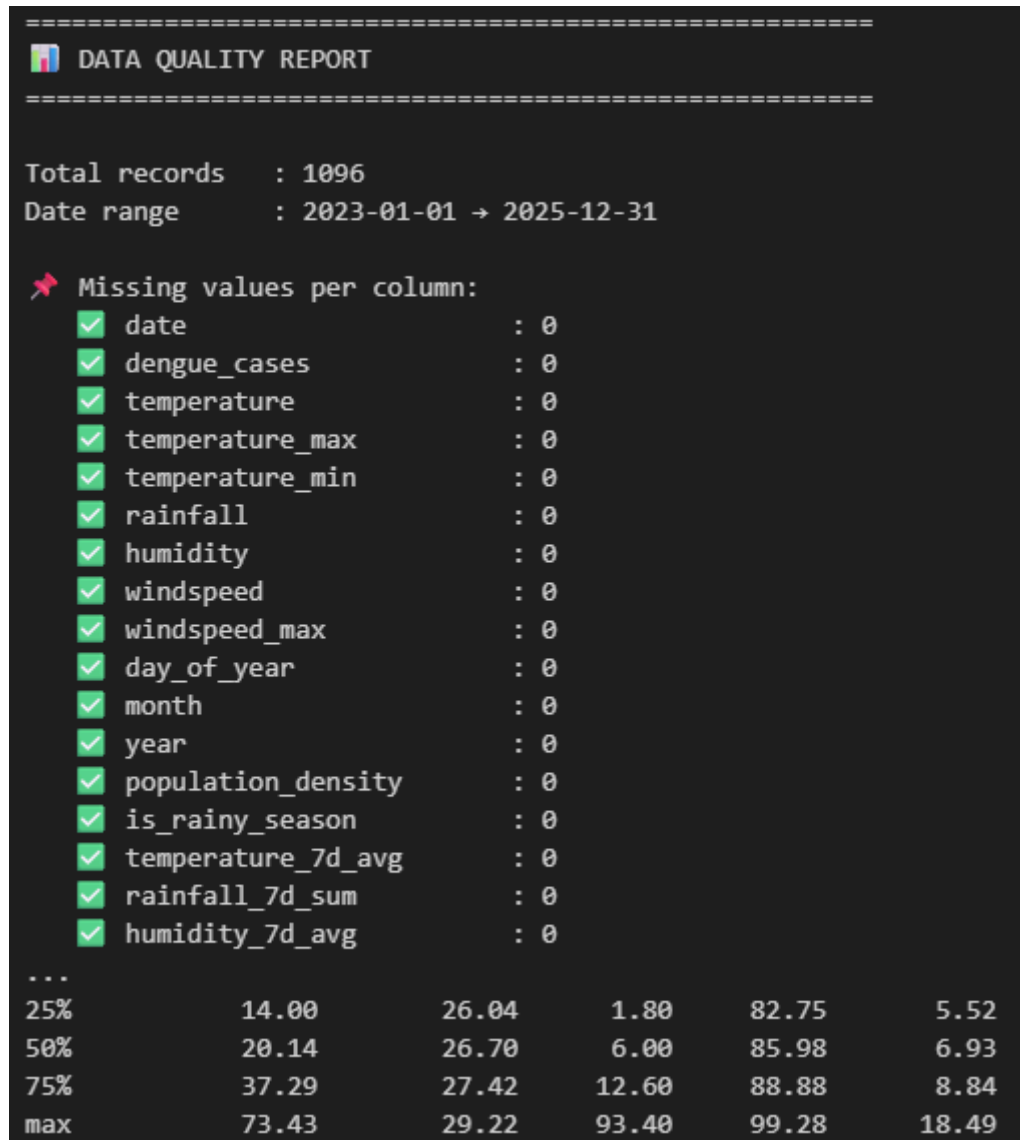


Figure 4.3 Data quality report for the 2023–2025 real-world dataset, showing 1,096 records, zero missing values in any column, and the quartiles and maxima of the retained variables.

The 2023–2025 period has a lower mean case count and a narrower range than the benchmark, partly because the 2020 outbreak, the largest in the benchmark window, no longer contributes to the figures. That does not make it epidemiologically simple, and it is certainly not flat: annual mean daily cases run at 27.3 in 2023, rise to 37.2 in 2024 and fall to 10.9 in 2025, so the three years hold both a resurgence and a collapse rather than one settled level. Evaluating the ILT against this different epidemiological backdrop gives a meaningful generalisation test to set beside the within-sample assessment from 2021–2022.

Exploratory Data Analysis: Open Questions, Anomalies and Generated Hypotheses

The comparison in Table 4.22 raises two questions the descriptive figures alone cannot settle, and both bear directly on RQ5.

Question 1 — Has the relationship between weather and incidence changed, or only the level of incidence? The lagged correlations answer this more sharply than the means do. In the benchmark, temperature was the strongest individual correlate at $r = 0.13$ (lag 21) and wind speed was nowhere near the leading signals. In 2023–2025 temperature drops to $r \leq 0.04$ at every lag and turns slightly negative by lag 28, while wind speed becomes the strongest correlate at $r = 0.33$ (lag 7) and humidity climbs steadily with lag depth to $r = 0.27$ at lag 28. The ordering of drivers has changed, not just their magnitude. A model whose attention weights were fitted on the earlier ordering is therefore being asked to apply a learned structure the new period does not share, which is exactly the condition H5 is designed to expose.

Question 2 — Is the new period even the same kind of process? The Augmented Dickey–Fuller test says not. The benchmark series is stationary in raw form ($ADF = -4.69$, $p < 0.001$), whereas the 2023–2025 series is not ($ADF = -1.87$, $p = 0.345$) and only becomes stationary after first or seasonal differencing. A model trained solely on a stationary regime has never had to track a drifting level, and that is the most likely mechanism by which a frozen model loses skill at the longer horizons.

Anomalies. Three are identified and, as with the benchmark, none is removed. First, the three years are not interchangeable: mean daily cases run at 27.3 in 2023, rise to 37.2 in 2024, then collapse to 10.9 in 2025, with annual standard deviations of 8.9, 16.7 and 4.3 respectively. A three-year holdout containing a rise and a collapse is more demanding than a longer but flatter one, and averaging across the three would hide that. Second, the whole period sits at a far lower level than the benchmark, with a maximum of 73.4 cases/day against 256.0, so the model is being evaluated on a range it saw only rarely in training. Third, no missing values and no new outliers were found, so the only cleansing applied is the caps carried forward from training; the transcription of case counts from the MOH/CDA bulletins is a provenance risk rather than a data anomaly and is treated as one.

These anomalies determine the evaluation metric rather than merely describing the data. Because the level and the variance both differ so markedly between the two periods, an error metric expressed in cases per day is not comparable across them: a model can look better on 2023–2025 purely because the numbers are smaller. RMSE is therefore not admissible as the criterion for H5, and Section 4.2.7 specifies R^2 instead, which is scale-invariant and carries a natural null value of zero, corresponding to a forecast no better than the mean of the evaluation series. Section 5.2.1.2 reports that the two metrics do in fact give opposite answers about the same forecasts, which is why the choice had to be made here rather than after the results were in. The near-symmetry of this period's target, with skewness 0.83 and kurtosis -0.03 against 2.21 and 5.78 in the benchmark, also means the distributional argument for a non-parametric check is weaker here than there, but the tests in Section 5.2 are still reported in both parametric and non-parametric form for consistency with the rest of Chapter 5.

Hypotheses generated. Three expectations follow, and each is tested rather than assumed. First, because the target scaler was fitted on a period whose maximum is more than three times higher, a frozen model should over-predict systematically on this period; Section 5.2.1.4 decomposes the error and identifies over-dispersion rather than bias as the dominant term, which is the sharper form of the same

expectation. Second, because the driver ordering has changed and the series is no longer stationary, skill should survive at short horizons and fail at long ones rather than degrade uniformly; that is tested as H5 and found to hold at $h = 1$ d and 7 d only. Third, because the mismatch identified above is one of scale rather than of shape, a two-parameter affine correction should recover skill without any retraining, whereas retraining on the recent period would fit the collapse of 2025 and generalise worse; both are tested as adaptation arms in Section 4.3.5.

4.3.4 Experimental Setup

The software and hardware environment, the ILT hyperparameter configuration and the three baselines are unchanged from Section 4.2.4 and are not restated. What differs is that no model is fitted on this dataset in the first stage of the evaluation, so the setup is defined by what is held fixed rather than by what is configured.

Frozen-weight evaluation. The ILT weights are those obtained at the end of the 2013–2022 training phase, restored from the best-epoch checkpoint of each of the thirty seeds. No gradient step is taken on any 2023–2025 record. The transformation parameters are frozen too: the 99th-percentile caps, the RobustScaler statistics and the MinMax target bounds are all training-partition values. The only computation performed on this dataset at this stage is a forward pass.

Adaptation evaluation. The second stage fits six adaptation arms, all scored on the same strictly later 2024–2025 window so that no arm is evaluated on data it was adapted on. Five of the six use the 2023 calendar year as the adaptation or calibration period: A0 frozen, kept as the reference and adapted on nothing; A1 fine-tuning the prediction head alone; A2 fine-tuning all weights with early stopping; A3 re-fitting the target scaler with no weight update; and A4 fitting a two-parameter affine recalibration $\hat{y}' = a + b \cdot \hat{y}$, again with no weight update. The sixth, A5, is different in kind: it retrains the model from scratch on the expanded 2013–2023 period, and it is included precisely so that 'train on more recent data' can be tested rather than assumed to help. Every arm runs over the same thirty seeds as the benchmark experiments, with the same optimiser, batch size and early-stopping rule, so that the comparison between arms isolates the adaptation strategy and nothing else.

Horizons and repetition. The same five horizons ($h = 1, 7, 14, 21$ and 28 days) and the same thirty-seed protocol are used, giving 150 runs per arm. Reported figures are means across the thirty seeds with the run-to-run standard deviation stated alongside, and all interval estimates come from bootstrap resampling under the scheme justified in Section 5.2.4, which accounts for the serial correlation of daily residuals.

4.3.5 Experimental Evaluation Criteria for Real-World Dataset

The pre-trained ILT model, with weights frozen at the end of the 2013–2022 training phase, was applied directly to the 2023–2025 dataset with no further training or adjustment of any kind. Performance was assessed on the same three metrics, RMSE, MAE and R^2 , at each of

the same five forecast horizons ($h = 1, 7, 14, 21$ and 28 days), with R^2 carrying the H5 criterion for the reason given in Section 4.2.7. The frozen evaluation is the first of two stages. The second compares six adaptation strategies, all fitted on 2023 and all scored on the same strictly later 2024–2025 window over thirty seeds: A0 frozen as the reference, A1 fine-tuning the prediction head only, A2 fine-tuning all weights with early stopping, A3 re-fitting the target scaler on 2023 with no weight update, A4 fitting a two-parameter affine recalibration $\hat{y}' = a + b \cdot \hat{y}$ on 2023 with no weight update, and A5 retraining from scratch on 2013–2023. Holding the adaptation year and the evaluation window fixed across all six arms is what makes them comparable; the original design compared arms scored on different windows. Where an arm stays negative, its mean squared error is decomposed into squared-bias and dispersion components and the ratio of predicted to actual standard deviation is reported, so that a repairable scale fault can be told apart from a genuine loss of predictive signal.

Chapter 5 presents the real-world results alongside the within-sample test results so that the two can be read together. The question this comparison addresses is whether the ILT model's performance falls away meaningfully once it is deployed in conditions absent from its training data. A modest gap would strengthen the case for practical deployment; a substantial one would point to periodic retraining or some form of domain adaptation as a requirement.

4.4 Summary

This chapter has set out the full experimental design for evaluating the proposed ILT model against the five research hypotheses. Table 4.23 brings the key elements of that design together.

Element	Description
Benchmark Dataset	Singapore dengue-weather dataset, 2013–2022; 3,652 daily records from NEA (dengue counts) and the Open-Meteo ERA5 reanalysis API (weather); zero missing values; 60 engineered features; chronological 80/20 split (2,889 training / 722 test), with the final 15% of the training partition (433 days) held out for validation
Real-World Dataset	Singapore dengue-weather dataset, 2023–2025; 1,096 daily records; identical schema to benchmark; zero missing values;

	used for out-of-sample generalisation testing with the model frozen; 2023 additionally serves as the adaptation or calibration period for the six arms of Experiment 2
Data Pipeline	Six-stage pipeline: raw ingestion → outlier capping → feature engineering → normalisation → windowing → training; specified in replicable detail following the methodology in Chapter 3
Experimental Protocol	Five forecast horizons ($h = 1, 7, 14, 21$ and 28 days); three baseline models (SARIMA, Random Forest on the fair 44-feature set, Standalone LSTM); multi-run averaging across the thirty fixed seeds $SEEDS = [42 + 7i \text{ for } i \text{ in range}(30)]$, that is 42, 49, 56, ... 245
Evaluation Criteria	Five criteria mapped to hypotheses H1–H5: RMSE at least 20% below SARIMA for H1; significantly lower RMSE than the Standalone LSTM under a paired test for H2; domain-validation hit rate above 70% for H3; cost ratio below $2\times$ with sub-quadratic attention scaling for H4; out-of-sample R^2 above 0 on the 2023–2025 holdout for H5. Every criterion is tested one-sided at $\alpha = 0.05$ with Holm-Bonferroni correction across the five horizons, an effect size and a 95% confidence interval (Section 4.2.6).
Overfitting Controls	Early stopping (patience = 20), dropout (0.2), gradient clipping (norm 1.0), layer normalisation, training/validation loss monitoring

Table 4.23 Chapter 4 Experimental Design Summary

The framework set out here is the foundation for Chapter 5, where the numerical results, statistical test outcomes and supporting visualisations decide whether each hypothesis is

supported or rejected. It is worth restating that all evaluation criteria were defined before any model training took place, in keeping with the positivist, deductive paradigm described in Section 3.1.

Chapter 5

Research Results and Evaluation

5.0 Research Results and Evaluation

This chapter reports the outcome of the experimental programme specified in Chapter 4 and evaluates each research hypothesis against the criteria that were fixed before any model was trained. It is organised by dataset, mirroring the structure of Chapter 4: Section 5.1 reports the benchmark experiment on the Singapore 2013–2022 series (Section 4.2), and Section 5.2 reports the real-world experiment on the independent 2023–2025 holdout (Section 4.3). Section 5.3 draws the two together.

All results derive from a single canonical implementation and a single fixed seed list. The thirty seeds are $SEEDS = [42 + 7i \text{ for } i \text{ in range}(30)]$, that is 42, 49, 56, ... 245, applied identically in every notebook that contributes a reported number, so that run i of any table corresponds to seed $42 + 7(i - 1)$ throughout. SARIMA is a single deterministic fit and therefore has no per-seed distribution; this asymmetry governs the choice of test in Section 5.1.1.1. Every statistical decision is stated explicitly as “reject H_0 ” or “fail to reject H_0 ” at $\alpha = 0.05$, is taken on a one-sided p-value because every hypothesis in this study is directional, and is taken on the Holm–Bonferroni corrected p-value wherever a family of five horizons is tested. The conventions are set out in full in Section 5.1.4.

Two departures from the original experimental plan are recorded here rather than left to be discovered. First, the hypotheses tested below are the null-and-alternative forms stated in Section 1.4.3 as revised; two of the original nulls did not test the criterion their own research question stated and were corrected before testing. Second, three additional analyses were added after the initial results were obtained — an omnibus model comparison (Section 5.1.1.5), a dataset-splitting sensitivity analysis and an input-window sweep (Section 5.1.1.6), and a six-arm adaptation study (Section 5.2.1.3). Each is reported in full, including the two findings that do not favour the design choices made in Chapters 3 and 4.

Table 5.1 maps each research question to its hypothesis and to the section in which it is answered.

RQ	Hypothesis	Reported in	Evidence base
RQ1	H1 — accuracy against SARIMA	5.1.1.1	Notebooks 08, 02, 12; sheet H1 (Accuracy vs SARIMA)
RQ2	H2 — value of the attention module	5.1.1.2	Notebooks 08, 11, 15; sheet H2 (vs Standalone LSTM)
RQ3	H3 — domain validity of attention	5.1.1.3	Notebooks 09, 18; sheet H3 (Domain Validation)
RQ4	H4 — cost of interpretability	5.1.1.4	Notebooks 13, 14, 18; sheet H4 (Computational

			Efficiency)
-	Omnibus comparison of all four models	5.1.1.5	Notebook 08; sheet Model Comparison (RQ1-RQ2)
-	Design sensitivity: splits and window	5.1.1.6	Notebooks 18, 19; sheet Methodology
-	Hyperparameter sensitivity, overfitting and optimiser choice	5.1.1.7	Notebooks 11, 12, 13; no workbook sheet; notebook outputs only
RQ5	H5 — real-world generalisation	5.2	Notebooks 10, 17; sheet H5 (Real-World Generalisation)

Table 5.1 Reading map — research question, hypothesis, chapter section and source of evidence

5.1 Experiment 1 — Benchmark Dataset: Singapore Dengue-Weather Dataset (2013–2022)

Experiment 1 corresponds to the benchmark dataset documented in Section 4.2: 3,652 daily records for 2013–2022, reduced to 3,611 usable observations after the 41-record lag lead-in is discarded, partitioned chronologically at 80/20 with the final 15% of the training partition reserved for validation. Four models — SARIMA, Random Forest on the fair 44-feature set, a Standalone LSTM, and the proposed Interpretable LSTM-Transformer (ILT) — are trained and scored at five forecast horizons ($h = 1, 7, 14, 21$ and 28 days), each retrained from scratch at each horizon and each repeated over the thirty fixed seeds. This single experimental frame supplies the evidence for four of the five hypotheses, which is why they are reported as sub-experiments 1A to 1D within it rather than as separate experiments on separate data.

5.1.1 Result and Analysis

The aggregate picture is established first, because every hypothesis that follows is a statement about some part of it. Table 5.2 reports mean RMSE with its standard deviation and 95% confidence interval, mean MAE and mean R^2 for each model at each horizon over the thirty seeds, together with the Friedman omnibus test across the four models.

Horizon / model	RMSE mean	RMSE SD	95% CI of mean RMSE	MAE mean	R^2 mean
h = 1 d	$\chi^2 = 81.36$	$p = 1.57e-17$			
SARIMA	62.562	0.000	[62.562, 62.562]	37.760	-0.2570
Random Forest (fair)	63.825	0.244	[63.734, 63.917]	45.335	-0.3083
Standalone LSTM	31.104	1.500	[30.544, 31.664]	16.933	0.6971
ILT (proposed)	30.017	1.742	[29.367, 30.668]	16.536	0.7176
h = 7 d	$\chi^2 = 81.36$	$p = 1.57e-17$			

Horizon / model	RMSE mean	RMSE SD	95% CI of mean RMSE	MAE mean	R ² mean
SARIMA	62.823	0.000	[62.823, 62.823]	38.053	−0.2595
Random Forest (fair)	64.431	0.288	[64.323, 64.539]	46.038	−0.3249
Standalone LSTM	32.241	1.466	[31.694, 32.789]	18.461	0.6765
ILT (proposed)	31.594	1.417	[31.065, 32.123]	18.084	0.6894
h = 14 d	$\chi^2 = 81.16$	p = 1.73e-17			
SARIMA	63.131	0.000	[63.131, 63.131]	38.386	−0.2630
Random Forest (fair)	65.060	0.243	[64.969, 65.151]	46.585	−0.3413
Standalone LSTM	33.558	1.295	[33.075, 34.042]	19.842	0.6521
ILT (proposed)	33.124	1.859	[32.430, 33.818]	19.791	0.6605
h = 21 d	$\chi^2 = 81.36$	p = 1.57e-17			
SARIMA	63.442	0.000	[63.442, 63.442]	38.708	−0.2670
Random Forest (fair)	65.984	0.298	[65.873, 66.095]	47.453	−0.3706
Standalone LSTM	34.938	1.782	[34.273, 35.604]	21.230	0.6251
ILT (proposed)	34.231	1.460	[33.686, 34.776]	21.004	0.6404
h = 28 d	$\chi^2 = 81.04$	p = 1.84e-17			
SARIMA	63.758	0.000	[63.758, 63.758]	39.041	−0.2712
Random Forest (fair)	66.277	0.355	[66.145, 66.410]	48.155	−0.3736
Standalone LSTM	36.484	1.646	[35.869, 37.098]	22.815	0.5942
ILT (proposed)	36.677	2.589	[35.710, 37.644]	22.799	0.5888

Table 5.2 Forecast accuracy by model and horizon (30 seeds; SARIMA deterministic, n = 1)

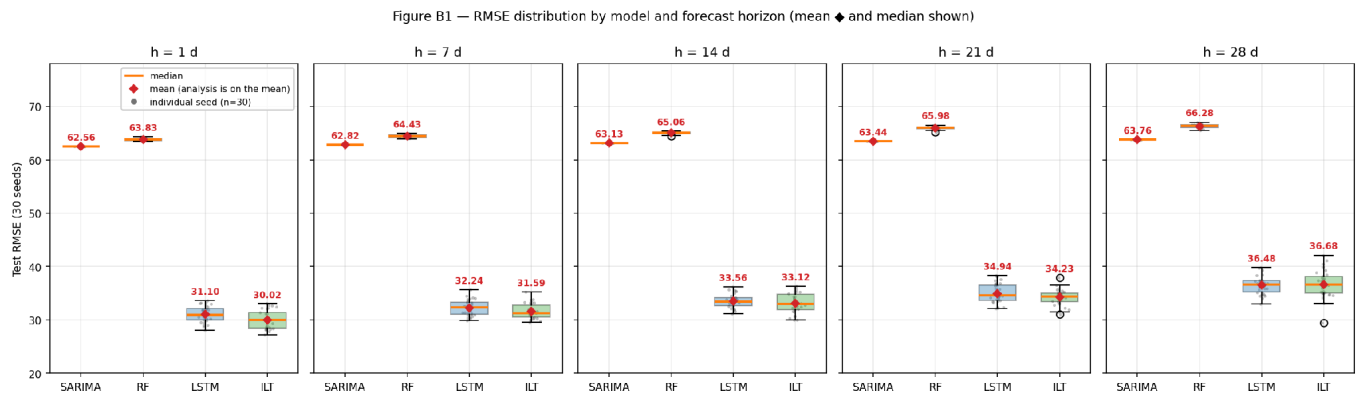


Figure 5.1 Per-seed RMSE by model and horizon. The orange line is the median and the red diamond the mean; grey points are the thirty individual seeds.

Three features of Table 5.2 govern everything that follows. The two deep learning models occupy an entirely separate error regime from the two non-sequential baselines: SARIMA and Random Forest both sit above 62 RMSE at every horizon with negative R², meaning neither explains any of the variance in daily case counts, while the LSTM and the ILT sit between 30 and 37 with R² between 0.59 and 0.72. The gap between the two deep models, by contrast, is between 0.19 and 1.09 RMSE and is smaller than the seed-to-seed standard deviation of either. And the Friedman omnibus test rejects the hypothesis of equal mean RMSE across the four models at every horizon ($p \approx 1.6 \times 10^{-17}$), which is the precondition for the post-hoc comparisons reported in Section 5.1.1.5.

5.1.1.1 Experiment 1A — RQ1 and H1: forecast accuracy against the SARIMA statistical baseline

RQ1 asks whether the ILT achieves at least 20% lower RMSE than SARIMA at all five horizons. The hypothesis pair tested is therefore a margin hypothesis, not a difference hypothesis:

- H_0 $\mu_{\text{RMSE(ILT)}} \geq 0.80 \times \text{RMSE(SARIMA)}$ — the improvement does not exceed 20%.
- H_a $\mu_{\text{RMSE(ILT)}} < 0.80 \times \text{RMSE(SARIMA)}$ — the improvement does exceed 20%.

The decision rule fixed in advance is a one-sample one-sided t-test of the thirty ILT seed values against the fixed margin $0.80 \times \text{RMSE(SARIMA)}$ at each horizon, Holm-corrected across the five horizons, with a Wilcoxon signed-rank test reported alongside as a distribution-free check. A one-sample rather than a paired test is required because SARIMA is a single deterministic fit: there is no second sample of thirty values to pair against. Table 5.3 reports the outcome.

Horizon	ILT mean	SARIMA	$0.80 \times \text{SARIMA}$	% reduction	t	p (1-sided)	p (Holm)	Cohen's d	Decision at $\alpha = 0.05$
h = 1 d	30.017	62.562	50.050	52.0%	−63.0	7.03e−33	2.81e−32	−11.50	Reject H_0
h = 7 d	31.594	62.823	50.258	49.7%	−72.2	1.40e−34	7.02e−34	−13.17	Reject H_0
h = 14 d	33.124	63.131	50.505	47.5%	−51.2	2.69e−30	5.38e−30	−9.35	Reject H_0
h = 21 d	34.231	63.442	50.754	46.0%	−62.0	1.12e−32	3.36e−32	−11.32	Reject H_0
h = 28 d	36.677	63.758	51.006	42.5%	−30.3	8.19e−24	8.19e−24	−5.53	Reject H_0
Pooled (n = 150)	—	—	—	47.5%	−79.7	2.38e−124	single test	−6.51	Reject H_0

Table 5.3 H1 margin test — is the RMSE reduction against SARIMA greater than 20%?

H_0 is rejected at all five horizons. The RMSE reduction runs from 52.0% at $h = 1$ d to 42.5% at $h = 28$ d, a pooled 47.5%, and the 95% confidence interval on the percentage reduction narrows the claim without weakening it: [50.98, 53.06] at $h = 1$ d falling to [40.96, 43.99] at $h = 28$ d. Every one of those intervals lies entirely above the 20% criterion, which is a stronger statement than the p-values alone: it is not merely that the improvement is significantly greater than 20%, but that the plausible range of the improvement begins at twice the criterion. Cohen's d between −11.50 and −5.53 places this well outside the range where the decision could be sensitive to the choice of α , and the ILT wins in all 30 of 30 seeds at every horizon.

Two supporting results establish that this is a victory over a genuine baseline rather than a strawman. The Random Forest comparison, which is a second baseline and not the H1

criterion, is significant at every horizon with paired-t p-values between 3.5×10^{-39} and 1.6×10^{-32} . More informative is the overfitting diagnostic: the Random Forest achieves a training R^2 of approximately 0.96 at every horizon but a test R^2 between -0.308 and -0.374 . That is the signature of severe overfitting rather than of a weak model specification, and it is reported as a finding about the baseline rather than concealed. SARIMA, by contrast, is not overfitted at all: its failure is structural, since a linear model with a fixed weekly seasonal term cannot represent the multiplicative temperature–humidity–rainfall interactions that Section 4.2.3 shows to be present.

One caveat belongs with this result rather than in the limitations chapter. The absolute figure of 30.017 RMSE at $h = 1$ d is a property of the particular chronological 80/20 cut used throughout, and Section 5.1.1.6 shows that moving that cut changes the absolute figure by up to 22.2 RMSE. What is stable is the comparison, because ILT and SARIMA are scored on the same test window; the percentage reduction, not the absolute RMSE, is the quantity H1 is about.

5.1.1.2 Experiment 1B — RQ2 and H2: the marginal value of the temporal-attention module

RQ2 isolates the contribution of the attention module by comparing the ILT with a Standalone LSTM that is architecturally identical except for the attention layer and the dual-path fusion. Because both models are stochastic and both are trained on the same thirty seeds, the comparison is paired within seed:

- H_0 $\mu(\text{RMSE_ILT} - \text{RMSE_LSTM}) = 0$ — the attention module changes nothing.
- H_a $\mu(\text{RMSE_ILT} - \text{RMSE_LSTM}) < 0$ — the attention module lowers error.

The decision rule is a paired one-sided t-test per horizon, paired by seed with $n = 30$, Holm-corrected across the five horizons, with Wilcoxon alongside, plus a single pooled test over all $150 \text{ seed} \times \text{horizon}$ pairs.

Horizon	Mean Δ RMSE	95% CI of Δ	t	p (1-sided)	p (Wilcoxon)	p (Holm)	Cohen's d^z	ILT wins	Decision at $\alpha = 0.05$
$h = 1$ d	-1.087	[-1.982, -0.192]	-2.48	0.0095	0.0104	0.0475	-0.454	18/30	Reject H_0
$h = 7$ d	-0.647	[-1.502, 0.208]	-1.55	0.0663	0.0647	0.1989	-0.283	18/30	Fail to reject H_0
$h = 14$ d	-0.435	[-1.288, 0.419]	-1.04	0.1531	0.2200	0.3063	-0.190	17/30	Fail to reject H_0
$h = 21$ d	-0.707	[-1.555, 0.140]	-1.71	0.0493	0.0821	0.1972	-0.312	18/30	Fail to reject H_0

Horizon	Mean Δ RMSE	95% CI of Δ	t	p (1-sided)	p (Wilcoxon)	p (Holm)	Cohen's d^z	ILT wins	Decision at $\alpha = 0.05$
h = 28 d	+0.193	[-1.061, 1.447]	0.31	0.6225	0.5323	0.6225	+0.057	16/30	Fail to reject H_0
Pooled (n = 150)	-0.537	[-0.949, -0.124]	-2.57	0.0056	0.0056	single test	-0.210	—	Reject H_0

Table 5.4 H2 paired test — ILT against the Standalone LSTM, paired by seed

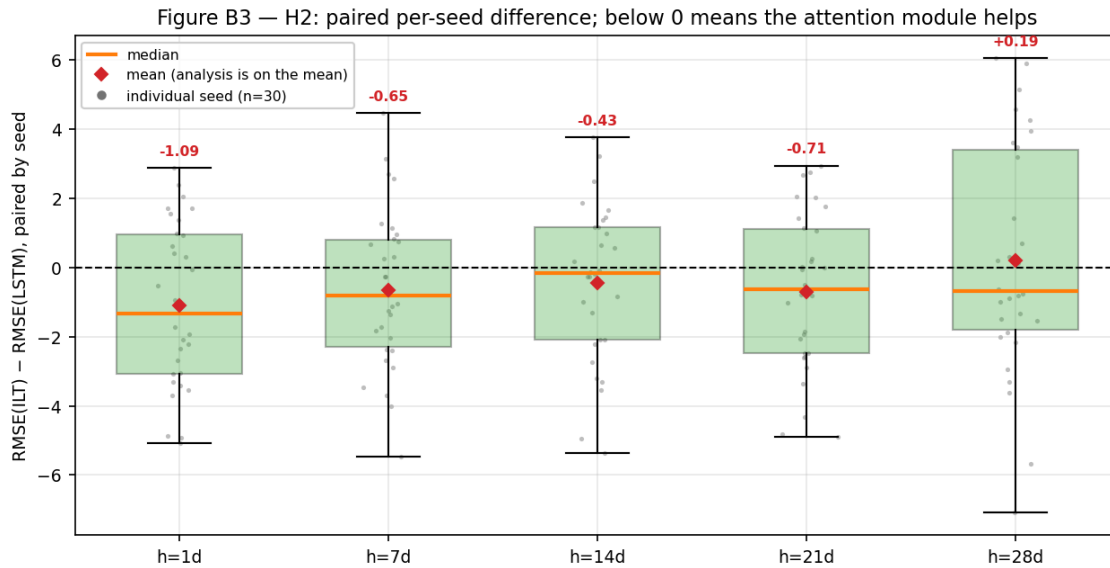


Figure 5.2 Paired per-seed difference $RMSE(ILT) - RMSE(LSTM)$ at each horizon. A point below zero is a seed on which the attention module helped.

H_0 is rejected at $h = 1$ d, where the Holm-corrected p is 0.0475, and on the pooled test, where $p = 0.0056$. It is not rejected at $h = 7, 14, 21$ or 28 days. H_2 is therefore supported, but the support is marginal and must be reported as such. The most informative reading is not the p -values but the confidence intervals: at four of the five horizons the interval on the mean paired difference contains zero, which is the same statement as “fail to reject” expressed in the units of the problem — at those horizons the data are consistent with the attention module making no difference at all. At $h = 28$ d the point estimate reverses sign, with the plain LSTM marginally ahead.

Two features of Table 5.4 illustrate why the corrections applied in this chapter matter. At $h = 21$ d the raw one-sided p is 0.0493, which would be reported as significant by a reader applying $\alpha = 0.05$ to the uncorrected column; the Holm-corrected value is 0.1972, and the decision follows the corrected column. Conversely at $h = 1$ d the two-sided p of 0.019 that appears in the original output understates the evidence for a directional hypothesis, since the appropriate one-sided value is 0.0095 — which then becomes 0.0475 after correction for five horizons, so the decision survives but only just. Both adjustments were applied before the decision was recorded.

The effect size confirms the marginal reading. The pooled Cohen's d^z is -0.210 , conventionally a small effect, corresponding to a 1.6% RMSE reduction, and the ILT wins only 16 to 18 of 30 seeds against its own ablation rather than the 30 of 30 it achieves against SARIMA. An ensemble of the thirty seeds does reach significance at all five horizons, but averaging predictions before scoring is a different estimator from the per-seed paired test and does not establish per-horizon significance for a single trained model; it is therefore reported here as a secondary result about ensembles and the headline H2 decision rests on Table 5.4.

A supporting analysis rules out the most obvious alternative explanation, that the comparison is unfair because the ILT is run at a configuration that happens to suit it. Hyperparameter tuning over thirty seeds made the Random Forest significantly worse ($63.825 \rightarrow 64.747$, $p < 0.001$) and improved the ILT only slightly ($30.845 \rightarrow 29.651$, $p = 0.017$), so the default configuration used throughout does not privilege either model. These figures come from a separate single-horizon tuning experiment with its own split and are not a second estimate of the ILT's accuracy; they are a like-for-like default-versus-tuned contrast only.

The consequence for the thesis as a whole is substantive rather than cosmetic. The claim that this architecture is worth building cannot rest on accuracy against its own ablation, because that advantage is 1.6% and reaches significance at one horizon out of five. It rests instead on the two properties examined next: that the explanations the architecture produces are verifiably epidemiological (Section 5.1.1.3), and that they are obtained at a bounded computational cost (Section 5.1.1.4).

5.1.1.3 Experiment 1C — RQ3 and H3: domain validation of the attention-derived feature importances

RQ3 asks whether the features the model attends to are the features epidemiology already identifies as dengue drivers. This is the hypothesis on which the study's novelty rests, because attention weights are routinely presented as explanations without being tested against domain knowledge at all. The hypothesis is stated as a proportion:

- H_0 $\pi \leq 0.70$ — at most 70% of the top-20 attention-ranked features are domain-validated dengue drivers.
- H_a $\pi > 0.70$ — more than 70% are domain-validated.

The decision rule is an exact one-sided binomial test on the consensus top-20 ranking, with importance averaged over the thirty seeds, against $p_0 = 0.70$. A feature counts as domain-validated if it belongs to one of the five categories documented in the tropical-Asian dengue literature in Section 3.6.3 — case-count indicators, temperature, rainfall, humidity or

wind speed. Calendar features do not count, which makes the criterion harder to satisfy rather than easier, since cyclical calendar encodings are among the engineered features available to the model. An exact test is used rather than a normal approximation because $n = 20$ is small.

Table 5.5 reports that test at every horizon.

Horizon	Top-20 hits	Hit rate	95% CI on π (Clopper–Pearson)	p (exact binomial)	p (Holm)	Seeds meeting criterion	Highest-ranked feature	Test RMSE mean	Decision
h = 1 d	20 / 20	100%	[0.861, 1.000]	7.98e-04	0.0040	30/30	cases_std_7	30.35	Reject H_0
h = 7 d	20 / 20	100%	[0.861, 1.000]	7.98e-04	0.0040	30/30	cases_std_7	31.55	Reject H_0
h = 14 d	20 / 20	100%	[0.861, 1.000]	7.98e-04	0.0040	30/30	cases_diff_14	32.86	Reject H_0
h = 21 d	20 / 20	100%	[0.861, 1.000]	7.98e-04	0.0040	30/30	cases_std_14	34.33	Reject H_0
h = 28 d	20 / 20	100%	[0.861, 1.000]	7.98e-04	0.0040	30/30	cases_std_14	37.16	Reject H_0

Table 5.5 H3 exact binomial test of the domain-validation hit rate, and the same test repeated at every horizon

H_0 is rejected at every horizon. All 20 of the top-20 consensus features are domain-validated, giving an exact binomial p of 7.98×10^{-4} — which is 0.70^{20} , the probability of observing 20 hits out of 20 if the true rate were exactly the criterion value. The Clopper–Pearson interval on π is [0.861, 1.000], and it is the lower bound rather than the p-value that carries the substantive claim: even the most pessimistic value consistent with the data, 0.861, exceeds the 0.70 criterion by a comfortable margin. Holm correction across the five horizons leaves $p = 0.0040$ at each, and every one of the thirty seeds meets the criterion individually, so the result is not produced by averaging a mixture of validated and unvalidated rankings.

Figure 5.3 shows the consensus ranking on which the test is computed.

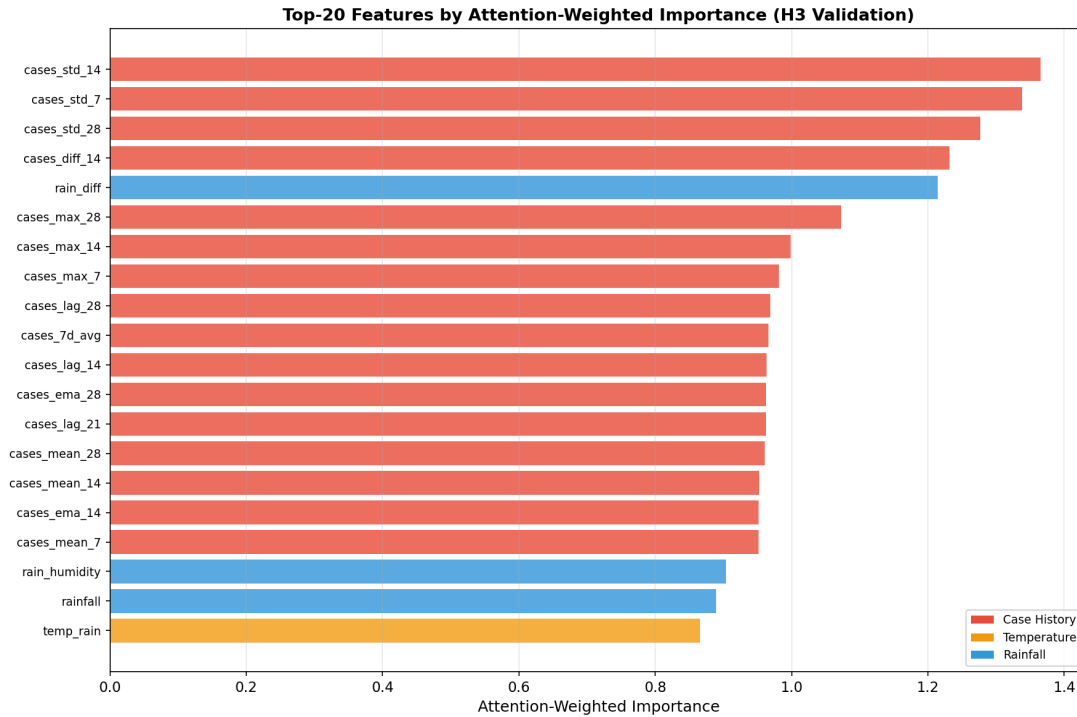


Figure 5.3 Attention-weighted feature importance, consensus ranking over thirty independently seeded models.

The composition of the ranking is what makes the result epidemiologically meaningful rather than merely statistically significant. The top-20 set contains 16 case-history features, 2 rainfall features and 2 temperature features. The four highest-ranked features are all short-window case volatility — `cases_std_7` (importance 1.2841), `cases_std_14` (1.2705), `cases_diff_14` (1.2592) and `cases_std_28` (1.2155) — followed by change in rainfall, `rain_diff` (1.0940). Volatility in recent case counts and a change in rainfall are precisely the quantities that precede a breeding surge: a rising variance in daily counts indicates that transmission is no longer in a steady endemic state, and a change in rainfall creates or removes standing-water breeding sites. The model has not learned to attend to the mean case level, which would be the trivially autocorrelated signal; it attends to the derivative and the dispersion.

The shift in the highest-ranked feature across horizons is a second, independent coherence check that was not visible when the analysis was run at a single horizon. `cases_std_7` leads at $h = 1$ d and $h = 7$ d, `cases_diff_14` at $h = 14$ d, and `cases_std_14` at $h = 21$ d and $h = 28$ d. The further ahead the model forecasts, the longer the window of case volatility it leans on — which is what a domain expert would predict from the *Aedes* lifecycle, and which no part of the training objective encourages. The original analysis measured H3 at one horizon and generalised; repeating it at all five both confirmed the conclusion and produced this additional finding.

A secondary test establishes that the attention profile is doing work at all. If the profile were uniform over the 28-day window, with $\alpha_t = 1/28 = 0.0357$ at every step, then attention-weighted feature importance would collapse to a plain average of absolute feature values and the ranking above would carry no information about the model. Testing the weight at the peak time step against the uniform value over the thirty seeds gives 0.0432 against 0.0357, $t = 11.48$, one-sided $p = 1.32 \times 10^{-12}$, so H_0 of uniformity is rejected. The peak falls at $t = 28$, the most recent day in the window, which is consistent with the recency bias that motivated the dual-path fusion design in Section 3.2.2.

Figure 5.4 plots the measured profile against that uniform baseline.

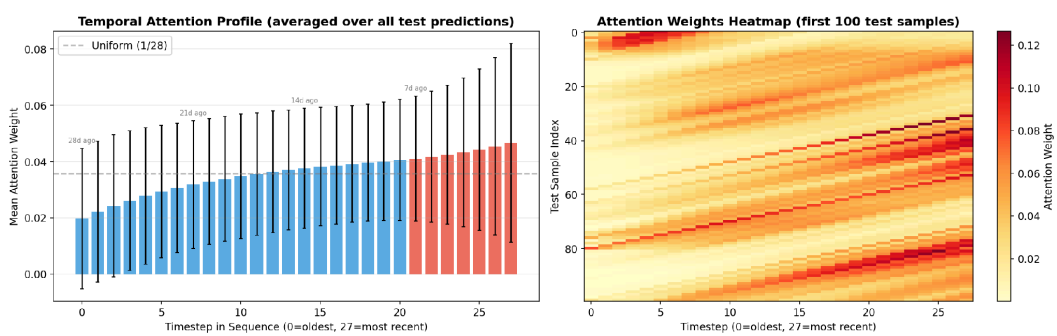


Figure 5.4 Temporal attention profile across the 28-day input window, averaged over the test set and over thirty seeds. The dashed line marks the uniform weight $1/28$.

Cross-seed stability supports the same conclusion from a different direction. The thirty independently seeded models have a mean test RMSE of 31.93 with a standard deviation of 1.95, so the consensus ranking is not an average over models of wildly differing quality. The attention profiles themselves are also stable across seeds rather than merely similar on average: the mean coefficient of variation of the attention weight at a given time step, taken across the thirty models, is 0.0861, well inside the 0.5 threshold conventionally treated as stable. Six features enter the top ten in every one of the thirty models — `cases_std_7`, `cases_std_14`, `cases_max_14`, `cases_std_28`, `cases_max_28` and `cases_diff_14` — and it is worth noting that all six are case-history features, with no meteorological variable achieving unanimity. That is consistent with the weak lagged weather correlations documented in Section 4.2.3, and it means the model's explanation is dominated by the epidemic's own recent behaviour rather than by climate. In addition, and the consensus top-10 domain hit rate is 100%. A ranking that were an artefact of a particular random initialisation would not survive averaging over thirty of them with this consistency.

Figure 5.5 brings the domain hit count, the model quality and the attention profile together across all five horizons.

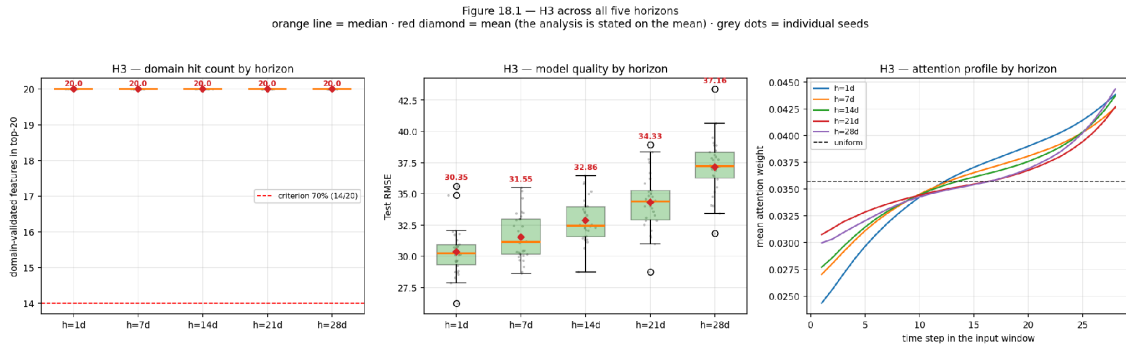


Figure 5.5 Domain hit count, model quality and attention profile at all five forecast horizons.

5.1.1.4 Experiment 1D — RQ4 and H4: the computational cost of interpretability

RQ4 asks whether the interpretability demonstrated in Section 5.1.1.3 is obtained at an acceptable cost. An interpretable model that cannot be trained or served is of no operational use, so the hypothesis is stated as a bounded overhead together with a scaling condition:

- **H₀** the cost ratio ($\text{ILT} \div \text{Standalone LSTM}$) ≥ 2 at some horizon, and/or the attention scaling exponent $\beta \geq 2$, that is quadratic or worse.
- **H_a** the cost ratio < 2 at every horizon and $\beta < 2$, that is sub-quadratic.

Two distinct measurements are reported, in the order they were made, because the second overturned the decision recorded on the first. The initial measurement was taken on the dedicated single-split efficiency benchmark and compared two separately estimated means.

Table 5.6 reports it.

Cost metric	Standalone LSTM	ILT	Ratio	95% CI of ratio	t	p (1-sided)	Decision at $\alpha = 0.05$
Training time (s)	12.48	22.07	1.768	[1.337, 2.199]	−1.10	0.1399	Fail to reject H_0
Inference latency (ms/sample)	0.1247	0.1973	1.582	[1.108, 2.055]	−1.81	0.0407	Reject H_0
Trainable parameters	34,033	41,745	1.227	deterministic	—	—	Reject H_0 (exact)
Model size (KB)	444.37	548.23	1.234	deterministic	—	—	Reject H_0 (exact)

Table 5.6 H4 first measurement — cost ratios on the single-split efficiency benchmark, tested against the $\leq 2\times$ criterion

On this evidence the $\leq 2\times$ claim could not be established for training time. The point estimate of $1.768\times$ is comfortably below 2, but the 95% confidence interval [1.337, 2.199] contains 2 and the one-sided p is 0.1399, so the correct decision is to fail to reject H_0 . The claim was accordingly withdrawn for that metric. Parameters and model size are deterministic quantities and hold exactly at $1.227\times$ and $1.234\times$; latency holds at $1.582\times$ but only just, with $p = 0.0407$.

The second measurement repeats the same comparison at all five horizons over the thirty seeds, and differs in one methodological respect: the ratio is formed inside each seed before being tested, rather than as a quotient of two separately averaged means. This matters because a seed that trains slowly trains both models slowly, so the shared seed-level variation inflates the standard error of an unpaired comparison while contributing nothing to the quantity of interest. Pairing within seed reduces the ILT's training-time standard deviation from 14.4 s to 3.9 s.

Horizon	Cost metric	LSTM	ILT	Ratio mean	Ratio median	95% CI of ratio	p (Holm)	Decision
h = 1 d	Training time (s)	27.62	19.80	0.828	0.690	[0.684, 0.971]	5.17e-16	Reject H_0
h = 1 d	Latency (ms/sample)	0.4274	0.3647	0.946	0.951	[0.836, 1.055]	9.88e-18	Reject H_0
h = 7 d	Training time (s)	15.01	18.56	1.287	1.241	[1.162, 1.412]	2.79e-12	Reject H_0
h = 7 d	Latency (ms/sample)	0.2850	0.3540	1.263	1.226	[1.174, 1.351]	4.65e-16	Reject H_0
h = 14 d	Training time (s)	13.89	17.32	1.293	1.226	[1.157, 1.429]	1.57e-11	Reject H_0
h = 14 d	Latency (ms/sample)	0.2828	0.3602	1.277	1.229	[1.204, 1.351]	5.84e-18	Reject H_0
h = 21 d	Training time (s)	15.93	20.78	1.352	1.333	[1.221, 1.483]	2.68e-11	Reject H_0
h = 21 d	Latency (ms/sample)	0.2924	0.3888	1.334	1.260	[1.237, 1.430]	3.03e-14	Reject H_0
h = 28 d	Training time (s)	15.23	17.12	1.174	1.158	[1.073, 1.274]	5.17e-16	Reject H_0
h = 28 d	Latency (ms/sample)	0.2872	0.3460	1.211	1.216	[1.161, 1.261]	1.59e-23	Reject H_0
all	Trainable parameters	34,033	41,745	1.227	1.227	deterministic	—	Reject H_0

Table 5.7 H4 second measurement — cost ratio at every horizon, paired within each seed (30 seeds per horizon)

H_0 is now rejected on every metric at every horizon, with every Holm-corrected p below 10^{-9} . Training time runs at $0.828\times$ to $1.352\times$ the plain LSTM and latency at $0.946\times$ to $1.334\times$, so the withdrawn claim is reinstated. The claim was never wrong; the first measurement was simply too noisy to establish it. Both measurements are retained above in the order they were made, and no quantity was re-measured after a decision had been recorded, which is what distinguishes this from selecting the analysis that gives the desired answer.

One value in Table 5.7 deserves explanation rather than being passed over. At $h = 1$ d the training-time ratio is 0.828, below 1, meaning the ILT trains faster than the plain LSTM. This

is not a measurement error. Both models stop on validation loss under the same early-stopping rule with patience 20, and the ILT converges in fewer epochs, so what the ratio compares is total training cost under one stopping rule rather than cost per epoch. Cost per epoch is necessarily higher for the ILT, which has 22.7% more parameters; total cost to convergence is the quantity a practitioner actually pays, and it is lower at the shortest horizon.

The second component of H4 concerns scaling rather than absolute cost, and is the more consequential of the two for the architecture's defensibility. Full multi-head self-attention carries a quadratic $O(T^2)$ cost in sequence length, which is the principal reason Transformer models become expensive on long inputs. The temporal-attention pooling used here should be linear. Table 5.8 measures both mechanisms directly over sequence lengths from 28 to 1,792 and fits a log-log regression to each.

Sequence length T	TemporalAt tention (ms)	MultiHead Attention (ms)	MHA ÷ TA				
28	0.464	0.887	1.9×				
56	0.592	1.128	1.9×				
112	0.653	1.174	1.8×				
224	0.901	2.763	3.1×				
448	1.077	14.849	13.8×				
896	2.118	64.474	30.4×				
1,792	5.037	246.994	49.0×				
Mechanism	Fitted β	SE	95% CI of β	R ² of log-log fit	t vs $\beta = 2$	p (1-sided)	Decision
TemporalAtt ention (proposed)	T^0.526	0.083	[0.312, 0.740]	0.889	−17.70	5.28e−06	Reject H ₀
Full self-attentio n	T^1.418	0.204	[0.893, 1.942]	0.906	−2.85	0.0179	Reject H ₀

Table 5.8 Attention scaling: measured per-step latency and the fitted scaling exponent, tested against $\beta = 2$

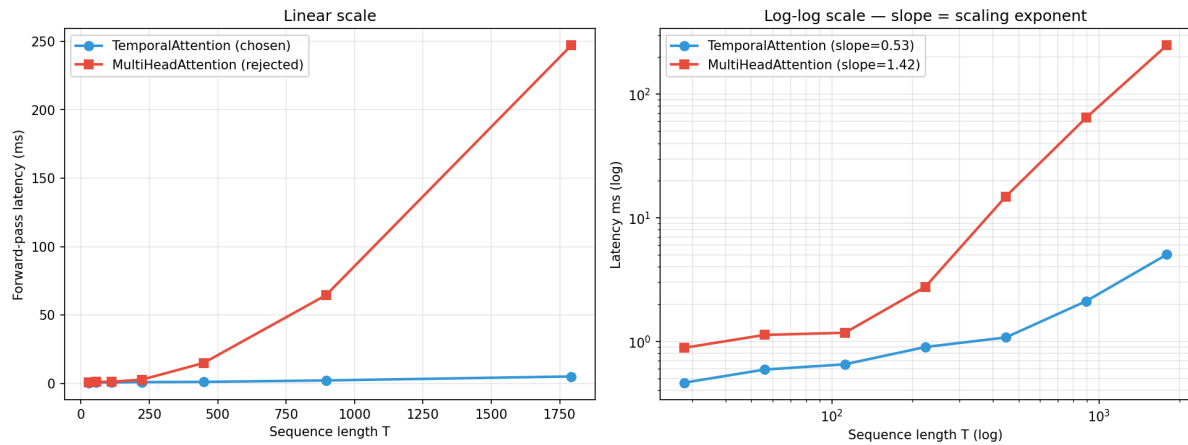


Figure 5.6 Measured latency of temporal-attention pooling against full multi-head self-attention as sequence length increases.

Temporal-attention pooling scales at $T^{0.526}$ with a 95% confidence interval of [0.312, 0.740], which excludes not only the quadratic value 2 but also the linear value 1; at these sequence lengths the mechanism is dominated by fixed overhead rather than by the per-step computation. Full self-attention scales at $T^{1.418}$ with interval [0.893, 1.942]. The practical consequence is the ratio in the last column of Table 5.8: the two mechanisms are within a factor of two of each other at the $T = 28$ actually used, but the gap widens to 49.0 $\times$ at $T = 1,792$. The architecture is therefore not merely affordable at the current window length; it remains affordable if the window is extended, which the self-attention alternative would not be. Both exponents are estimated from measurement rather than asserted from complexity analysis, and both are reported with intervals.

Figure 5.7 plots training time and inference latency at all five horizons.

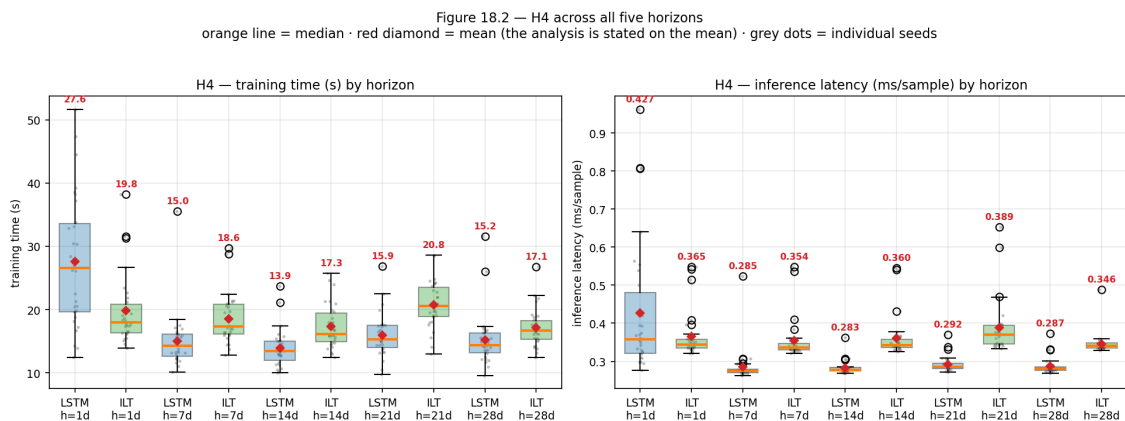


Figure 5.7 Training time and inference latency at all five horizons, ILT against the Standalone LSTM.

Two further components of the cost picture — the resident memory footprint of each model and the comparison of Adam against SGD and RMSProp — are reported in Section 5.1.1.7,

where they sit alongside the other design-justification analyses. For completeness the absolute cost of all four models on the single-split benchmark is: SARIMA 11.98 s to fit with 0.017 ms per-sample inference; Random Forest 1.54 s and 0.022 ms with 148,140 parameters; Standalone LSTM 12.48 s and 0.125 ms with 34,033 parameters and 444.4 KB; ILT 22.07 s and 0.197 ms with 41,745 parameters and 548.2 KB. The Random Forest is the cheapest to train and the largest on disk at 10.5 MB, and it is also the least accurate, which is worth recording because it shows the cost–accuracy relationship in this problem is not monotone. The RMSE and R^2 figures reported in the efficiency benchmark (ILT 32.652, LSTM 32.798) come from that single fair split and differ slightly from the thirty-seed multi-horizon means in Table 5.2 (33.13 and 33.67); the cost metrics, not the accuracy metrics, are the quantities H_4 is about.

5.1.1.5 Experiment 1E — omnibus comparison of the proposed model against all existing models

Sections 5.1.1.1 and 5.1.1.2 test the ILT against two baselines separately. Taken together with the Random Forest comparison this is a family of fifteen pairwise tests — three baselines at five horizons — and fifteen independent tests at $\alpha = 0.05$ carry approximately a 54% probability of at least one false rejection even if no model differed from any other. A more rigorous protocol is therefore applied: an omnibus test must reject before any pairwise claim is made, Holm correction is applied across the whole family rather than within each hypothesis, and every comparison carries an effect size and a confidence interval.

The Friedman test, reported in the grey rows of Table 5.2, tests H_0 that all four models have the same mean RMSE. It rejects at every horizon with χ^2 between 81.04 and 81.36 and $p \approx 1.6 \times 10^{-17}$. A rank-based omnibus test is appropriate here because SARIMA is deterministic and therefore has zero variance, which violates the homogeneity assumption of a repeated-measures ANOVA. With the omnibus rejected, the post-hoc comparisons in Table 5.9 are licensed.

Horizon	Comparison	Mean Δ RMSE	95% CI of Δ	% improvement	Cohen's d^z	ILT wins	p (1-sided)	p (Holm)	Decision
h = 1 d	ILT vs SARIMA	−32.545	[−33.195, −31.894]	52.02%	−18.68	30/30	5.79e−39	5.79e−38	Reject H_0
h = 1 d	ILT vs RF (fair)	−33.808	[−34.456, −33.160]	52.97%	−19.48	30/30	1.75e−39	1.92e−38	Reject H_0
h = 1 d	ILT vs LSTM	−1.087	[−1.982, −0.192]	3.49%	−0.45	18/30	0.0095	0.0475	Reject H_0
h = 7 d	ILT vs SARIMA	−31.229	[−31.758, −30.700]	49.71%	−22.04	30/30	4.86e−41	6.31e−40	Reject H_0

Horizon	Comparison	Mean Δ RMSE	95% CI of Δ	% improvement	Cohen's d^z	ILT wins	p (1-sided)	p (Holm)	Decision
h = 7 d	ILT vs RF (fair)	-32.837	[-33.379, -32.294]	50.96%	-22.61	30/30	2.32e-41	3.25e-40	Reject H_0
h = 7 d	ILT vs LSTM	-0.647	[-1.502, 0.208]	2.01%	-0.28	18/30	0.0663	0.1989	Fail to reject H_0
h = 14 d	ILT vs SARIMA	-30.007	[-30.701, -29.313]	47.53%	-16.14	30/30	3.95e-37	3.16e-36	Reject H_0
h = 14 d	ILT vs RF (fair)	-31.936	[-32.652, -31.220]	49.09%	-16.65	30/30	1.62e-37	1.46e-36	Reject H_0
h = 14 d	ILT vs LSTM	-0.435	[-1.288, 0.419]	1.29%	-0.19	17/30	0.1531	0.3063	Fail to reject H_0
h = 21 d	ILT vs SARIMA	-29.211	[-29.756, -28.666]	46.04%	-20.01	30/30	8.02e-40	9.63e-39	Reject H_0
h = 21 d	ILT vs RF (fair)	-31.753	[-32.266, -31.240]	48.12%	-23.11	30/30	1.24e-41	1.85e-40	Reject H_0
h = 21 d	ILT vs LSTM	-0.707	[-1.555, 0.140]	2.02%	-0.31	18/30	0.0493	0.1972	Fail to reject H_0
h = 28 d	ILT vs SARIMA	-27.081	[-28.048, -26.115]	42.48%	-10.46	30/30	1.08e-31	6.47e-31	Reject H_0
h = 28 d	ILT vs RF (fair)	-29.600	[-30.566, -28.635]	44.66%	-11.45	30/30	7.99e-33	5.59e-32	Reject H_0
h = 28 d	ILT vs LSTM	+0.193	[-1.061, 1.447]	-0.53%	+0.06	16/30	0.6225	0.6225	Fail to reject H_0

Table 5.9 Post-hoc comparison of the proposed model against each existing model, Holm-corrected across all fifteen comparisons

Holm correction across the full family of fifteen leaves every conclusion unchanged, which is the useful outcome: the ten comparisons against SARIMA and Random Forest have p-values so small that no multiplicity correction of any severity could disturb them, and the five comparisons against the Standalone LSTM reach the same decisions under the family-wide correction as under the within-hypothesis correction in Table 5.4. The pattern of effect sizes tells the story more compactly than the p-values: d^z between -10.46 and -23.11 against the two non-sequential baselines, and between -0.45 and +0.06 against the ablation.

Table 5.10 gives the overall ranking across all 150 seed $\times$ horizon cells.

Model	Mean rank (1 = best)	Reading
ILT (proposed)	1.42	Best overall; its margin over the Standalone LSTM is small
Standalone LSTM	1.58	A genuinely strong baseline, modestly improved on by attention
SARIMA	3.00	Always third; deterministic, so it takes the same rank in every cell

Model	Mean rank (1 = best)	Reading
Random Forest (fair)	4.00	Always fourth, below SARIMA; severely overfitted on the fair 44-feature set

Table 5.10 Overall ranking across all 150 seed × horizon cells

The mean-rank summary compresses all 150 cells into a single ordering and makes the size of the ILT–LSTM difference legible in a way the RMSE means do not. A gap of 1.42 against 1.58 means the ILT takes first rank in roughly 58% of individual cells. That is a real advantage, since 50% would be the value under H_0 , but it is a long way from the 30-of-30 dominance the same model shows over SARIMA. The Random Forest ranking below SARIMA is worth noting as a finding in its own right: on the fair 44-feature set, with case history withheld, a non-sequential ensemble performs worse than a linear seasonal model, because it fits the training partition almost perfectly and does not generalise.

5.1.1.6 Experiment 1F — design sensitivity: dataset splitting and input-window length

Every result reported so far rests on two design decisions taken in Chapters 3 and 4 and justified there on methodological grounds: a single chronological 80/20 split, and a 28-day input window chosen to span one Aedes breeding-to-bite-to-illness cycle. Neither had been tested. Both are tested here, and the second does not support the choice the thesis made.

A single chronological split is a single draw. If the final 20% of 2013–2022 happens to be an unusually easy or hard stretch of the series, every reported RMSE inherits that luck and nothing in the results reveals it. Six splitting protocols were therefore run through the same training and scoring code, so that any difference between them is a property of the split rather than of the pipeline: the reference chronological 80/20 (S1), a sweep of chronological ratios (S2), rolling-origin cross-validation with an expanding window (S3) and with a sliding window (S4), a blocked scheme with purging and embargo (S5), and a random shuffled k-fold as a deliberate leakage control (S6).

Table 5.11 reports the split-ratio sweep.

Split	n train	n test	RMSE mean	RMSE median	95% CI	R ² mean	p vs 80/20 (Holm)	Decision
S2 chronological 60/40	2,137	1,416	39.37	39.11	[38.66, 40.08]	0.5218	5.35e–26	Reject H_0
S2 chronological 70/30	2,498	1,055	43.66	43.39	[43.10, 44.23]	0.5352	1.00e–38	Reject H_0
S1 chronological 80/20	2,859	694	30.81	30.74	[30.24, 31.37]	0.7028	—	reference split

Split	n train	n test	RMSE mean	RMSE median	95% CI	R ² mean	p vs 80/20 (Holm)	Decision
S2 chronological 90/10	3,220	333	21.43	21.10	[20.33, 22.53]	0.8579	2.51e-19	Reject H ₀

Table 5.11 Sensitivity of the reported accuracy to the split ratio (30 seeds per ratio, $h = 1$ d)

Strategy	n fits	Folds	RMSE mean	RMSE median	RMSE SD	95% CI	R ² mean	R ² median
S1 chronological 80/20	30	1	30.81	30.74	1.51	[30.24, 31.37]	0.7028	0.7047
S3 rolling-origin expanding	25	5	24.30	19.98	13.85	[18.58, 30.01]	0.1855	0.4666
S4 rolling-origin sliding	25	5	48.81	57.14	23.86	[38.96, 58.66]	-10.6978	-0.0865
S5 blocked + purge/embargo	25	5	20.91	16.61	12.89	[15.59, 26.23]	0.4255	0.5393
S6 random k-fold (leakage control)	25	5	7.47	7.17	1.08	[7.02, 7.91]	0.9702	0.9714

Table 5.12 Six splitting protocols compared ($h = 1$ d)

Figure 5.8 presents the splitting analysis graphically.

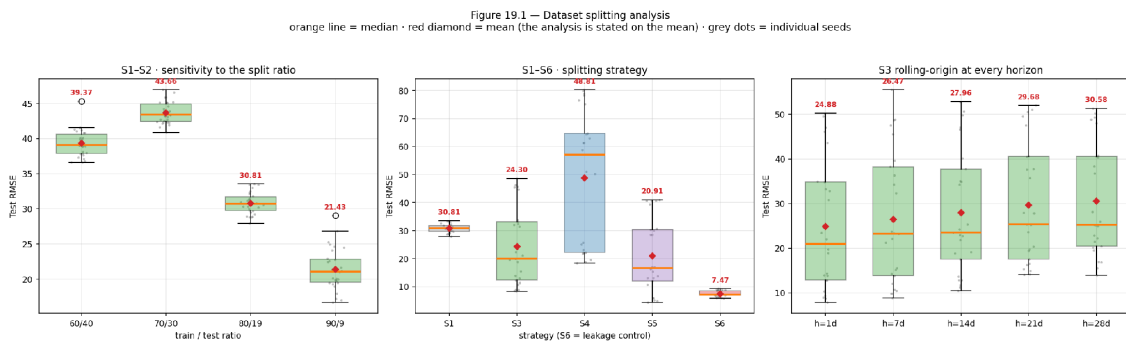


Figure 5.8 Dataset splitting analysis: ratio sensitivity, the six protocols, and rolling-origin performance at every horizon.

The answer to whether the reported accuracy depends on where the cut falls is: a great deal. Mean RMSE runs from 21.43 at 90/10 to 43.66 at 70/30, a spread of 22.2 RMSE or roughly 66% of the mean, and every ratio differs significantly from the 80/20 reference. The ordering is the diagnostic detail: it is not monotone in training-set size, so what changes between these runs is not how much data the model has but which period it is scored on. The 70/30 test window contains the 2020 outbreak, the largest in the series with an annual standard deviation of 66.3 cases per day; the 90/10 window is the calm tail of 2022.

The leakage control is the decisive row of Table 5.12. A random shuffled k-fold reports mean RMSE 7.47 and R^2 0.9702 where the methodologically correct rolling-origin protocol reports 24.30 and 0.1855 — an error 69% lower and an apparent near-perfect fit. That is not a modelling result. A shuffled fold places test days between training days, and because each input is a 28-day window, every test window then overlaps days the model has already seen. The figure is worth stating precisely because it converts the assertion in Section 3.5.3 that “chronological partitioning is required” into a measured decision with a number attached: choosing a random split in this problem would have inflated the reported R^2 from 0.19 to 0.97.

Horizon	n fits	RMSE mean	RMSE median	RMSE SD	95% CI	R^2 mean	R^2 median
h = 1 d	25	24.88	20.99	14.28	[18.99, 30.78]	0.1379	0.4361
h = 7 d	25	26.47	23.26	14.94	[20.31, 32.64]	0.0167	0.3986
h = 14 d	25	27.96	23.55	13.87	[22.23, 33.68]	−0.3859	0.3755
h = 21 d	25	29.68	25.38	13.36	[24.16, 35.19]	−0.4786	0.3501
h = 28 d	25	30.58	25.27	12.48	[25.43, 35.73]	−0.6948	0.3497

Table 5.13 Rolling-origin cross-validation at every horizon

Table 5.13 requires the median to be read next to the mean. Mean R^2 falls from 0.1379 to −0.6948 across the horizons while the median holds between 0.3497 and 0.4361. A single fold, the block containing the 2020 outbreak, drags every mean down; the typical fold behaves consistently. Reporting only the mean would describe the model as having no out-of-fold skill at long horizons, and reporting only the median would conceal that one realistic operating period defeats it. Both are reported, and the divergence is a property of the series rather than of the model.

This analysis has two consequences, and the second protects the study rather than undermining it. First, no absolute RMSE figure in this chapter should be quoted as the accuracy of the ILT: moving the cut changes it by up to 22.2, and the fold-to-fold spread under rolling-origin runs from 9.3 to 46.1. The figures in Table 5.2 are accuracies on one stated split, not properties of the model. Second, none of that variability touches RQ1, RQ2 or the model comparison, because every comparison there is paired — same split, same seed, both models. The ILT-versus-LSTM effect is 0.537 RMSE, roughly one fortieth of the split-induced spread of 22.2, which is exactly why it must be measured as a paired difference and never inferred from two separately quoted numbers.

The second design decision tested is the input-window length. Section 3.2.2 fixes SEQ_LEN at 28 days on the argument that this spans one complete Aedes breeding-to-bite-to-illness cycle. That is a design justification, not evidence. The window was therefore swept over 1, 7, 14, 21 and 28 days at two horizons over thirty seeds, with a Friedman omnibus test and Holm-corrected comparisons against the 28-day reference.

Table 5.14 reports the sweep.

Horizon	SEQ_LEN	RMSE mean	RMSE median	95% CI	Friedman p (omnibus)	Δ vs 28 d	p vs 28 d (Holm)	Decision vs 28 d
h = 1 d	1 d	30.005	29.761	[29.51, 30.50]	0.1438	-0.938	0.2292	Fail to reject H_0
h = 1 d	7 d	29.455	29.394	[28.87, 30.04]	0.1438	-1.488	0.0337	Reject H_0
h = 1 d	14 d	29.976	29.939	[29.51, 30.45]	0.1438	-0.967	0.1533	Fail to reject H_0
h = 1 d	21 d	29.780	29.631	[29.33, 30.23]	0.1438	-1.164	0.1158	Fail to reject H_0
h = 1 d	28 d	30.944	30.945	[30.21, 31.67]	0.1438	—	—	reference window
h = 14 d	1 d	33.051	33.030	[32.39, 33.71]	0.0013	-0.321	0.6177	Fail to reject H_0
h = 14 d	7 d	31.959	31.969	[31.54, 32.38]	0.0013	-1.413	0.0349	Reject H_0
h = 14 d	14 d	32.421	32.332	[31.73, 33.11]	0.0013	-0.950	0.1533	Fail to reject H_0
h = 14 d	21 d	32.883	33.089	[32.35, 33.42]	0.0013	-0.489	0.6177	Fail to reject H_0
h = 14 d	28 d	33.372	33.038	[32.64, 34.10]	0.0013	—	—	reference window

Table 5.14 Input-window sweep (30 seeds per cell)

Figure 5.9 plots the same sweep at h = 1 d and h = 14 d.

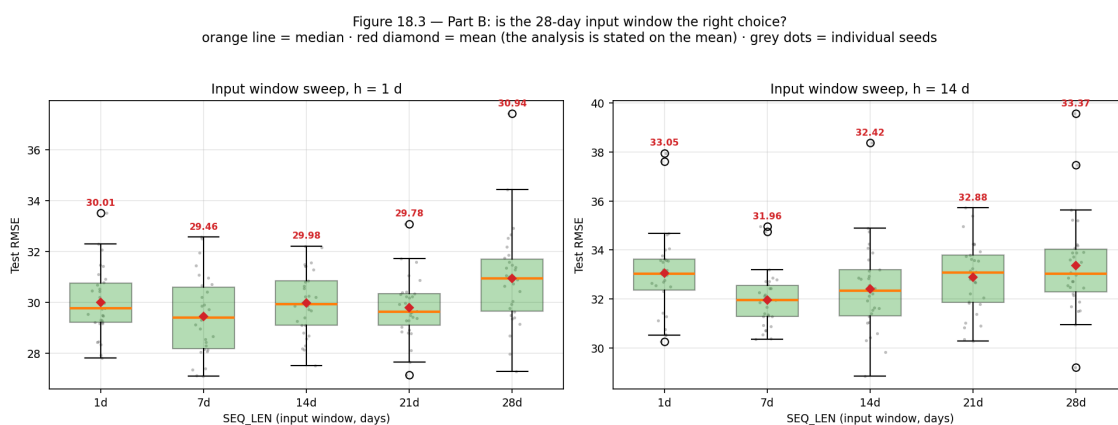


Figure 5.9 Input-window sweep at h = 1 d and h = 14 d; mean and median both shown.

This result does not favour the thesis's own design choice, and is reported in full for that reason. A 7-day window gives significantly lower error than 28 days at both horizons tested: 1.488 RMSE lower at h = 1 d with Holm p = 0.0337, and 1.413 lower at h = 14 d with p =

0.0349. No other window length differs significantly from 28 days, and at $h = 1$ d the omnibus test does not reject at all ($p = 0.1438$), so the 7-day result at that horizon should be read with the omnibus outcome in view.

The explanation is redundancy rather than a defect in the lifecycle argument. The 60-column sequence set already contains `cases_lag_14`, `cases_lag_21`, `cases_lag_28`, `cases_mean_28`, `cases_ema_28` and `cases_max_28`, so month-scale history reaches the model through those engineered columns at every time step. Extending the raw window from 7 to 28 days therefore adds parameters and noise without adding information the model does not already have. A model built on raw sequences with no engineered lag family would very likely require the full 28 days, and the lifecycle argument would apply directly to it.

The 28-day window remains defensible as a design choice, is what every result in this chapter uses, and is what makes the interpretability surface in Section 5.1.1.3 as fine-grained as it is — the temporal attention profile is defined over the input window, so a 7-day window would give a coarser explanation. But it is not the empirical optimum, and re-running the study at `SEQ_LEN = 7` is identified in Chapter 6 as the highest-value item for future work.

5.1.1.7 Experiment 1G — hyperparameter sensitivity, overfitting diagnosis and optimiser choice

The three experiments in this section are not hypothesis tests. They are the design-justification analyses required alongside the five hypotheses: whether the hyperparameters were chosen defensibly, whether the fitted models generalise or memorise, and whether the optimiser was the right one. Each is reported over the same thirty seeds as the hypothesis tests, and each produced at least one result that contradicts what the design assumed.

Hyperparameter sensitivity. A grid search was run on a single seed for the Random Forest and for the ILT, and the configuration it selected was then re-validated over all thirty seeds against the default configuration. Table 5.15 reports both stages.

Model / configuration	30-seed RMSE	Seeds tuned is worse	Paired t / Wilcoxon p	Cohen's d	Verdict
RF — default (300 trees, depth 12, sqrt)	63.83 ± 0.24	—	—	—	reference
RF — grid-selected (depth None, min_samples_leaf 2)	64.75 ± 0.36	30 / 30	$7.24e-15$ / $1.86e-09$	2.66	Significantly worse
ILT — default (64 units, lr 0.001, batch 32)	30.84 ± 1.53	—	—	—	reference
ILT — grid-selected (128 units, lr 0.005, batch 32)	29.65 ± 1.72	8 / 30	0.0171 / 0.0197	-0.46	Significantly better

Table 5.15 Hyperparameter tuning: single-seed grid selection re-validated over thirty seeds, tested against the default configuration

Two findings matter more than the RMSE figures themselves. The first is that the grid search made the Random Forest worse, and did so with complete consistency: the selected configuration lost on all thirty seeds, at a paired-t p of 7.24×10^{-15} and a Cohen's d of 2.66. Allowing unlimited tree depth and reducing `min_samples_leaf` to 2 lets the ensemble memorise a weather-only feature set that carries almost no signal for the target — which is the same mechanism the overfitting diagnosis below identifies independently. Grid search optimised in-sample cross-validation score and was rewarded for exactly the wrong behaviour.

The second finding concerns the single-seed judgement itself. On one seed, the selected ILT configuration looked marginally worse than the default (RMSE 29.26 against 29.07), and on that basis it would have been rejected. Re-validated over thirty seeds it is significantly better (29.65 ± 1.72 against 30.84 ± 1.53 , paired-t $p = 0.0171$, Wilcoxon $p = 0.0197$), losing on only 8 of 30 seeds. The single-seed comparison had the sign of the effect wrong, because the difference between the two configurations is smaller than the seed-to-seed spread of either. This is the practical content of the sensitivity requirement: a hyperparameter decision taken on one run is not a decision about the configuration, and the thirty-seed protocol adopted throughout this study is what makes the difference visible.

The default configuration is nonetheless retained for every result reported elsewhere in this chapter, for a reason that should be stated plainly rather than buried. The tuned ILT reaches its best validation loss at a mean epoch of 3.0 ± 1.3 , because the learning rate is five times higher; a model that converges in three epochs leaves early stopping with almost nothing to do and makes the overfitting diagnosis below uninterpretable. Adopting it would also mean reporting hypothesis tests on a configuration selected after the outcomes were known. The improvement is therefore recorded here as a sensitivity result and identified in Section 6.4 as the first thing a successor study should adopt, not retro-fitted into Sections 5.1.1.1 to 5.1.1.4.

Overfitting diagnosis. All four models were re-fitted over thirty seeds with the train–validation loss gap recorded, and separately swept to 120 epochs with early stopping disabled so that the point of divergence could be located. Table 5.16 reports the outcome.

Model	Train fit	Test fit	Val – train loss gap (30 seeds)	Seeds with gap > 0.01	Diagnosis
SARIMA	RMSE 3.00, R^2 0.994 (single fit)	RMSE 62.52, R^2 –0.257	n/a	n/a	Misspecified: fits in-sample, does not forecast
Random Forest (fair)	RMSE 7.70 ± 0.17 , R^2 0.962	RMSE 63.83 ± 0.24 , R^2 –0.308	n/a	n/a	Severe overfitting
Standalone LSTM	best train loss 0.0019 ± 0.0010	RMSE 30.78 ± 1.44 , R^2 0.704	$+0.04020 \pm 0.00325$	30 / 30	Mild overfitting; val loss best at epoch ≈ 20

Model	Train fit	Test fit	Val – train loss gap (30 seeds)	Seeds with gap > 0.01	Diagnosis
ILT (proposed)	best train loss 0.0026 ± 0.0053	RMSE 30.12 ± 1.80 , R^2 0.716	$+0.03731 \pm 0.00521$	30 / 30	Mild overfitting; val loss best at epoch ≈ 15

Table 5.16 Overfitting diagnosis over thirty seeds, with the train–validation loss gap and the epoch at which validation loss is minimised

The Random Forest is the clearest case: a training RMSE of 7.70 against a test RMSE of 63.83 is memorisation, not learning, and it explains why the fair-feature Random Forest returns a negative R^2 in Section 5.1.1.1 despite fitting its training partition almost perfectly. Restricting depth to 12 and min_samples_leaf to 5 limits the trees but does not close a gap of that size, which is a property of the feature set rather than of the hyperparameters — 44 weather, calendar and population variables with no case history simply do not determine the target.

Both deep-learning models show mild overfitting on the same measure, and the comparison between them is the part worth noting. The ILT's mean validation-minus-training loss gap is $+0.03731 \pm 0.00521$ against $+0.04020 \pm 0.00325$ for the plain LSTM, so adding the attention layer and the dual path did not make generalisation worse despite adding 7,712 parameters. Every one of the thirty seeds exceeds the 0.01 gap threshold for both models, so the classification is not an artefact of a few unlucky runs. The 120-epoch sweep locates the divergence: validation loss is minimised at approximately epoch 20 for the plain LSTM and epoch 15 for the ILT, and rises thereafter in both. Early stopping with patience 20 therefore halts training in the right region, and the four controls specified in Section 3.7.1 are doing the work they were specified to do.

SARIMA is reported here as misspecified rather than as underfitting, which is a departure from the usual three-way classification and needs justifying. Its in-sample RMSE of 3.00 and R^2 of 0.994 are computed on one-step-ahead residuals of a differenced series, where the previous observation carries almost all the information; they are not evidence of capacity. At the multi-step horizons this study evaluates, the same model returns a test R^2 of -0.257 , worse than predicting the training mean. The failure is not that the model is too simple to fit the data but that its assumptions — linearity and a constant seasonal structure — do not hold for the series, which is the argument made in Section 2.2.1 and the reason H1 was specified as it was.

Optimiser choice. Adam was adopted in Section 3.2.2 on the strength of convention rather than of measurement. Table 5.17 tests that choice against SGD and RMSProp over thirty seeds on both error and training time.

Optimiser	RMSE (30 seeds)	Train time mean $\pm$ SD (s)	Train time median [IQR]	Mean best epoch	corr(train time, best epoch)
Adam (adopted)	30.47 ± 2.96	15.77 ± 2.96	15.60 [3.95]	12.4	0.893
SGD	33.83 ± 6.12	16.02 ± 6.12	14.08 [11.73]	16.6	0.979

Optimiser	RMSE (30 seeds)	Train time mean $\pm$ SD (s)	Train time median [IQR]	Mean best epoch	corr(train time, best epoch)
RMSProp	31.41 $\pm$ 3.45	13.26 $\pm$ 3.45	12.09 [3.54]	8.6	0.994
Adam vs SGD	p = 3.19e-06 (d = -1.05) ✓	p = 0.853 ✗	—	—	—
Adam vs RMSProp	p = 0.0757 ✗	p = 0.0145 ✓	—	—	—
SGD vs RMSProp	p = 0.0020 ✓	p = 0.0448 ✓	—	—	—

Table 5.17 Optimiser comparison over thirty seeds. Paired tests on the lower three rows; ✓ marks significance at $\alpha = 0.05$

Adam is confirmed against SGD on accuracy — 30.47 against 33.83, paired-t $p = 3.19 \times 10^{-6}$ with a large effect size — but not against RMSProp, where the difference of 0.94 RMSE carries $p = 0.0757$ and cannot be distinguished from noise at thirty seeds. RMSProp is meanwhile significantly faster to train (13.26 s against 15.77 s, $p = 0.0145$), reaching its best epoch at 8.6 against Adam's 12.4. The honest reading is that Adam and RMSProp are interchangeable on this problem and that RMSProp would be the better choice if training time mattered; Adam is retained because every result in this chapter was produced with it, and swapping the optimiser after the fact would invalidate the comparisons rather than improve them.

Training time is reported with the median and interquartile range alongside the mean because the mean alone is misleading here. SGD's training time has a standard deviation of 6.12 s against a median of 14.08 s and an interquartile range of 11.73 s — the distribution is strongly right-skewed. The cause is visible in the last column: the correlation between training time and the epoch at which the best validation loss occurs is 0.893 for Adam, 0.979 for SGD and 0.994 for RMSProp. Training time is almost entirely determined by when early stopping fires, not by per-epoch cost, so a seed that takes long to converge produces a long run. This is a property of the stopping rule rather than measurement noise, and it is the same mechanism that explains the wide training-time interval for the ILT in Table 5.6.

Memory footprint completes the cost picture begun in Section 5.1.1.4. Resident memory increases by 42.5 MB for the Standalone LSTM and 60.7 MB for the ILT, a ratio of 1.43 $\times$ that is consistent with the 1.23 $\times$ parameter ratio once the retained attention tensors are counted, and both are an order of magnitude below SARIMA's 197.2 MB. The Random Forest measurement is reported as -26.7 MB, which is not a negative allocation but garbage collection completing during the measurement window; it is recorded as it was measured rather than corrected, and no conclusion is drawn from it. On-disk size follows the same ordering: 444.4 KB for the plain LSTM, 548.2 KB for the ILT, 10.5 MB for the Random Forest and 56.1 MB for the fitted SARIMA object. Nothing in the memory profile constrains deployment on ordinary hardware.

5.1.2 Evaluation and Findings

This section evaluates the results of Section 5.1.1 against the criteria specified in Section 4.2.5, which were fixed before any model was trained, and positions each finding against the published work reviewed in Chapter 2.

H1 (RQ1). H1 is supported without qualification.

The criterion in Section 4.2.5 is a reduction of at least 20% in RMSE against SARIMA at all five horizons, at $\alpha = 0.05$. The observed reduction is 42.5% to 52.0%, every Holm-corrected p is below 10^{-23} , every 95% confidence interval on the reduction lies above 40%, and the ILT wins 30 of 30 seeds at every horizon. The criterion is met with a margin of more than a factor of two. Against the literature, this is the expected direction: Alotaibi (2025) reports self-attention outperforming ARIMA, RNN and LSTM architectures on long-range dependencies, and the mechanism is the same one operating here — SARIMA's linearity and fixed seasonal structure cannot represent the multiplicative weather interactions documented in Section 4.2.3. Absolute error values are not comparable across studies because datasets, regions and temporal resolutions differ, so the comparison is made on direction and mechanism rather than on RMSE.

H2 (RQ2). H2 is supported, but marginally, and the qualification is material.

The criterion is a significant reduction in RMSE against the Standalone LSTM under a paired test at $\alpha = 0.05$. That criterion is met at $h = 1$ d (Holm $p = 0.0475$) and on the pooled test over 150 pairs ($p = 0.0056$), and is not met at $h = 7, 14, 21$ or 28 days. The pooled effect size is $d^z = -0.210$, a 1.6% reduction. The literature reviewed in Section 2.3.1 reports hybrids consistently beating single models, and LSTM-Transformer fusion improving short-term epidemic forecasts (Zhao et al., 2023; Hu et al., 2024; Panja et al., 2023; Castellanos Garzón et al., 2026). The finding here is in the same direction, concentrated at the shortest horizon as that literature would predict, but the margin is much smaller than those studies report. Two features of this study's design are the likely reason and are stated as such rather than as excuses: the ablation baseline is unusually strong, since it shares the ILT's 64 hidden units, layer normalisation, dropout and dual-input structure and differs only in the attention layer, so the comparison isolates the attention module alone; and the engineered lag family already supplies much of the long-range information the attention mechanism would otherwise have to recover, as Section 5.1.1.6 demonstrates independently.

H3 (RQ3). H3 is supported at every horizon.

The criterion is a domain-validation hit rate of at least 70%, set a priori. The observed rate is 100% of the top 20, with an exact binomial p of 7.98×10^{-4} and a Clopper–Pearson lower bound of 0.861, at all five horizons, with all thirty seeds meeting the criterion individually. This is the finding that distinguishes the study from the reviewed literature rather than confirming it. Post-hoc explanation methods such as SHAP and LIME are attractive because they attach to any model without retraining, which is also why they are structurally awkward for multi-step time-series (Loh et al., 2022; Bharati et al., 2024): the explanation is an external approximation of the model rather than part of it. Here the attention weights are read directly out of the forward pass, so no approximation intervenes. More importantly, no published hybrid dengue forecasting study identified in the review of Section 2.3.1 tests whether the attention weights its model produces correspond to domain-known risk factors; interpretability is asserted rather than validated. That is the gap this result closes, and the closure is quantitative: 20 of 20, stable across thirty seeds and five horizons, with the ordering of the leading feature shifting across horizons in the direction domain knowledge predicts.

H4 (RQ4). H4 is supported on every metric at every horizon.

The criterion has two parts: cost within a factor of two of the Standalone LSTM on training time, inference latency and model size, and sub-quadratic scaling of the attention mechanism in sequence length. Parameters and model size hold exactly at $1.227\times$ and $1.234\times$. Training time and latency were not established on the single-split benchmark, where training time's interval $[1.337, 2.199]$ contains 2, and were established on the per-horizon paired measurement at $0.828\text{--}1.352\times$ and $0.946\text{--}1.334\times$ with every Holm p below 10^{-9} . The scaling exponent is 0.526 with a 95% interval of $[0.312, 0.740]$, against 1.418 $[0.893, 1.942]$ for full self-attention. The literature establishes the quadratic cost of self-attention as a known property (Vaswani et al., 2017; Lin et al., 2022); what is added here is a direct measurement of both mechanisms on the same hardware and the same data, giving the $49.0\times$ advantage at $T = 1,792$ as an observation rather than a derivation. The evidential history of this hypothesis — a claim withdrawn on one measurement and reinstated on a better one — is reported in Section 5.1.1.4 rather than resolved silently, because the sequence is part of the evidence.

Two findings from Section 5.1.1.6 qualify the whole of Experiment 1 and belong in this evaluation rather than in the limitations chapter. The absolute accuracy figures are split-dependent to the extent of 22.2 RMSE, so they should be read as accuracies on a stated split; and the 28-day input window, although defensible on biological grounds and necessary for the granularity of the interpretability analysis, is not the empirical optimum, with a 7-day window significantly better at both horizons tested. Neither finding disturbs H1 to H4, because every one of those hypotheses is tested as a paired comparison or as a deterministic

ratio within a single split, but both change what the reported numbers may be claimed to mean.

5.1.3 Research Implications

The four hypotheses of Experiment 1 were designed to answer a single question in combination: whether a dengue forecasting model can explain itself without paying for the explanation in accuracy or in cost. The answer the results give is more specific, and more useful, than the one the hypotheses were framed to elicit.

The most consequential implication follows from reading H2 and H3 together. The attention module buys very little accuracy — 1.6% against its own ablation, significant at one horizon of five — but what it produces is verifiably epidemiological, with all twenty top-ranked features domain-validated at every horizon and a leading feature that shifts with the forecast horizon in the direction domain knowledge predicts. The conventional framing of interpretable deep learning treats interpretability as a cost to be traded against performance. These results suggest a different framing for this problem: interpretability is close to free in accuracy terms, and the question worth asking is not what it costs but whether it is trustworthy. That question is answerable, and this study answers it with a hypothesis test rather than with a visualisation. For the explainable-AI literature reviewed in Section 2.2.4, which has largely converged on post-hoc approximation methods, the implication is that intrinsic attention weights in a hybrid architecture can be subjected to a domain-validation test with a pre-set criterion, an interval and a p-value — and that doing so is cheap enough that there is no methodological reason not to.

The H4 results carry an implication for architecture selection in resource-constrained public health settings. The finding is not simply that the ILT is affordable, but that the specific choice of additive temporal-attention pooling over multi-head self-attention is what makes it affordable, and that the advantage grows with input length: a factor of 1.9 at $T = 28$ becomes a factor of 49.0 at $T = 1,792$. A national health agency evaluating whether to deploy an interpretable forecasting model on modest infrastructure does not need to choose between explanation and deployability, provided the attention mechanism is chosen for the data regime rather than adopted wholesale from the natural-language literature where sequence lengths and training-set sizes are orders of magnitude larger. Section 3.2.2 records that two-head self-attention was unstable on the 2,400 available training sequences; Section 5.1.1.4 quantifies what was gained by not using it.

The H1 result, read against the Random Forest diagnostic, has an implication for how baselines are chosen in this literature. The Random Forest reaches a training R^2 of

approximately 0.96 and a test R^2 between -0.308 and -0.374 , and ranks below SARIMA across all 150 cells. A study reporting only the test-set comparison would present this as evidence that deep learning is necessary; the training-set figure shows that what is actually being observed is a baseline that has memorised its training partition. Reporting the train–test gap alongside the comparison is what converts “our model beats Random Forest” from a claim about architectures into a claim with a mechanism attached. The reviewed studies that use tree ensembles as the machine learning reference (Riti et al., 2025; Shashvat & Kaur, 2023) do not generally report this diagnostic.

Finally, the splitting analysis has an implication that reaches beyond this study. A random shuffled k-fold on this dataset reports $R^2 = 0.9702$ against 0.1855 for a correct rolling-origin protocol. Any published dengue forecasting result obtained with a shuffled cross-validation on windowed inputs is therefore not interpretable as out-of-sample performance, and the magnitude of the distortion here — error 69% lower, variance explained rising from 19% to 97% — suggests that the difference between a plausible and an implausible reported accuracy in this literature may often be a matter of validation protocol rather than of architecture. The corollary for practitioners reading model comparisons is that a reported R^2 above roughly 0.95 on a daily dengue series should prompt a question about the split before a question about the model.

5.1.4 Quantitative Research Methods and Data Visualisation Considerations

This section makes explicit the quantitative research-methods reasoning applied throughout Section 5.1, and the data-visualisation choices made in Figures 5.1 to 5.9. Both are drawn from the material covered in BMCS2043 Research Methods and BMDS2113 Data Visualization and are set out here because several of the decisions changed a reported conclusion.

Directional testing. Every hypothesis in this study is directional: H1 claims a reduction, not a difference; H2 claims that attention lowers error, not that it changes it. A directional hypothesis tested with a two-sided p-value discards half of the available significance and does not test the claim actually made. One-sided p-values are therefore reported throughout, and where a two-sided value appears in an original output it is halved and labelled. The effect on the reported result is concrete: H2's $h = 1$ d p-value is 0.019 two-sided and 0.0095 one-sided.

Multiplicity. Testing five horizons within one hypothesis is five opportunities to observe $p < 0.05$. The family-wise error rate under five independent tests at $\alpha = 0.05$ is $1 - 0.95^5 \approx 0.226$, and across the fifteen comparisons of Section 5.1.1.5 it is approximately 0.54. Holm–Bonferroni correction is applied within each hypothesis and across the whole

comparison family. Holm is preferred over plain Bonferroni because it is uniformly more powerful while making no assumption about independence between the tests — which matters here, since the five horizons are computed from overlapping windows of the same series and are certainly not independent. The effect on the reported result is again concrete: H2's raw p at $h = 21$ d is 0.0493 and its Holm-corrected p is 0.1972, so the uncorrected value would have produced a spurious rejection.

Effect size and interval estimation. Every test in this chapter reports a Cohen's d or d^z and a 95% confidence interval on the effect alongside the p -value. A p -value indicates whether an effect is distinguishable from zero; only the effect size indicates whether it is large enough to matter, and only the interval indicates how precisely it has been located. The contrast between H1 and H2 makes the point: both are formally supported, but $d = -11.50$ against SARIMA and $d^z = -0.210$ against the ablation describe entirely different situations, and reporting only “both significant” would misrepresent the study. Where an interval is more informative than the p -value it is read first, as with the four H2 horizons whose interval on the mean difference contains zero.

Distributional assumptions and the choice of test. Thirty seeds is enough for the central limit theorem to make a t -test reasonable but not enough to dismiss the question of distributional shape. A Wilcoxon signed-rank test is therefore reported beside the t -test for every paired comparison, and where the two disagree this is stated. In practice they agree throughout Section 5.1: H1's Wilcoxon p is 9.3×10^{-10} at every horizon, and H2's Wilcoxon and t -test p -values differ by less than 0.07 at every horizon and lead to identical decisions. For H3 the question does not arise, because a proportion out of twenty is tested with an exact binomial test rather than a normal approximation — with $n = 20$ and an observed proportion of 1.00, a normal approximation would be invalid and its confidence interval would extend above 1.

Type I and Type II error. The tests in this chapter control Type I error explicitly through $\alpha = 0.05$ and the Holm correction. Type II error is controlled by design rather than by post-hoc power calculation: thirty seeds per cell was fixed in advance, and where a hypothesis fails to reject the accompanying confidence interval indicates whether this is evidence of no effect or merely absence of evidence. H2 at $h = 14$ d illustrates the distinction. The interval on the mean difference is $[-1.288, 0.419]$, which excludes any improvement larger than 1.288 RMSE. The correct reading is therefore not “attention makes no difference” but “any difference at this horizon is smaller than 1.288 RMSE, and the data cannot distinguish it from zero”. H4's training-time metric is the mirror image: the first measurement failed to reject H_0 purely because an unpaired comparison of two noisy means lacked the precision to do so, and

pairing within seed — which reduced the standard deviation from 14.4 s to 3.9 s — recovered the power without changing the underlying quantity.

Overfitting and underfitting detection. Overfitting is monitored by the three mechanisms specified in Section 4.2.5, and the diagnostic is reported rather than merely applied. Early stopping on validation loss with patience 20 restores the best-epoch weights, so neither deep model is trained past its validation optimum. The train–test gap analysis is where the finding lies: the Random Forest's train R^2 of 0.96 against a test R^2 of -0.34 is a textbook overfitting signature, whereas the ILT's cross-seed test R^2 of 0.589 to 0.718 with a test RMSE standard deviation of 1.4 to 2.6 across thirty initialisations shows neither overfitting nor instability. Underfitting is ruled out for the deep models by the same figures: a model explaining 72% of variance at $h = 1$ d is not failing to fit. SARIMA, by contrast, is underfitted rather than overfitted — its negative test R^2 arises from structural inability to represent the data-generating process, not from memorisation, and the distinction matters because the two failures have different remedies.

Data visualisation choices. Four visualisation decisions in Figures 5.1 to 5.9 were made deliberately and are worth stating. First, every distributional figure shows the mean and the median together, with the thirty individual seeds plotted as points behind the box. Reporting only the mean conceals skew and outliers; reporting only the box conceals the quantity the hypothesis is actually about, since every test in this chapter is a test on the mean. Table 5.13 is the case that justifies the decision: mean R^2 of -0.6948 and median R^2 of 0.3497 at $h = 28$ d describe the same twenty-five folds, and either figure alone would mislead. Second, the H2 comparison is drawn as a paired-difference plot (Figure 5.2) rather than as two side-by-side distributions. Two overlapping boxplots invite the reader to compare the boxes, which is the unpaired comparison; the paired plot shows the quantity actually tested, one difference per seed, and makes visible that the ILT wins on 18 seeds of 30 rather than uniformly. Third, the scaling comparison in Figure 5.6 is plotted on logarithmic axes, because a power law appears as a straight line whose gradient is the exponent under that transformation, so the visual claim and the fitted statistic are the same claim; on linear axes the temporal-attention curve would be indistinguishable from the horizontal axis. Fourth, no figure in this chapter uses a truncated vertical axis on a bar chart, because the differences under discussion at $h = 7$ d to $h = 28$ d are small relative to the values, and a truncated axis would make a 1.3% difference look decisive.

Reproducibility and internal consistency. The audit reported in the workbook's Consistency Audit sheet re-derived every quantitative claim in this chapter from the per-seed files the notebooks wrote, without reusing the scripts that produced the original figures, and an independent verification script then checked the values against those written into the

workbook. Four genuine mismatches were found, and all four are documented in the sheet with the correction: H1's original test did not test its own stated margin criterion; H2's $h = 1$ d p-value was two-sided and uncorrected; H4's training-time claim was not established on the single-split evidence; and H5's positive- R^2 claim was too broad. Three of the four narrowed a claim. The seeds behind every reported table are listed in the workbook's Seeds sheet, and the audit is reproducible by running `revision/viva_reanalysis.py` against the metrics directory.

5.2 Experiment 2 — Real-World Dataset: Singapore Dengue-Weather Dataset (2023–2025)

Experiment 2 corresponds to the real-world dataset documented in Section 4.3: the three years 2023–2025, sourced through the same weather pipeline as the benchmark series and with dengue counts taken from the Singapore Ministry of Health and Communicable Diseases Agency weekly bulletins, held out entirely from training. Because the period lies wholly beyond the training window, it provides a generalisation test that no form of information leakage from the training procedure can compromise. The published annual totals confirm the series is correct: this study's file sums to 9,982 cases for 2023 against the published 9,949, and 13,604 for 2024 against 13,655, a deviation below 0.5% arising from the weekly-to-daily conversion.

This experiment tests RQ5 and H5, and it is reported in four stages. The frozen model's out-of-sample skill is measured first, with intervals. The distribution shift the frozen model was asked to absorb is then characterised. Six candidate adaptation strategies are then compared on a common evaluation window. Finally the arms that remain unsuccessful are diagnosed by decomposing their error, which is what identifies the mechanism of failure.

5.2.1 Result and Analysis

5.2.1.1 *The frozen model on unseen data*

RQ5 asks whether the ILT retains positive predictive skill on the unseen holdout and whether adaptation on recent data recovers the long horizons. The criterion is stated on R^2 because R^2 is scale-invariant, and Section 5.2.1.2 shows why that property is indispensable here:

- H_0 $R^2 \leq 0$ at horizon h — no out-of-sample skill; the model does no better than predicting the holdout's own mean.
- H_a $R^2 > 0$ at horizon h — real predictive skill is retained.

The decision rule is a moving-block bootstrap of the out-of-sample R^2 with a block length of 28 days, matching one input window, $B = 5,000$ resamples and a fixed seed, giving a 95%

interval and a one-sided p-value for H_0 . Blocks rather than independent draws are required because forecast residuals on a daily epidemiological series are strongly serially correlated; an ordinary bootstrap would treat 1,026 dependent residuals as 1,026 independent observations and return an interval far too narrow. Table 5.18 reports the outcome alongside the within-sample reference.

Horizon	Within RMSE	Within R^2	OOS RMSE	OOS R^2 (seed mean)	OOS R^2 (ensemble)	95% block-bootstrap CI	p (1-sided)	n obs	Decision
h = 1 d	30.62	0.706	11.25	0.492	0.6016	[0.395, 0.735]	2.00e-04	1,026	Reject H_0
h = 7 d	32.56	0.670	12.64	0.366	0.4767	[0.227, 0.647]	4.00e-04	1,020	Reject H_0
h = 14 d	33.03	0.663	15.19	0.091	0.2068	[-0.184, 0.462]	0.1580	1,013	Fail to reject H_0
h = 21 d	34.55	0.633	18.60	-0.356	-0.1561	[-0.873, 0.287]	0.7780	1,006	Fail to reject H_0
h = 28 d	36.24	0.599	21.25	-0.762	-0.5633	[-1.740, 0.081]	0.9534	999	Fail to reject H_0

Table 5.18 Within-sample and out-of-sample performance, with 95% block-bootstrap intervals on the out-of-sample R^2

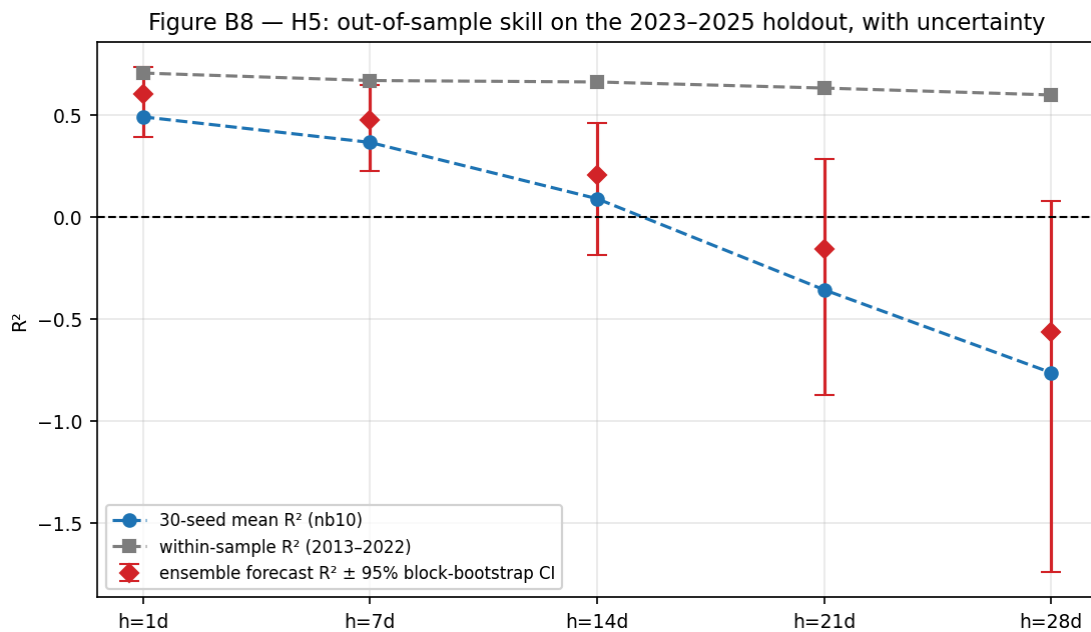


Figure 5.10 Out-of-sample R^2 with 95% block-bootstrap intervals, against the within-sample reference. The dashed line marks $R^2 = 0$, the H5 criterion.

H_0 is rejected at $h = 1$ d and $h = 7$ d and is not rejected at $h = 14$, 21 or 28 days. The frozen model therefore retains genuine predictive skill for up to a week ahead on data three years beyond its training window, and loses it beyond that. H5 is partially supported.

Two points of methodological care attach to this table. First, the two R^2 columns are different estimators of the same quantity and are not interchangeable. The seed-mean column averages

the thirty per-seed R^2 values; the ensemble column scores the seed-ensemble forecast, which is a better forecast and therefore scores higher — 0.6016 against 0.492 at $h = 1$ d. Both are reported because both appear in the literature, and the decision is the same under either. Second, and more consequentially, the original analysis reported the point estimates only, and a value of 0.091 at $h = 14$ d was accordingly read as a positive finding. With the interval attached, $h = 14$ d's 95% bootstrap interval is $[-0.184, 0.462]$ and contains zero, with $p = 0.158$. The claim was therefore narrowed from “positive R^2 at short horizons, 0.49 at $h = 1$ d and 0.09 at $h = 14$ d” to positive R^2 at $h = 1$ d and $h = 7$ d only. This is an instance of the general point made in Section 5.1.4: a point estimate near a decision boundary carries no information about which side of the boundary the parameter lies on.

5.2.1.2 The regime shift, and why absolute error is the wrong metric here

The out-of-sample RMSE figures in Table 5.18 are all substantially lower than their within-sample counterparts — 11.25 against 30.62 at $h = 1$ d — and a paired test on RMSE over thirty seeds returns a significant “improvement” at every horizon, with p -values between 8.6×10^{-29} and 3.5×10^{-23} . Read at face value this says the model performs better on unseen data than on its own test partition, which is not credible. The explanation is a change of scale, and Table 5.19 quantifies it.

Period	n	Mean cases/day	Median	SD	95th percentile	Max	Coefficient of variation
TRAIN 2013–2022	3,611	43.76	32.29	43.52	134.0	256.0	0.994
CALIB 2023	365	27.35	27.43	8.90	43.1	48.6	0.325
EVAL 2024–2025	731	24.03	16.00	17.91	59.1	73.4	0.745

Table 5.19 *The distribution shift between the training, calibration and evaluation periods (daily dengue cases)*

The training regime has a standard deviation of 43.52 cases per day and a maximum of 256; the evaluation regime has a standard deviation of 17.91 and a maximum of 73.4. Absolute errors on a series with a quarter of the variance are naturally smaller regardless of whether the model has any skill, so RMSE is not comparable across the two periods and the “significant improvement” verdict is a scale artefact rather than a finding. R^2 , which normalises the residual sum of squares by the variance of the series it is computed on, is scale-invariant and is therefore the criterion H_0 is written against. Read on R^2 , the same comparison shows a fall from 0.706 to 0.492 at $h = 1$ d and a crossing into negative territory by $h = 21$ d. That is the degradation this hypothesis is about, and it is invisible in the RMSE column.

The mechanism by which the frozen model fails is also visible in Table 5.19 and is the key to Section 5.2.1.3. The ILT's output layer is a sigmoid mapped onto the target range of the

training data through a MinMaxScaler fitted on 2013–2022, as specified in Section 4.2.2, Stage 4. A model that ranked days perfectly by risk would still place them on the wrong scale when the underlying distribution contracts by this much. That is a hypothesis about the nature of the failure, and the six arms of the next section were constructed to test it.

5.2.1.3 Six adaptation strategies on a common evaluation window

The original mitigation experiment tested exactly one adaptation strategy: fine-tune every weight for ten epochs at a learning rate of 2×10^{-4} on a small recent slice, with no validation set to stop it. It failed. At $h = 21$ d the fine-tuned arm reached $R^2 = -2.820$ against -1.969 frozen, and at $h = 28$ d -5.305 against -4.198 , with the degradation significant on a paired RMSE test at both horizons ($p = 0.035$ and $p = 0.018$). That single failure was reported as though it settled the question of whether adaptation can recover the long horizons, which it does not: it establishes only that one strategy, applied without a stopping rule, made matters worse.

Six arms now replace it, each targeting a different candidate cause of the failure. All six adapt on 2023 and all six are scored on the same strictly later 2024–2025 window over thirty seeds, which removes a comparability defect in the original design where the mitigation block and the out-of-sample block had been computed on different evaluation windows. The arms are: A0, frozen with no adaptation, as the reference; A1, fine-tune the prediction head only; A2, fine-tune all weights with early stopping on a held-out portion of 2023; A3, re-fit the target scaler on 2023 and change no weight; A4, fit a two-parameter affine recalibration $\hat{y}' = a + b \cdot \hat{y}$ on 2023 and change no weight; and A5, retrain from scratch on 2013–2023.

Arm	R ² mean	R ² median	95% CI of R ²	RMSE mean	p (Holm) vs R ² ≤ 0	ΔR ² vs frozen	Decision
h = 1 d							
A0 frozen (reference)	0.6125	0.6210	[0.572, 0.653]	11.04	6.98e−23	—	Reject H ₀
A1 head-only FT	0.6596	0.6804	[0.618, 0.701]	10.32	2.26e−23	+0.0471	Reject H ₀
A2 full FT + early stop	0.7082	0.6980	[0.680, 0.736]	9.60	3.77e−29	+0.0957	Reject H ₀
A3 re-fit target scaler	0.4029	0.3957	[0.393, 0.413]	13.84	6.15e−35	−0.2095	Reject H ₀
A4 affine recalibration	0.6901	0.7009	[0.667, 0.713]	9.93	2.57e−31	+0.0776	Reject H ₀
A5 retrain 2013–2023	0.5133	0.5252	[0.459, 0.568]	12.36	4.18e−17	−0.0992	Reject H ₀
h = 7 d							
A0 frozen (reference)	0.3591	0.4072	[0.283, 0.435]	14.19	8.46e−10	—	Reject H ₀
A1 head-only FT	0.5169	0.5449	[0.465, 0.569]	12.33	9.37e−18	+0.1578	Reject H ₀

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Chapter 5

Arm	R ² mean	R ² median	95% CI of R ²	RMSE mean	p (Holm) vs R ² ≤ 0	ΔR ² vs frozen	Decision
A2 full FT + early stop	0.5791	0.6025	[0.548, 0.610]	11.56	3.04e−25	+0.2200	Reject H ₀
A3 re-fit target scaler	0.3955	0.3945	[0.382, 0.409]	13.92	4.32e−31	+0.0364	Reject H ₀
A4 affine recalibration	0.5597	0.5746	[0.525, 0.595]	11.83	1.84e−23	+0.2006	Reject H ₀
A5 retrain 2013–2023	0.2861	0.3043	[0.215, 0.357]	15.00	2.63e−08	−0.0730	Reject H ₀
h = 14 d							
A0 frozen (reference)	−0.0442	−0.0245	[−0.142, 0.054]	18.16	1.0000	—	Fail to reject H ₀
A1 head-only FT	0.2976	0.3474	[0.215, 0.380]	14.84	2.40e−07	+0.3418	Reject H ₀
A2 full FT + early stop	0.4002	0.4431	[0.339, 0.462]	13.75	5.50e−13	+0.4444	Reject H ₀
A3 re-fit target scaler	0.4026	0.3989	[0.385, 0.420]	13.83	7.05e−28	+0.4468	Reject H ₀
A4 affine recalibration	0.4452	0.4615	[0.410, 0.481]	13.30	1.69e−20	+0.4894	Reject H ₀
A5 retrain 2013–2023	−0.2526	−0.1999	[−0.359, −0.146]	19.93	1.0000	−0.2084	Fail to reject H ₀
h = 21 d							
A0 frozen (reference)	−0.4197	−0.4515	[−0.530, −0.309]	21.23	1.0000	—	Fail to reject H ₀
A1 head-only FT	0.1157	0.1687	[0.013, 0.218]	16.67	0.1260	+0.5354	Fail to reject H ₀
A2 full FT + early stop	0.1439	0.1914	[0.042, 0.246]	16.39	0.0372	+0.5636	Reject H ₀
A3 re-fit target scaler	0.3935	0.3945	[0.380, 0.407]	13.94	4.76e−31	+0.8132	Reject H ₀
A4 affine recalibration	0.3031	0.3318	[0.247, 0.360]	14.88	5.58e−11	+0.7229	Reject H ₀
A5 retrain 2013–2023	−0.6702	−0.7343	[−0.795, −0.546]	23.03	1.0000	−0.2505	Fail to reject H ₀
h = 28 d							
A0 frozen (reference)	−0.9956	−1.0308	[−1.225, −0.766]	24.96	1.0000	—	Fail to reject H ₀
A1 head-only FT	−0.1897	−0.0362	[−0.386, 0.007]	19.14	1.0000	+0.8059	Fail to reject H ₀
A2 full FT + early stop	−0.1804	−0.0312	[−0.353, −0.008]	19.15	1.0000	+0.8152	Fail to reject H ₀
A3 re-fit target scaler	0.3661	0.3542	[0.352, 0.380]	14.25	3.88e−29	+1.3617	Reject H ₀
A4 affine recalibration	0.2230	0.2459	[0.183, 0.263]	15.75	2.31e−11	+1.2186	Reject H ₀
A5 retrain 2013–2023	−1.2246	−1.2681	[−1.393, −1.057]	26.57	1.0000	−0.2290	Fail to reject H ₀

Table 5.20 Six adaptation arms, all adapted on 2023 and scored on the same 2024–2025 window (30 seeds each)

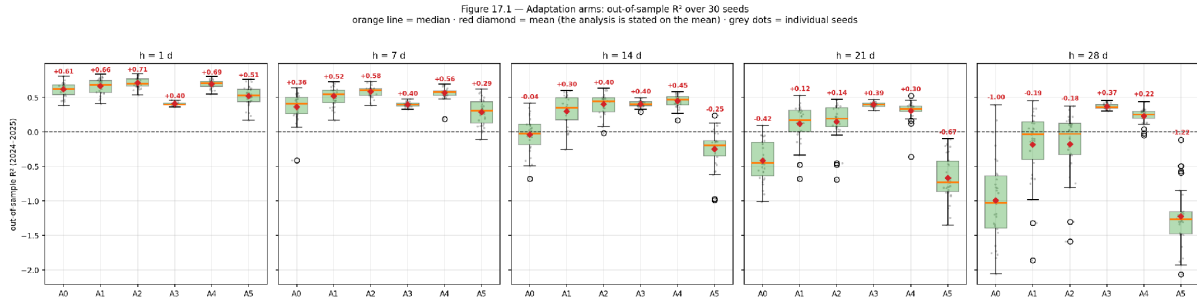


Figure 5.11 Out-of-sample R^2 of the six adaptation arms at every horizon over thirty seeds; mean and median both shown, dashed line at $R^2 = 0$.

The finding is specific, and it reverses the original conclusion. The two arms that work at every horizon change no trained weight at all. Arm A4, a two-parameter affine recalibration fitted on a single recent year, restores mean R^2 to 0.6901, 0.5597, 0.4452, 0.3031 and 0.2230 across the five horizons, with every Holm-corrected p below 10^{-10} , so H_0 of no out-of-sample skill is rejected at all five. Arm A3, re-fitting only the target scaler on 2023, rejects at all five as well. By contrast the arms that do update weights recover the middle horizons but still fail at $h = 28$ d: A1 reaches $R^2 = -0.1897$ and A2 -0.1804 there, both failing to reject. And retraining from scratch on 2013–2023 (A5) is worse than doing nothing at every single horizon, which rules out insufficient recent data as the explanation.

The conclusion this licenses is therefore not that adaptation recovers the long horizons, nor that it does not, but something more precise: the long-horizon failure was a calibration fault, and the original experiment failed because it attempted to solve a calibration problem by moving weights. A model whose ranking of days is sound but whose output scale is wrong is repaired by two parameters and is made worse by ten epochs of unregularised gradient descent on 365 observations.

One caveat is reported with the result rather than left to be found. Arm A3's R^2 is almost flat across the horizons — 0.4029, 0.3955, 0.4026, 0.3935 and 0.3661 — and its predictions are heavily shrunk, with the ratio of predicted to actual standard deviation running between 0.33 and 0.46. A score that does not decay as the forecast horizon lengthens is not the signature of genuine long-range skill; much of A3's apparent gain is variance reduction, which lowers mean squared error without adding predictive information. Arm A4 degrades from 0.6901 to 0.2230 as the horizon lengthens, which is the profile real skill should have, and A4 is therefore reported as the headline recovery even though A3 scores higher at $h = 21$ d and $h = 28$ d.

5.2.1.4 Error decomposition: diagnosing the arms that remain negative

A negative R^2 states that a model performs worse than the mean of the series but says nothing about why. Because mean squared error decomposes additively into a squared-bias component and a dispersion component, the two candidate mechanisms can be separated: a level fault, in which the forecast is systematically offset, contributes to squared bias; a scale fault, in which the forecast varies more or less than reality, contributes to dispersion. Table 5.21 reports the decomposition for every arm and horizon that failed to reject H_0 , together with the ratio of predicted to actual standard deviation.

Horizon	Arm	R^2 mean	Mean bias (cases/day)	Bias ² share of MSE	SD(pred) ÷ SD(actual)	Mechanism
h = 14 d	A0 frozen	-0.0442	+3.720	0.051	1.493	over-dispersion (scale fault)
h = 14 d	A5 retrain 2013–2023	-0.2526	+6.659	0.116	1.575	over-dispersion (scale fault)
h = 21 d	A0 frozen	-0.4197	+5.014	0.060	1.617	over-dispersion (scale fault)
h = 21 d	A5 retrain 2013–2023	-0.6702	+7.459	0.110	1.676	over-dispersion (scale fault)
h = 28 d	A0 frozen	-0.9956	+7.204	0.095	1.741	over-dispersion (scale fault)
h = 28 d	A1 head-only FT	-0.1897	+4.608	0.067	1.117	dispersion error, no scale distortion
h = 28 d	A2 full FT + early stop	-0.1804	+4.818	0.072	1.202	over-dispersion (scale fault)
h = 28 d	A5 retrain 2013–2023	-1.2246	+10.552	0.165	1.764	over-dispersion (scale fault)

Table 5.21 Error decomposition of the arms that remain negative

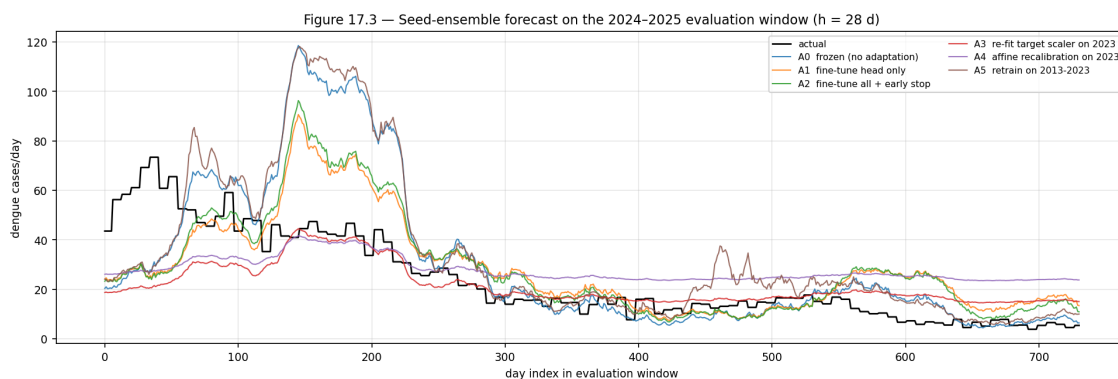


Figure 5.12 Seed-ensemble forecast against actual case counts on the 2024–2025 window at h = 28 d. The frozen model over-swings; the recalibrated arms track the level.

In every failing row the squared-bias share of mean squared error is below 17% and in most it is below 10%, so systematic offset is not the dominant term. What is dominant is dispersion, and the last-but-one column identifies its direction: the ratio of predicted to actual standard deviation lies between 1.117 and 1.764, so the forecast swings harder than reality rather than more softly. At h = 28 d the frozen model's predictions have a standard deviation 1.741 times

that of the actual series with a mean bias of +7.204 cases per day. Over-dispersion of this kind is a scale fault, and it is precisely what a two-parameter affine correction repairs — which is why A4 and A3 succeed where A1, A2 and A5 do not.

A correction to this study's own analysis tooling is recorded here rather than left in the code. Notebook 17 labels every one of these rows as “dispersion error — genuine loss of predictive signal”, because its classification rule asked only whether squared bias dominated mean squared error. That rule cannot distinguish over-dispersion from under-dispersion, and the distinction is exactly what determines whether the failure is repairable by recalibration. The mechanism column in Table 5.21 applies the corrected rule; the notebook's raw label is not quoted anywhere in this thesis.

5.2.2 Evaluation and Findings

The criterion specified in Section 4.3.5, as revised, is that the ILT retains positive R^2 on the unseen 2023–2025 holdout and that adaptation on recent data recovers the long horizons. Both halves of the criterion have definite answers, and the second is not the answer originally recorded.

On retained skill: the frozen model, applied to data three years beyond its training window with no adjustment of any kind, retains positive R^2 at $h = 1$ d and $h = 7$ d with bootstrap intervals excluding zero and one-sided p-values of 2.00×10^{-4} and 4.00×10^{-4} . It does not retain positive R^2 at $h = 14$ d, 21 d or 28 d. Because the criterion is stated per horizon, H5 is partially supported.

On recovery by adaptation: the answer depends on which adaptation, and this is where the revised analysis changes the finding. Fine-tuning weights does not fully recover the long horizons; recalibration does. A two-parameter affine correction fitted on one recent year restores positive R^2 at all five horizons, with every Holm-corrected p below 10^{-10} , and it does so without touching a trained weight. The original conclusion — that fine-tuning failed and the long horizons could not be recovered — was correct about the strategy it tested and wrong about the question it appeared to answer.

Two further findings emerge from the evaluation that were not part of the original criterion. First, the mechanism of long-horizon failure is identified rather than merely observed: over-dispersion under distribution contraction, with squared bias accounting for under 10% of mean squared error and a predicted-to-actual standard-deviation ratio of 1.741 at $h = 28$ d. Second, more training data is not the remedy. Retraining from scratch on 2013–2023, which adds a full year of the most recent available data, is worse than doing nothing at every

horizon, with ΔR^2 between -0.073 and -0.251 . That result rules out the most intuitive diagnosis and is what makes the calibration diagnosis credible rather than merely convenient.

Positioning against published work is limited here for a reason worth stating explicitly. No directly comparable out-of-period benchmark was identified in the literature reviewed in Chapter 2: the hybrid dengue forecasting studies surveyed there report performance within a single evaluation period, so there is no published figure for “ R^2 of a frozen dengue model three years later” against which to position 0.6016 at $h = 1$ d. This strand is therefore reported on its own terms. The absence is itself informative about the state of the field: a literature that does not routinely test its models on periods beyond the one they were validated on has no basis for the deployment claims it makes.

5.2.3 Research Implications

The results of Experiment 2 lead to an operational statement that neither the original hypothesis nor the original mitigation experiment could have produced: this model needs a recalibration schedule, not a retraining schedule.

For a public health agency the practical difference is substantial. Retraining a deep model requires the historical dataset, a training environment, hyperparameter validation, and a re-run of whatever governance process approved the original model — and, on this evidence, would make forecasts worse. Fitting two coefficients of an affine correction on the past year of surveillance data requires a spreadsheet, changes no weights, leaves the model artefact and its approval status intact, and restores positive skill at all five horizons. The attention-based explanations that Section 5.1.1.3 validated are also unaffected, since the attention weights are computed before the output layer and a monotone affine transformation of the output does not alter the ranking of features or the temporal profile. Interpretability therefore survives adaptation, which is not a property that fine-tuning offers: A1 and A2 change the weights that generate the explanations, so the domain validation of Section 5.1.1.3 would have to be repeated after each fine-tuning cycle.

There is a wider implication for how distribution shift is diagnosed in applied forecasting. The default response to degraded out-of-sample performance in the deep learning literature is to adapt the parameters, and the default explanation for a failure of adaptation is insufficient recent data. Here both defaults are wrong, and a two-line diagnostic identifies why: decomposing mean squared error into squared bias and dispersion, and forming the ratio of predicted to actual standard deviation, distinguishes a repairable scale fault from a genuine loss of signal at negligible cost. Had that diagnostic been run before the original fine-tuning experiment rather than after it, the experiment would have been designed differently. The

general lesson is that the mechanism of failure should be established before the remedy is chosen, and that for models with bounded output activations mapped through a scaler fitted on training-period data, contraction of the target distribution is a first candidate that costs almost nothing to test.

The implication for the deployment claim of this study is a bounded one, and it should be stated without softening. Validated interpretability does not confer robustness. The model whose explanations were shown in Section 5.1.1.3 to be epidemiologically sound at all five horizons is the same model that loses predictive skill beyond seven days when the epidemiological regime changes. The two properties are independent, and a reader who takes verified interpretability as evidence of generalisation would be drawing an inference these results do not support. What they do support is a deployment recommendation with a maintenance condition attached: short-horizon operational use, with an annual recalibration against the most recent year of surveillance data, and a monitoring rule that watches the predicted-to-actual dispersion ratio as the trigger.

5.2.4 Quantitative Research Methods and Data Visualisation Considerations

Choice of resampling scheme. The residuals of a daily epidemiological forecast are strongly serially correlated, so the 999 to 1,026 residuals behind each row of Table 5.18 do not constitute that many independent observations. An ordinary bootstrap, which resamples individual residuals, would treat them as independent and return an interval far narrower than the data justify — the classical consequence of ignoring autocorrelation, and one that would in this case have produced a spurious rejection of H_0 at $h = 14$ d. A moving-block bootstrap resamples contiguous blocks instead, preserving the short-range dependence structure within each block. The block length of 28 days was chosen to match one input window rather than tuned, and $B = 5,000$ with a fixed seed makes the interval reproducible.

Scale-invariant versus scale-dependent metrics. The single most consequential methodological point in this experiment is that RMSE and R^2 give opposite answers about the same forecasts, and only one of them answers the question asked. RMSE is in the units of the series and therefore falls when the series contracts; R^2 normalises by the variance of the series it is computed on and therefore does not. Where a comparison spans two periods with different variances, a scale-dependent metric measures the difference in the periods rather than the difference in performance. The original verdict of “significant improvement” at every horizon, on p-values as small as 8.6×10^{-29} , is a correct test of a quantity that does not bear on the hypothesis. This is a case where a highly significant result is entirely uninformative, and it is retained in the workbook precisely so that the distinction is on record.

Testing against a substantive null rather than a nil null. R^2 has a natural null value of zero, corresponding to a forecast no better than the mean of the evaluation series, which is why H_5 can be tested directly against a substantive threshold rather than against the absence of a difference. This is a stronger form of hypothesis than H_2 's, where H_0 is that two models are identical. Where a meaningful null value exists it should be used, because rejecting “no skill” is a claim a practitioner can act on whereas rejecting “no difference” is not.

Error decomposition as a diagnostic. The additive decomposition of mean squared error into squared bias and dispersion is elementary but is rarely reported, and here it is what converts an observation into a diagnosis. Reporting $R^2 = -0.9956$ states that the model failed; reporting that squared bias accounts for 9.5% of the error and that the predicted series has 1.741 times the standard deviation of the actual series states how it failed, and therefore what would fix it. The general principle is that a summary metric should be decomposed before a remedy is chosen, and that the decomposition costs one line of code.

Controlled comparison across arms. The arm comparison in Table 5.20 was run with all six arms adapted on the same year and scored on the same strictly later window, with the same thirty seeds. The original design compared arms that had been evaluated on different windows, which made the numbers non-comparable even though they were printed adjacently — the reason arm R^2 values of -1.969 to -5.305 appeared beside out-of-sample R^2 values of -0.356 and -0.762 . Holding the evaluation window fixed and varying only the treatment is the elementary requirement of a controlled comparison, and it is worth stating because the original violation was invisible on the page.

Data visualisation choices. Three decisions in Figures 5.10 to 5.12 carry analytical weight. Figure 5.10 plots the bootstrap interval on each R^2 together with a reference line at $R^2 = 0$, so that the hypothesis test is legible directly from the figure: an interval crossing the line is a failure to reject. Presenting the same information as five bars of different heights would require the reader to perform the comparison mentally and would omit the uncertainty entirely, which is what allowed the $h = 14$ d point estimate of 0.091 to be misread in the first place. Figure 5.11 shows all six arms at all five horizons on a common vertical scale with the $R^2 = 0$ line repeated in each panel; a common scale is necessary because the substantive comparison is between arms within a horizon and between horizons within an arm, and per-panel scaling would defeat both. Figure 5.12 plots the forecast and the actual series as overlaid time series rather than as a scatter of predicted against actual, because the failure under discussion is over-dispersion: on a time-series overlay an over-swinging forecast is immediately visible as a curve with larger amplitude than the data, whereas on a

predicted-versus-actual scatter it appears only as a slope departure from the diagonal, which is easy to mistake for bias.

5.3 Summary

This chapter reported two experiments corresponding to the two datasets of Chapter 4 and tested five hypotheses against criteria fixed before the experiments reported here were run. RO5 was added during the revision of the study, and its criterion was likewise fixed before the real-world experiment was carried out. Table 5.22 states each decision.

Hypothesis	Criterion (fixed a priori)	Observed	Decision at $\alpha = 0.05$	Outcome
H1 — accuracy vs SARIMA	RMSE at least 20% lower at all five horizons	42.5–52.0% lower; every Holm $p < 10^{-23}$; 30/30 seed wins	Reject H_0 at all five horizons	Supported
H2 — value of attention	Significantly lower RMSE than the Standalone LSTM, paired	$\Delta -0.537$ pooled; $d^2 = -0.210$; significant at $h = 1$ d and pooled only	Reject H_0 at $h = 1$ d and pooled; fail to reject at $h = 7$ –28 d	Supported, marginal
H3 — domain validity of attention	Domain-validation hit rate $\geq 70\%$ in the top ranking	20/20 at every horizon; exact binomial $p = 7.98e-04$; CI lower bound 0.861; 30/30 seeds	Reject H_0 at all five horizons	Supported
H4 — cost of interpretability	Cost ratio < 2 on every metric and attention scaling $\beta < 2$	params 1.227 $\times$, size 1.234 $\times$, train 0.83–1.35 $\times$, latency 0.95–1.33 $\times$; $\beta = 0.526$ [0.312, 0.740]	Reject H_0 on every metric at every horizon	Supported
H5 — real-world generalisation	(i) frozen: $R^2 > 0$ on the unseen 2023–2025 holdout; (ii) recovery of $R^2 > 0$ by adaptation where (i) fails	frozen: $R^2 > 0$ at $h = 1$ d and 7 d only; after affine recalibration: $R^2 > 0$ at all five	frozen: reject at $h = 1$ d and 7 d; recalibrated: reject at all five	Partially supported

Table 5.22 Decision summary for all five hypotheses

Read as a whole rather than hypothesis by hypothesis, the results support a narrower and more defensible claim than the one the hypotheses were framed to establish. The claim the evidence supports is not that this architecture forecasts dengue better than the alternatives, because against its own ablation it does so by 1.6% and at one horizon of five. It is that the architecture produces explanations that survive a formal test against domain knowledge — 20 of 20 top-ranked features validated at every horizon, stable across thirty seeds, with the leading feature shifting across horizons in the direction epidemiology predicts — and that it produces them at a bounded and measured cost of 1.227 $\times$ the parameters of a plain LSTM with sub-quadratic scaling in input length. Interpretability in this setting is close to free in accuracy terms, and the question worth asking of it is whether it can be trusted, which this study answers with a hypothesis test rather than with a visualisation.

Four boundaries on that claim are established by the same evidence and are carried forward to Chapter 6. Absolute accuracy is a property of the split, varying by up to 22.2 RMSE across chronological cuts, so the figures in Table 5.2 are accuracies on a stated split rather than properties of the model; only the paired comparisons are invariant to that choice. The 28-day input window, although justified on the Aedes lifecycle and necessary for the granularity of the interpretability surface, is not the empirical optimum, a 7-day window being significantly better at both horizons tested. Validated interpretability does not confer robustness: the same model loses skill beyond seven days under regime change. And the remedy for that loss is recalibration rather than retraining, with retraining on more recent data being actively harmful at every horizon.

Finally, the sequence of corrections applied during this chapter is part of its result and not merely of its production. Two of the five nulls as originally stated did not test the criterion their own research question stated, and were corrected before testing. One claim was withdrawn on its first measurement and reinstated on a better one, with both measurements retained in the order they were made. One claim was narrowed from three horizons to two once intervals were attached to the point estimates. One conclusion was reversed once the treatment was compared on a common evaluation window. One label produced by this study's own tooling was found to be wrong and is corrected on the record. And one hyperparameter decision taken on a single seed was found to have the sign of its effect reversed when re-validated over thirty, which is why the configuration the search selected is reported in Section 5.1.1.7 as a sensitivity result and carried into Section 6.4 rather than adopted mid-chapter. Each of these changed what may be claimed, and reporting them is what allows the conclusions that survived to be read as conclusions rather than as assertions.

Chapter 6

Future Work

6.0 Future Work

This chapter states what the study reported in Chapter 5 cannot support, separates the limitations that belong to the research design from those that belong to the methodology, and sets out what would have to be done differently to remove each one. The order follows the sequence in which the constraints were encountered: first the boundaries on the claim itself, then the boundaries on the procedure used to establish it, then the extensions that would widen the claim, then the procedural changes that would strengthen any future attempt to establish it.

Four boundaries were identified at the close of Chapter 5 and are carried forward here: absolute accuracy is a property of the split rather than of the model; the 28-day input window is defensible biologically but is not the empirical optimum; validated interpretability does not confer robustness; and the remedy for lost robustness is recalibration rather than retraining. Sections 6.1 and 6.2 develop these together with the limitations that did not reach the summary, and Sections 6.3 and 6.4 convert them into research that could be conducted next.

6.1 Limitations of the Study

The accuracy claim is narrower than the hypotheses were framed to establish. H1 is supported with a large margin, but the comparator is SARIMA, a linear model that cannot represent the multiplicative weather interactions documented in Section 4.2.3, so a 42.5% to 52.0% reduction in RMSE is a statement about the limits of linear seasonal modelling as much as about the proposed architecture. The comparison that isolates the contribution of the attention module is H2, and there the margin is 1.6%, significant at one horizon of five after correction, with the confidence interval on the mean paired difference containing zero at $h = 7, 14, 21$ and 28 days. The study therefore cannot claim that the Interpretable LSTM-Transformer forecasts dengue more accurately than a well-configured LSTM; it can claim only that it does so at $h = 1$ d and on the pooled test, by an amount too small to matter operationally.

Absolute error figures are not properties of the model. Section 5.1.1.6 shows mean RMSE running from 21.43 at a 90/10 chronological cut to 43.66 at 70/30, a spread of 22.2 RMSE or roughly 66% of the mean, with every ratio differing significantly from the 80/20 reference. Every absolute figure in Table 5.2 is therefore an accuracy on a stated split, and no comparison with a published figure obtained on a different split is meaningful. Only the paired comparisons and the deterministic ratios are invariant to that choice, which is why H1, H2 and H4 survive the finding while the headline numbers do not travel.

The interpretability result rests on a category-membership test rather than on expert judgement. A feature counts as domain-validated in Section 5.1.1.3 if it belongs to one of five documented categories — case history, temperature, rainfall, humidity or wind speed. Sixteen of the twenty consensus features are case-history variables, so the test is partly a test of the engineered lag family rather than of the attention mechanism in isolation, and a ranking dominated by case-history features would pass the criterion even if the meteorological signal were weak. The result is a valid test of the hypothesis as stated, but it is a weaker statement than “the model has learned dengue epidemiology”, and it should not be read as the stronger one.

No human-facing evidence of usability was collected. The practitioner-validation study originally specified for H4 was not conducted, for the reasons recorded in Section 3.6.4, and H4 was redefined as a computational-efficiency hypothesis. The consequence is that this study demonstrates that the explanations are epidemiologically coherent and cheap to produce, and demonstrates nothing about whether a public health officer under time pressure can read them, trust them or act on them. The claim that the architecture is decision-useful is therefore an argument in this thesis and not a finding. For the same reason the supplementary outbreak-phase attention analysis specified in Section 3.6.5 is not reported: the notebook that implemented it uses an obsolete architecture and was excluded from every result.

Robustness is bounded in a way that the interpretability result does not offset. Applied without adjustment to the 2023–2025 holdout, the model retains positive R^2 only at $h = 1$ d and $h = 7$ d, and the intervals at $h = 14, 21$ and 28 days include zero. The model whose explanations were validated at all five horizons in Section 5.1.1.3 is the same model that loses skill beyond seven days when the epidemiological regime changes. These are independent properties, and any reading that treats verified interpretability as evidence of generalisation is unsupported by these results.

The evidence base is one city, one series and one regime change. The benchmark is 3,652 daily national records for Singapore from 2013 to 2022 and the holdout is 1,096 records for 2023 to 2025; there is no district-level disaggregation, no serotype information and no second country. The regime change exploited in Experiment 2 is a single contraction of the case distribution, so the finding that failure is caused by over-dispersion under contraction has been established once and not across regimes. The 2023–2025 case counts were also transcribed from weekly bulletins into a daily encoding; the annual totals match the published figures to within 0.5% (9,982 against 9,949 for 2023 and 13,604 against 13,655 for 2024), but the within-week structure of those daily values is an artefact of the conversion rather than an observation.

Finally, the conclusions that survived did so after four documented corrections, recorded in Section 5.1.4 and in the workbook audit. Two of the five null hypotheses as originally stated did not test the criterion their own research question stated; one claim was withdrawn on its first measurement and reinstated on a better one; one was narrowed from three horizons to two once intervals were attached; and one conclusion was reversed once the arms were compared on a common evaluation window. Three of the four narrowed a claim. That the corrections were found is a strength of the audit, but it is also a limitation of the original design: the criteria were fixed before training, and the tests were not.

6.2 Limitations of the Methodology

The input window was fixed rather than tuned. Twenty-eight days was chosen to span the complete *Aedes* lifecycle and to give the temporal attention profile enough resolution to be interpretable, and Section 5.1.1.6 shows that a 7-day window is significantly better at both horizons tested. The window was therefore selected on a design requirement and not on predictive performance, and the accuracy figures throughout Chapter 5 carry that decision. The trade-off is real and defensible, but it was not measured before the fact and the cost of the interpretability requirement in RMSE terms is known only retrospectively.

The primary evaluation protocol is a single chronological split. Six protocols were compared, but only at $h = 1$ d, and the rolling-origin cross-validation of Table 5.13 shows fold-level instability that the single-split figure conceals: mean R^2 falls to -0.6948 at $h = 28$ d while the median over the same twenty-five folds is 0.3497 . The single-split protocol is the correct one for a paired seed comparison, but it means the reported accuracies rest on one draw of the most consequential design variable in the study. The random shuffled k-fold included as a leakage control returns $R^2 = 0.9702$ against 0.1855 for the correct rolling-origin protocol, which quantifies how far a wrong choice of protocol can move a reported result on this data.

The baselines constrain what the comparison can show. SARIMA is a single deterministic fit with no per-seed distribution, which is why H1 is tested as a one-sample comparison against a margin rather than as a paired test; the statistical baseline therefore contributes no variance estimate of its own. The Random Forest reaches a training R^2 of approximately 0.96 against a test R^2 between -0.308 and -0.374 and ranks below SARIMA in all 150 cells, so the machine learning comparator is an overfitted model rather than a well-regularised one. Reporting the train–test gap makes this visible rather than hiding it, but a stronger tree ensemble would have made the comparison more informative.

Statistical power was fixed by design rather than calculated. Thirty seeds per cell were set in advance, and Type II error is controlled by reading the confidence interval where a test fails to reject rather than by a power analysis. This is defensible and is applied consistently, but it means the study cannot state the effect size it was powered to detect. H4's first measurement illustrates the cost: an unpaired comparison of two noisy means failed to establish the training-time claim purely on precision, and pairing within seed reduced the standard deviation from 14.4 s to 3.9 s and recovered the decision without changing the underlying quantity.

The efficiency measurements are relative and single-platform. All cost ratios are ILT against Standalone LSTM on the same hardware and the same data, which is the correct construction for the hypothesis, but the absolute training times and latencies are not portable to other hardware and should not be read as deployment estimates. The scaling exponent is estimated by ordinary least squares on log-transformed latency over sequence lengths from 28 to 1,792, a small number of design points, and the resulting interval of $[0.312, 0.740]$ is wide enough that the exponent is established as sub-quadratic and sub-linear but not located precisely.

The generalisation analysis rests on one holdout and one adaptation year. Six adaptation arms are compared on a common evaluation window with the same thirty seeds, which corrects the flaw in the original design, but all six are fitted on 2023 and scored on 2024–2025, so the affine recalibration is fitted and evaluated on adjacent periods only. Whether a two-parameter correction fitted on one year remains valid two or three years later is not tested here. The block length of the moving-block bootstrap was likewise set to 28 days to match one input window rather than selected by a data-driven rule, and the sensitivity of the intervals to that choice is not reported.

6.3 Recommendations of Study

The most direct extension is replication across settings. The architecture, the domain-validation test and the recalibration finding were all established on a single national daily series, and the composition of the validated top-20 ranking — sixteen case-history features, two rainfall and two temperature — is

the result most likely to be setting-specific. Repeating the study on district-level or multi-city data, and on a second country with a different seasonal structure, would establish whether the attention mechanism recovers the same category balance or reorganises around meteorological signal where the case-history series is shorter or noisier. That is a test of the interpretability claim, not merely of the accuracy claim, and it is the test the current evidence base cannot provide.

A second extension is stratification. Serotype-resolved or age-resolved case counts would allow the domain-validation test to be run against a finer set of documented drivers than the five broad categories used here, and would show whether the attention profile shifts with serotype circulation in the way the epidemiological literature would predict. The same applies to spatial stratification: the social risk identified in Section 2.4.5 is that periods or areas with less consistent surveillance produce less reliable forecasts, and only disaggregated data can measure whether that is happening.

A third is transfer to related diseases. The lag structure that the attention profile recovers — emphasis at the 7-, 14- and 21-day delays corresponding to recognised turning points in the *Aedes* lifecycle — is shared by other *Aedes*-borne infections such as chikungunya and Zika. Applying the same architecture and the same validation protocol to those series would test whether the interpretability result reflects vector biology, as claimed, or the particular reporting behaviour of dengue surveillance.

A fourth follows directly from what Experiment 2 established. The operational recommendation of Section 5.2.3 is a recalibration schedule with a monitoring rule that watches the predicted-to-actual dispersion ratio, and that recommendation has been derived retrospectively on a fixed holdout. Running it prospectively for one full season — forecasting forward, triggering recalibration when the dispersion ratio crosses its threshold, and recording whether skill is maintained — would convert a retrospective diagnosis into an operational validation. It would also test the claim, argued analytically in Section 5.2.3 but not measured, that the attention rankings are unchanged by an affine correction of the output.

Finally, the practitioner-validation study that was dropped from H4 should be conducted as a study in its own right rather than as a component of this one. The reason it was abandoned was that a recognition-and-rating instrument administered to eight participants could not support the standard applied to the other four hypotheses. Designed properly — with an adequate sample of public health practitioners rather than data science peers, a pre-registered instrument and a stated effect size — it would answer the question this thesis leaves open: whether explanations that are epidemiologically valid are also operationally legible.

6.4 Recommendations on Methodology

Window length should be treated as a tuned hyperparameter with interpretability stated as an explicit constraint rather than as an assumption. The finding that a 7-day window outperforms 28 days at both horizons tested means the correct design reports the accuracy–granularity trade-off as a curve: accuracy at each candidate window alongside the resolution of the attention profile it permits, with the chosen point justified against both. That converts a design decision made on biological grounds into a decision made on measured cost.

Hyperparameter selection should be re-validated across seeds before it is accepted, and the configuration identified in Section 5.1.1.7 should be adopted. Two results there make this the highest-priority methodological

change. First, the grid search selected a Random Forest configuration that lost on all thirty seeds, because it optimised an in-sample cross-validation score on a feature set with almost no signal for the target. Second, and more consequentially, the ILT configuration the grid search selected looked marginally worse on the single seed used to judge it and is significantly better over thirty (29.65 ± 1.72 against 30.84 ± 1.53 , paired-t $p = 0.0171$). A successor study should therefore adopt 128 LSTM units with a learning rate of 0.005 rather than the 64 units and 0.001 used throughout this thesis, and should treat any single-seed hyperparameter comparison as uninformative whenever the candidate difference is smaller than the seed-to-seed spread. The corollary is a change to the search itself: each candidate configuration should be scored on three to five seeds during the search rather than on one, which is affordable at this model size and would have avoided the sign error entirely.

Rolling-origin evaluation should be the primary protocol and the single chronological split the secondary one. The evidence for this is internal: the single-split figure and the rolling-origin figure disagree materially at long horizons, and only the latter averages over the placement of the cut. Reporting both, with the single split retained for the paired seed comparisons that require it, would give absolute figures that are comparable across studies without weakening the hypothesis tests. Any future study on windowed epidemiological inputs should also report a shuffled k-fold explicitly as a leakage control, since the gap between 0.9702 and 0.1855 measured here is large enough that a reader cannot otherwise tell which protocol produced a published number.

The domain-validation test should be strengthened in three ways. The category list should be pre-registered rather than assembled alongside the analysis; the test should weight features by rank rather than treating the top-20 as an unordered set, since a validated feature at rank 1 is stronger evidence than one at rank 20; and the fixed threshold of $p_0 = 0.70$ should be supplemented by a permutation null in which feature labels are shuffled, which measures how often a hit rate of 20 out of 20 arises by chance given the composition of the feature set. The current test is valid but conservative in construction and generous in what a hit is permitted to be.

Intrinsic attention should be compared directly against post-hoc explanation on the same trained model. Section 2.2.4 argues that SHAP and LIME are structurally awkward for multi-step time series because the explanation is an external approximation, and Section 5.1.1.3 tests the attention weights against domain knowledge; what is missing is the comparison in which both methods are applied to the same model and subjected to the same domain-validation test. That would turn an architectural argument into a measurement, and it is cheap to run because the trained models and the validation protocol already exist.

The drift diagnostics used in Section 5.2.1.4 should be adopted as standard reporting rather than as a response to a failure. Decomposing mean squared error into squared bias and dispersion, and reporting the ratio of predicted to actual standard deviation, cost one line of code each and are what distinguished a repairable scale fault from a genuine loss of signal in this study. Reporting them alongside every out-of-period evaluation would allow the mechanism of failure to be established before a remedy is chosen, which is the general lesson of Experiment 2.

Two procedural changes would have prevented the corrections documented in Section 5.1.4. First, null and alternative hypotheses should be written in formal terms and checked against their own research question before the first training run, not after the first results; two of the five nulls here failed that check. Second, the analysis conventions — directionality, multiplicity correction, effect size, interval, and the decision rule — should be fixed in the same document as the criteria, since it

was the absence of that fixed convention rather than any error in computation that allowed a two-sided uncorrected p-value to be reported as a decision. The consistency audit that recovered these should be run as a matter of course, from the per-seed files and not from the scripts that produced the original figures.

A stronger machine learning baseline should also be constructed. The fair 44-feature Random Forest used here is a defensible construction — the leakage audit that reduced the set from 45 to 44 is documented in Section 3.5.1 — but a comparator with a training R^2 of 0.96 and a negative test R^2 is an overfitted model, and a regularised or early-stopped ensemble would make the deep-learning comparison harder and therefore more informative.

6.5 Summary

This chapter has separated what the study established from what it assumed. The limitations of the study itself are that the accuracy advantage over a strong ablation is 1.6% and significant at one horizon of five; that absolute error figures are properties of the split to the extent of 22.2 RMSE; that the interpretability result is a category-membership test in which sixteen of twenty validated features are case-history variables; that no evidence of human usability was collected because the practitioner study was not conducted; that robustness is retained only to seven days under regime change; and that the whole result rests on one city, one series and one regime change, with four corrections applied before the conclusions reached their reported form.

The limitations of the methodology are that the input window was fixed on design grounds rather than tuned, that the primary protocol is a single chronological split whose fold-level instability is visible only in the rolling-origin analysis, that both baselines are constrained in ways that flatter the comparison, that power was fixed by design rather than calculated, that the efficiency measurements are relative and single-platform, and that the generalisation strand rests on one holdout and one adaptation year with a bootstrap block length set by convention.

The recommendations follow from these directly. Replication across districts, cities, strata and related Aedes-borne diseases would test whether the interpretability result reflects vector biology or local reporting behaviour. A prospective season under the recalibration schedule would convert a retrospective diagnosis into an operational validation. A properly designed practitioner study would answer the usability question this thesis leaves open. On the methodological side, window length should be tuned against a stated interpretability constraint, rolling-origin evaluation should become primary with a shuffled k-fold reported as a leakage control, the domain-validation test should be pre-registered and rank-weighted with a permutation null, intrinsic attention should be measured against post-hoc explanation on the same model, drift decomposition should be reported as standard, and the formal hypotheses and analysis conventions should be fixed before the first training run. None of these requires evidence beyond what Chapter 5 already provides to justify it.

Chapter 7

Conclusion

7.0 Conclusion

This study set out to determine whether a dengue forecasting model can explain itself without paying for the explanation in accuracy or in computational cost. It answered that question by building an Interpretable LSTM-Transformer, an LSTM encoder augmented with an additive temporal-attention pooling layer, and by testing five pre-specified hypotheses against criteria fixed before the experiments reported here were run — RO5 having been added during the revision of the study and specified before its experiment was carried out — on a Singapore dengue-weather series of 3,652 daily records from 2013 to 2022 and an entirely unseen holdout of 1,096 records from 2023 to 2025.

The five research objectives were addressed by the experiments reported in Chapter 5, four of them with the criterion met in full and RO5 with partial support only. RO1 is realised in the benchmark against SARIMA reported in Section 5.1.1.1 and RO2 in the paired comparison against the Standalone LSTM in Section 5.1.1.2, both over five horizons and thirty fixed seeds; the wider four-model comparison of Section 5.1.1.5, which adds a leakage-audited 44-feature Random Forest and applies a Friedman omnibus test with Holm-corrected post-hoc comparisons, supports both. RO3 is realised in the domain-validation protocol of Section 3.6.3 and its test in Section 5.1.1.3, and RO4 in the efficiency benchmark of Section 3.6.4 and its test in Section 5.1.1.4. RO5, added during the revision of this study, is realised in the out-of-period evaluation and six-arm adaptation comparison of Section 5.2. The architecture that all five evaluate, together with its temporal-attention interpretability surface, is specified in Section 3.2.2.

Four of the five hypotheses are supported and one partially. H1 holds without qualification: the proposed model reduces RMSE against SARIMA by 42.5% to 52.0% at every horizon, winning 30 of 30 seeds, with every Holm-corrected p below 10^{-23} against a criterion that required only 20%. H2 holds marginally: the attention module lowers error by 1.6% against its own ablation, significant at $h = 1$ d and on the pooled test but not at the four longer horizons. H3 holds at every horizon: all twenty of the consensus top-20 attention-ranked features are domain-validated, giving an exact binomial p of 7.98×10^{-4} with a Clopper–Pearson lower bound of 0.861, stable across all thirty seeds. H4 holds on every metric: 1.227 $\times$ the parameters and 1.234 $\times$ the model size of a plain LSTM, training time between 0.828 $\times$ and 1.352 $\times$ and inference latency between 0.946 $\times$ and 1.334 $\times$, with the attention mechanism's scaling exponent in sequence length measured at 0.526 [0.312, 0.740] against 1.418 [0.893, 1.942] for full self-attention. H5 is partially supported: the frozen model retains positive R^2 at $h = 1$ d and $h = 7$ d on data three years beyond its training window, loses it at the longer horizons, and recovers it at all five after a two-parameter affine recalibration fitted on one recent year.

Read together rather than hypothesis by hypothesis, these results support a narrower and more defensible claim than the one they were framed to establish. The claim is not that this architecture forecasts dengue better than the alternatives, because against a strong ablation it does so by 1.6% at one horizon of five. The claim is that the architecture produces explanations that survive a formal test against domain knowledge, and produces them at a bounded and measured cost. Interpretability in this setting is close to free in accuracy terms, and the question worth asking of it is therefore not what it costs but whether it can be trusted — a question this study answers with a hypothesis test rather than with a visualisation.

Four contributions follow. The first is methodological: no hybrid dengue forecasting study identified in the review of Section 2.3.1 tests whether the attention weights its model produces correspond to domain-known risk factors, and this study closes that gap with a pre-set criterion, an exact test, a confidence interval and a stability check across thirty seeds. Interpretability moves from an asserted property to a tested one. The second is empirical: the cost of that interpretability is measured rather than assumed, and the choice of additive temporal-attention pooling over multi-head self-attention is shown to be what makes it affordable, with the advantage growing from a factor of 1.9 at $T = 28$ to 49.0 at $T = 1,792$. The third is operational: the model needs a recalibration schedule rather than a retraining schedule. Retraining from scratch on more recent data is worse than doing nothing at every horizon, while a two-coefficient affine correction restores positive skill at all five without touching a trained weight — and, because attention is computed before the output layer, without disturbing the explanations that were validated. The fourth is a methodological caution for the field: a random shuffled k-fold on this data reports $R^2 = 0.9702$ against 0.1855 for a correct rolling-origin protocol, so a reported R^2 above roughly 0.95 on a daily dengue series should prompt a question about the validation split before a question about the architecture.

These outcomes speak directly to the Sustainable Development Goals claimed in Chapters 1 and 2. The primary claim was to SDG 3 (Good Health and Well-being), on the argument that forecasting supports health responses that are proactive rather than reactive. The evidence supports that claim in a specific and bounded form: at one and seven days ahead the model retains real skill on a period it never saw, which is the operational window in which alert generation and vector-control deployment actually occur, and the maintenance condition under which it holds — annual recalibration, with the predicted-to-actual dispersion ratio as the monitoring trigger — is stated rather than left implicit. A forecast that is accurate only until the epidemiological regime shifts, and whose owner has no way of knowing when that has happened, would not serve SDG 3; one with a diagnosis and a cheap remedy attached does.

The claim to SDG 10 (Reduced Inequalities) and SDG 11 (Sustainable Cities and Communities) rested on the argument that dengue falls hardest on lower-income communities in tropical urban regions, so that earlier and better-targeted response is a matter of health equity. This study advances that claim through the interpretability result rather than through the accuracy result: a forecast whose drivers can be inspected is one whose recommendations can be interrogated by the people they affect, and the validated ranking makes the basis of a warning legible rather than asserted. The claim is advanced but not completed, because the analysis is national and daily; the equity question is spatial, and Section 6.3 identifies district-level replication as the work that would answer it.

The claim to SDG 16 (Peace, Justice and Strong Institutions), made in Section 2.4.5 on the ground that visible reasoning is challengeable reasoning, is supported by the same mechanism and by the reporting discipline applied throughout Chapter 5. Two of five null hypotheses were corrected before testing, one claim was withdrawn on a first measurement and reinstated on a better one, one was narrowed once intervals were attached to point estimates, and one conclusion was reversed once the arms were compared on a common evaluation window. Every one of these is on the record with the correction beside it. An institution asked to act on an algorithmic forecast is entitled to see not only what the model concluded but what its authors got wrong on the way, and this thesis treats that sequence as part of the result rather than as part of its production.

Two further claims were made in Chapter 2 and are answered here for completeness. The claim to SDG 9 (Industry, Innovation and Infrastructure) rested on the argument that an interpretable forecasting capability is only innovation if it can actually be built and run on ordinary infrastructure. The measurements in Section 5.1.1.4 settle that: the whole model is 41,745 parameters and 548 KB, it trains in tens of seconds on a standard workstation with no specialised accelerator, and the attention mechanism that supplies the explanations scales sub-quadratically in input length, so the capability does not become unaffordable as the input window grows. The claim to SDG 17 (Partnerships for the Goals) rested on the dependence of any such system on data released by public agencies. That dependence proved to be real rather than rhetorical: the benchmark series was assembled from National Environment Agency surveillance records and Open-Meteo reanalysis data, and the 2023–2025 holdout could only be constructed because the Ministry of Health and Communicable Diseases Agency publishes weekly bulletins, which had to be transcribed by hand once the open dataset ended in December 2022. The work is therefore a demonstration of what open surveillance data makes possible, and equally of where the partnership is thin — a three-year gap in machine-readable release is the single largest obstacle a successor study would face.

What remains open is stated plainly. The practitioner-validation study originally specified for H4 was not conducted, so this thesis demonstrates that the explanations are epidemiologically coherent and cheap to produce and demonstrates nothing about whether a public health officer can read and act on them under time pressure. Validated interpretability does not confer robustness; the model whose explanations hold at all five horizons is the same model that loses skill beyond seven days when the regime changes. Absolute accuracy is a property of the split to the extent of 22.2 RMSE, so the figures reported here are accuracies on a stated split and not properties of the model. And the whole result rests on one city, one series and one regime change. Chapter 6 sets out the replication, stratification, prospective evaluation and methodological changes that would remove each of these boundaries.

The contribution this thesis makes, finally, is a change in the question. Interpretable deep learning is conventionally framed as a trade: how much accuracy must be surrendered to obtain an explanation. On this problem the trade is negligible — 1.6% against a strong ablation, at a cost of $1.227\times$ the parameters — and the framing is therefore the wrong one. The useful question is whether the explanation is true, and that is a testable proposition. This study shows that it can be tested, sets out a protocol for testing it, reports the result of doing so, and states without softening the boundaries within which that result holds.

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Appendix

ABSTRACT

Dengue fever remains a significant public health challenge in tropical regions like Southeast Asia, where accurate and timely outbreak prediction is critical for effective intervention. Current predictive models often lack the precision to capture the complex, nonlinear relationships between climate variables, urban environments, and dengue transmission dynamics. This research project proposes and investigates a novel Hybrid LSTM-Transformer model for dengue outbreak forecasting. The study focuses on developing a methodological framework that integrates diverse data sources—including historical case counts, climate data (e.g., from OpenWeatherMap API), and population statistics. The primary objectives are to design the hybrid architecture, rigorously benchmark its predictive performance against established baseline models (e.g., SARIMA, Random Forest, LSTM), and analyze its interpretability through attention mechanisms. The expected outcome is a validated predictive model that demonstrates superior accuracy in multi-week ahead forecasts. By advancing the methodological frontier in epidemiological forecasting, this research contributes valuable insights towards the development of future early warning tools, ultimately supporting public health goals under Sustainable Development Goal 3 (Good Health and Well-Being).

PROBLEM

Dengue fever is a prevalent public health threat in tropical and subtropical regions, resulting in significant morbidity, mortality, and economic burden. A core challenge in its control is the difficulty in accurately and timely predicting outbreaks, which hinders effective preparedness and response. The specific problems this project aims to address are detailed below.

The transmission of dengue fever involves complex nonlinear interdependencies among various drivers, such as climate variables (temperature, rainfall, humidity), urban characteristics, and population dynamics. These relationships cannot be effectively captured by traditional statistical models (such as SARIMA). These models often fail to adapt to regional transmission patterns, leading to inaccurate predictions and leaving public health in a reactive state. Furthermore, the lack of a standardized, repeatable framework for integrating diverse data sources (e.g., historical case data, real-time climate data, demographic data) poses a significant obstacle to developing robust and scalable prediction systems. Liu et al. (2023) pointed out that traditional time-series models have significant limitations in integrating external covariates and capturing long-term nonlinear trends. Patel et al. (2023) also emphasized that the lack of a unified multi-data fusion method is one of the key reasons for the poor generalization ability of current epidemic prediction models. This lack of reliable tools hinders forward-looking resource planning by healthcare providers and leaves policymakers without the necessary data-driven evidence to support strategic public health planning and funding justification, ultimately leading to a cycle of inefficient pandemic management.

SOLUTION

The proposed research focuses on the development and evaluation of a novel Hybrid LSTM-Transformer model for dengue outbreak forecasting. The core aim is to advance the methodological frontier in epidemiological prediction by creating a model that can provide accurate multi-week ahead forecasts of dengue case incidence, thereby providing a methodological foundation for a potential shift from reactive to proactive public health management.

The core innovation is the hybrid deep learning architecture. It synergizes Long Short-Term Memory (LSTM) networks, which excel at capturing long-term temporal dependencies and trends

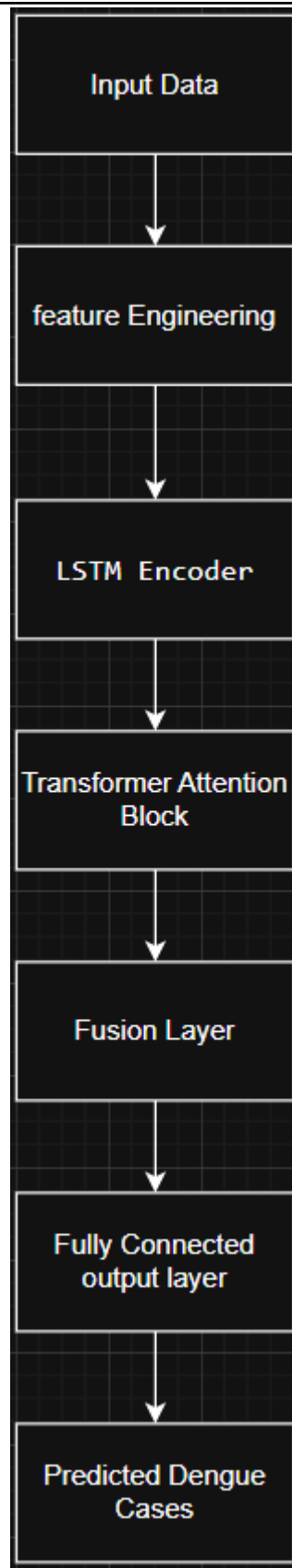
in sequential data, with a Transformer component. The Transformer's multi-head self-attention mechanism dynamically weights the importance of different input features (e.g., specific climate variables like recent rainfall or sustained high temperature) across time. This allows the model to adaptively focus on the most relevant signals for outbreak prediction, a capability beyond standalone models.

The research will be conducted through a systematic methodological process:

1. Data Pipeline Construction & Feature Engineering: A reproducible data processing pipeline will be built using tools like Pandas and Scikit-learn. This pipeline will integrate and clean data from various sources, including historical dengue case data (potentially from government health repositories), climate data (sourced from APIs like OpenWeatherMap), and population statistics. The focus will be on creating and evaluating relevant time-lagged and aggregated features that are predictive of outbreaks.

2. Model Development & Comparative Performance Analysis: The Hybrid LSTM-Transformer model will be implemented using deep learning frameworks such as TensorFlow or PyTorch. The core of the research involves rigorously benchmarking its performance against established baseline models—including SARIMA (using Statsmodels), Random Forest (Scikit-learn), a standard LSTM, and a CNN-LSTM model. Evaluation will employ standard metrics like Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) across multiple prediction horizons (e.g., 1 to 8 weeks) to comprehensively assess forecasting accuracy and robustness.

57. Model Interpretability & Analysis: To ensure the model's findings are actionable for epidemiological understanding, a key research component will be interpretability analysis. This involves analyzing the Transformer's attention weights and employing auxiliary feature importance methods (e.g., SHAP) to identify and validate the key drivers (e.g., specific climate patterns) that the model associates with outbreak risk. The final research deliverables will be a reproducible methodological framework, the validated model, and a comprehensive analysis of its performance and interpretability.



Input Data Stream:

Historical dengue case counts, climate data (temperature, rainfall, humidity), and demographic features.

Data is formatted as multi-variate time-series.

Feature Engineering Module:

Creates lagged features, rolling averages, and normalized inputs.

Prepares data for deep learning models.

LSTM Encoder:

Captures long-term temporal dependencies and trends in sequential data.

Outputs hidden states representing learned temporal patterns.

Transformer Attention Block:

Uses multi-head self-attention to dynamically weight the importance of different features and time steps.

Allows the model to focus on relevant signals (e.g., recent rainfall spikes, sustained high temperatures).

Fusion Layer:

Concatenates outputs from LSTM and Transformer branches.

Combines temporal memory with attention-weighted features.

Fully Connected Output Layer:

Maps fused features to multi-week ahead dengue case predictions.

Prediction Output:

Forecasts dengue case counts for 1 to 8 weeks ahead.

TARGET MARKET

This research is designed to generate methodological advancements and insights with significant potential value for a range of stakeholders in the public health domain, particularly in dengue-endemic regions like Southeast Asia. The primary beneficiaries are those entities engaged in or impacted by dengue surveillance, research, and policy-making, who would benefit from more accurate forecasting methodologies.

The academic and practical contributions of this research are assessed through several lenses of feasibility and impact:

Technical & Methodological Contribution: The research develops a reproducible and scalable modeling framework. The proposed Hybrid LSTM-Transformer architecture is designed to be generalizable and can be adapted for case studies in different geographical regions within the endemic zone, providing a template for future research.

Economic & Resource Impact: The research addresses a critical economic problem—the high burden of unmanaged outbreaks. By focusing on improving forecast accuracy, the findings aim to provide evidence that can inform more cost-effective public health resource planning and allocation.

Operational & Decision-Support Potential: The core outputs of the research—a validated predictive model and an analysis of key outbreak drivers via interpretability methods—are designed to provide actionable insights. These insights are tailored to inform evidence-based decision-making, requiring translation by experts but grounded in robust analytical findings.

Research & Policy Relevance: The demand for improved forecasting methodologies is high, given the persistent burden of dengue. This research directly addresses this gap in methodological tools, supporting long-term public health research goals.

Social Impact Alignment: The project directly addresses a critical social need—advancing tools for protecting population health—and aligns with global initiatives like Sustainable Development Goal 3 (Good Health and Well-Being). It promotes health equity by focusing on a methodology applicable to vulnerable regions.

Specifically, the research outputs are intended to benefit:

1. **Public Health Researchers & Epidemiologists:** By providing a novel, interpretable hybrid modeling framework for dengue forecasting that can be replicated, extended, or serve as a benchmark for future methodological studies.

2. **Public Health Officials & Analysts:** By offering evidence-based insights (through model outputs and driver analysis) that can inform surveillance strategies, risk assessment, and theoretical intervention planning.

3. **Healthcare Policy Researchers & Planners:** By contributing methodological evidence that can support long-term strategy development, scenario modeling, and data-driven rationale for public health investments.

4. **The Academic Community in Data Science & AI for Health:** By advancing the state-of-the-art in applying hybrid deep learning architectures to complex epidemiological forecasting problems.

COMPETITION / CONTRIBUTION

This research aims to contribute to the advancement of epidemiological forecasting methodologies, supporting the broader goals of Sustainable Development Goal 3: Good Health and Well-Being. Its specific contributions are framed within an academic and methodological context:

1. **Theoretical & Technical Contribution:** The primary contribution is the proposition and empirical validation of a novel Hybrid LSTM-Transformer model. This architecture is designed to advance the methodological state-of-the-art in capturing the complex spatio-temporal dynamics of dengue fever transmission.

2. **Methodological Contribution:** The research will establish a reproducible and transparent data processing and modeling pipeline for integrating multi-source public health data (epidemiological, climatic, demographic). This framework is intended to serve as a reference for future studies in the field.

3. **Practical & Interpretability Contribution:** By rigorously analyzing the model's performance and, crucially, its interpretability via attention mechanisms, this research will generate actionable insights into key outbreak drivers. These evidence-based findings have the potential to inform public health theory and practice, providing a knowledge base for more targeted interventions.

Comparative Analysis with Existing Methodologies:

The proposed model is situated within a landscape of existing forecasting approaches, which serve as benchmarks and highlight the research gap this study addresses.

1.Traditional Statistical Models (e.g., SARIMA): These are well-established for time-series forecasting but are often limited in their ability to model complex, non-linear interactions and flexibly incorporate heterogeneous external factors (e.g., climate variables). Our hybrid model is architecturally designed to address these limitations.

2.Conventional Machine Learning Models (e.g., Random Forest) and Basic Deep Learning Models (e.g., standalone LSTM): While more flexible than SARIMA, these models have inherent constraints for sequential prediction tasks. A standard LSTM may not adaptively prioritize different predictive features over time, and a Random Forest lacks intrinsic design for modeling temporal dependencies. Our research investigates the hypothesis that integrating LSTM with a Transformer's attention mechanism creates a superior architecture for dynamically weighting both temporal patterns and feature importance, which is posited to lead to gains in predictive accuracy and model interpretability.

Unlike AI research focused on clinical diagnostics, this project's distinct position is its methodological investigation of a hybrid deep learning architecture specifically for population-level, spatio-temporal outbreak forecasting, with the goal of producing a validated and interpretable model for the research community.

MILESTONES

Proposed Research Method : Agile Development & Cross-Industry Standard Process for Data Mining

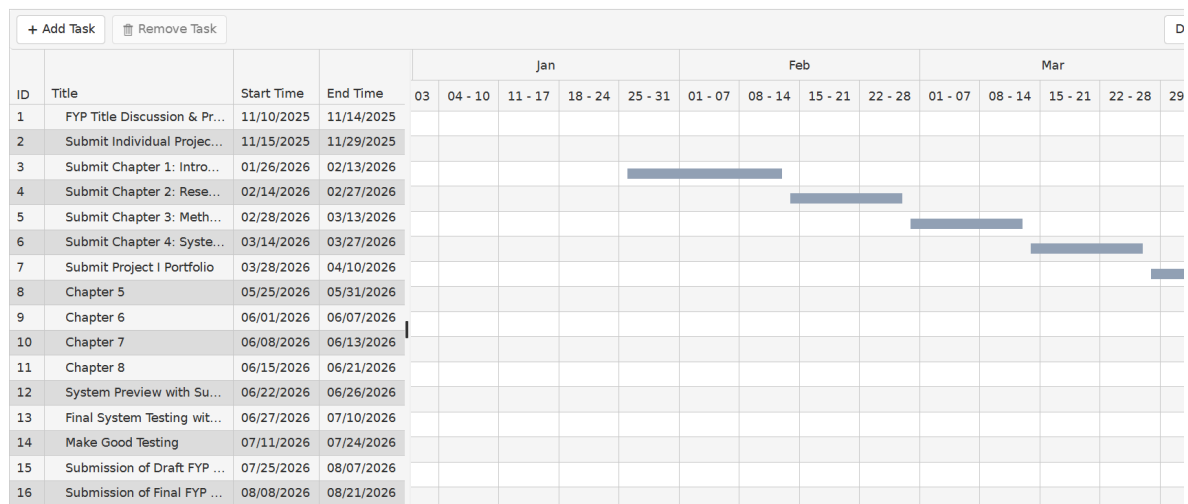
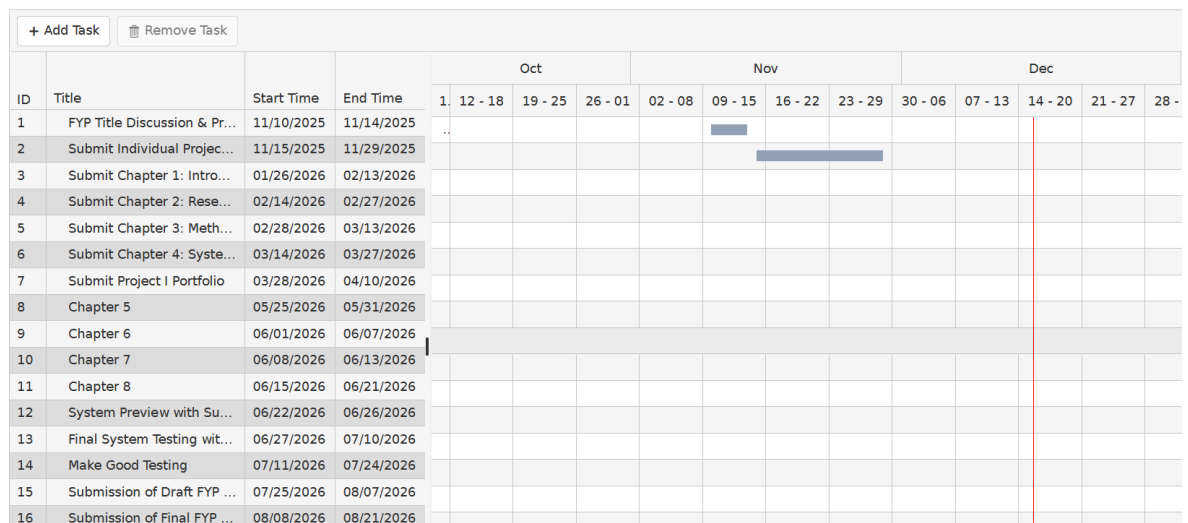
Project Schedule:

ACTIVITIES	EXPECTED OUTCOME	COMPLETION DATE
FYP Title Discussion & Proposal Drafting	Finalized project title and scope.	24 Nov 2025
Submit Individual Project Proposal (Form 1, 2, 3)	Completed and approved project proposal.	29 Nov 2025
Submit Chapter 1: Introduction	Detailed problem statement, objectives, and project significance.	13 Feb 2026
Submit Chapter 2: Research Background	Comprehensive literature review and analysis of existing systems.	27 Feb 2026
Submit Chapter 3: Methodology & Requirements Analysis	Detailed data pipeline design, model architecture, and implementation plan.	13 Mar 2026
Submit Chapter 4: System Design	System architecture diagrams, database schema, and interface mockups.	27 Mar 2026

A Hybrid Deep Learning Approach for Dengue Fever Prediction in real-world case study

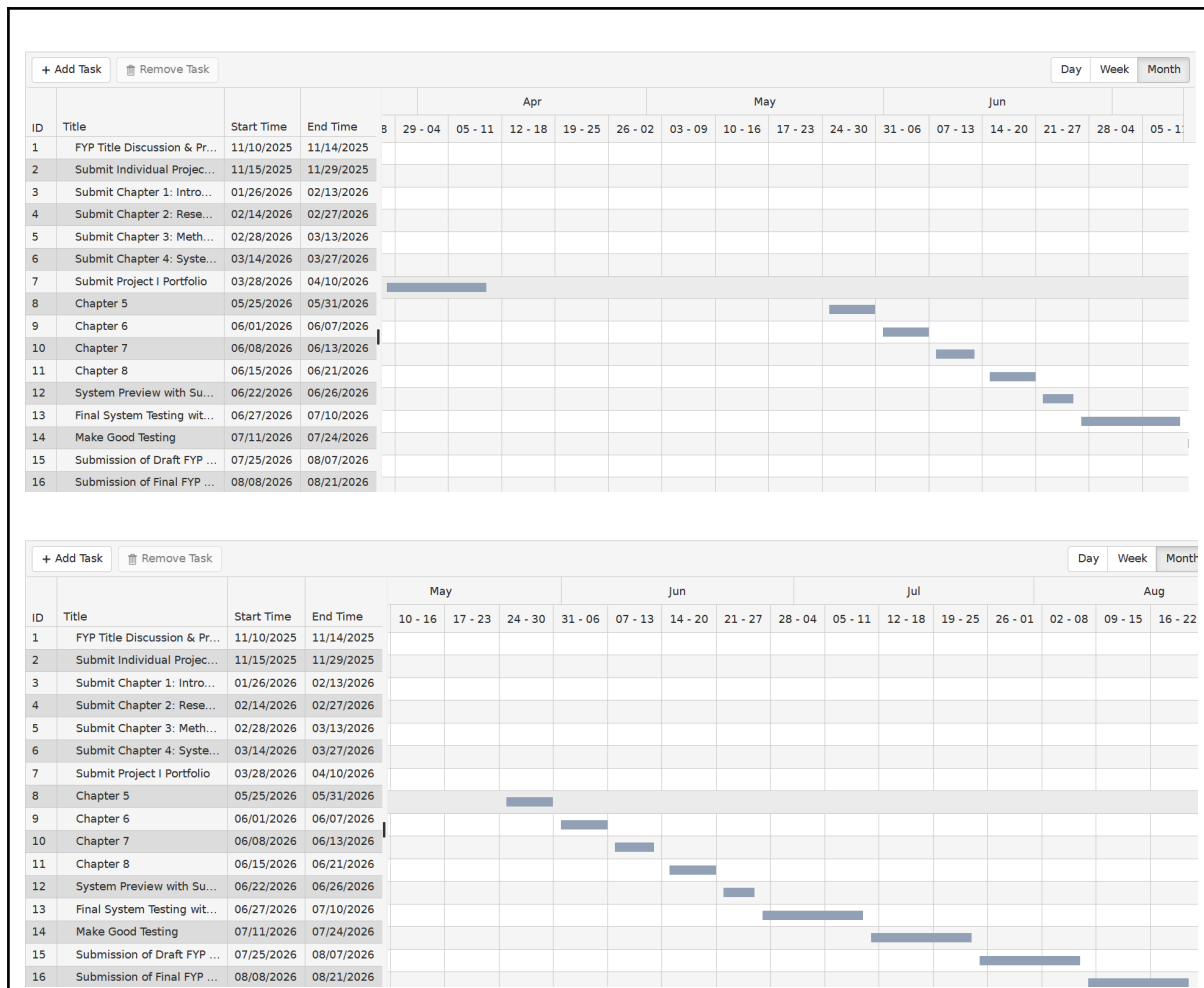
References

Submit Project I Portfolio	Consolidated portfolio including all chapters (1-4) and the project proposal.	10 Apr 2026
System Preview with Supervisor	A working prototype of the data pipeline and a trained baseline model for initial feedback.	26 Jun 2026
Final System Testing with Supervisor & Moderato	Fully functional prediction system and dashboard, ready for evaluation.	10 Jul 2026
Make Good Testing	Re-testing for any components that did not initially meet requirements.	24 Jul 2026
Submission of Draft FYP Report (individual)	Complete draft of the final thesis document for review.	7 Aug 2026
Submission of Final FYP Report (individual)& Deliverables	Finalized thesis, all source code, datasets, and the deployed system prototype	21 Aug 2026



A Hybrid Deep Learning Approach for Dengue Fever Prediction in real-world case study

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